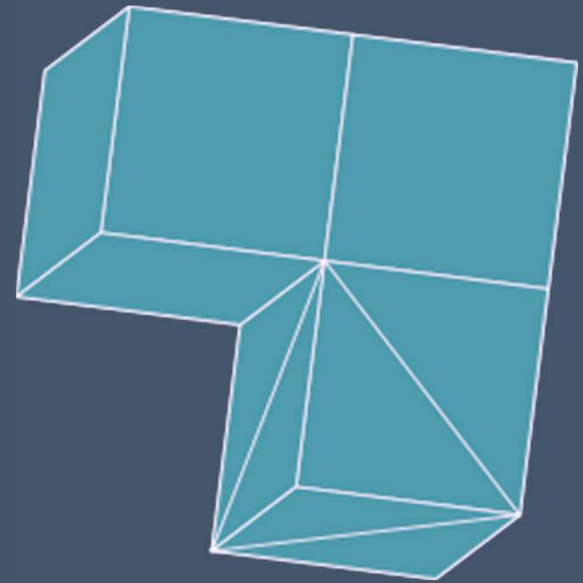


An Introduction to Kratos Multiphysics

Pooyan Dadvand

Riccardo Rossi



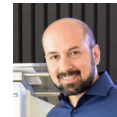
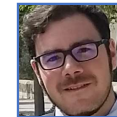
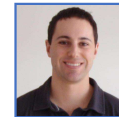
Multi-Disciplinary Is Teamworking!



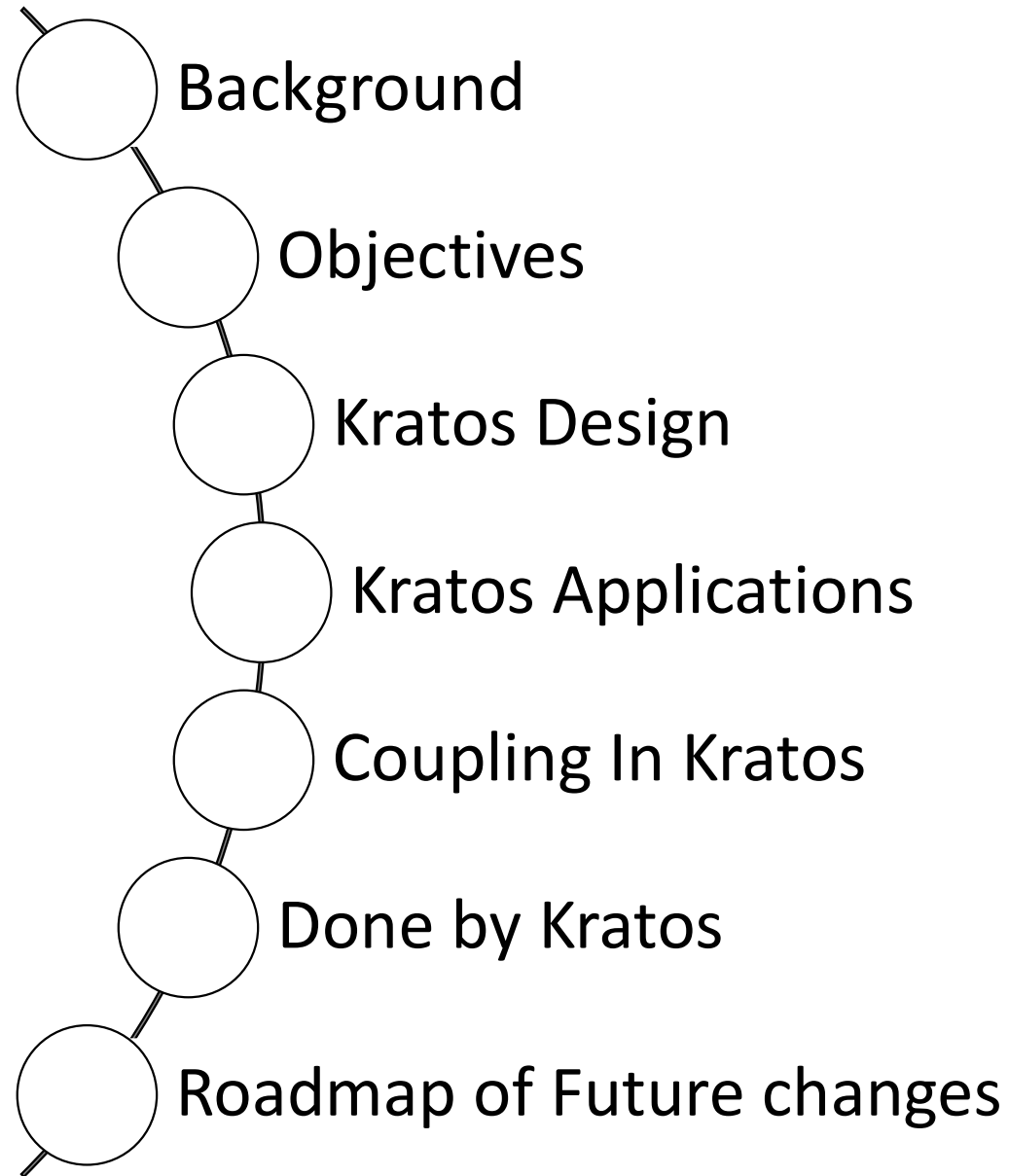
Institut für Statik
und Dynamik

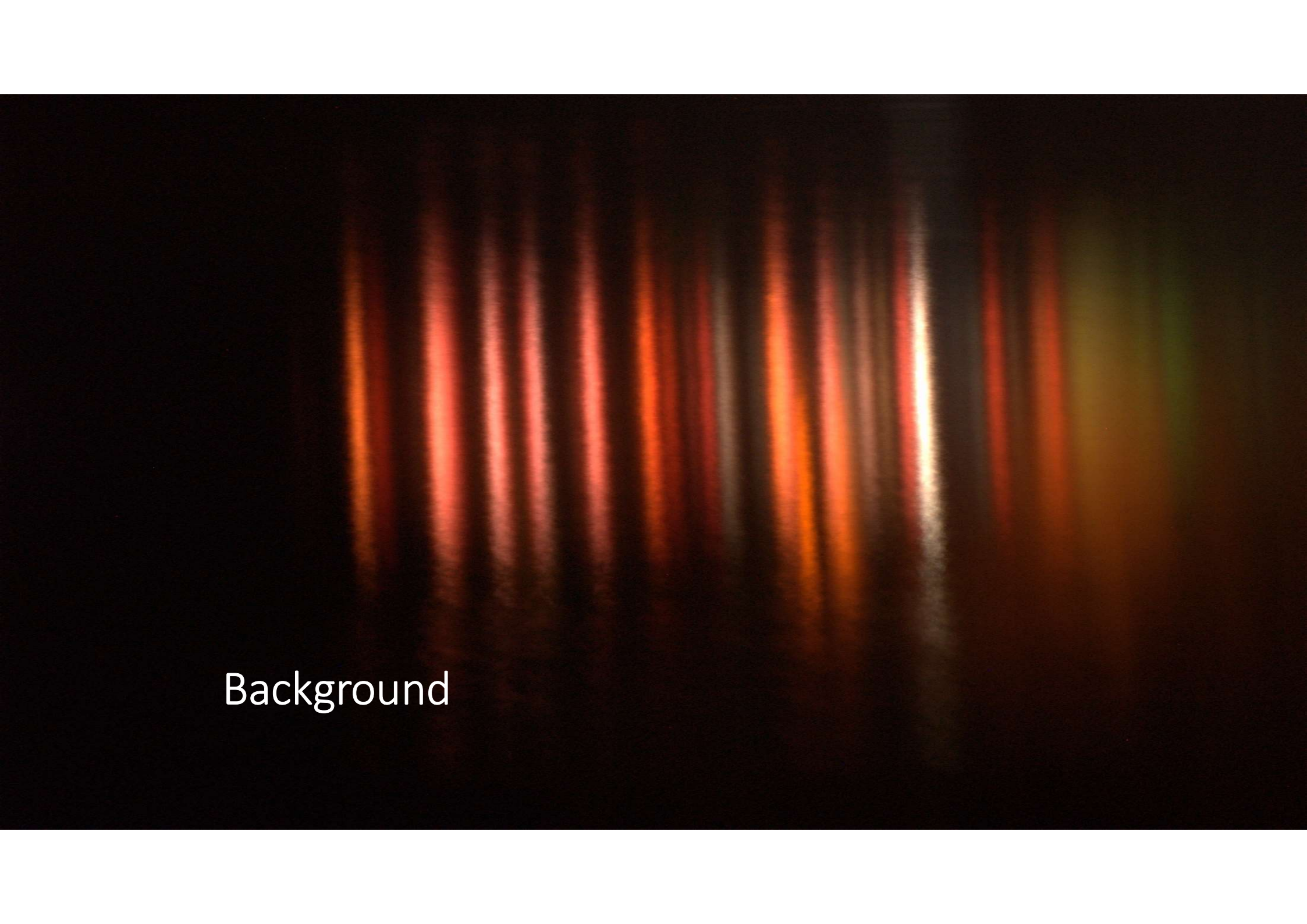


Deltares



Outline



The background of the slide is a dark, almost black, field. It is populated with numerous vertical streaks of light. These streaks vary in width and intensity, creating a sense of depth and movement. The colors of the streaks are primarily warm, ranging from deep reds and oranges to bright yellows and whites, particularly in the center-right area where the light is most intense. The overall effect is reminiscent of light trails from a long-exposure photograph or perhaps reflections on a dark, rippling surface.

Background

EVOLUTION: FROM IDEA TO PRODUCT

$\int d\Omega$

First Idea

- May lead to a breakthrough solution
- High Risk
- Academic Task (Post-Doc, PhD or master projects)

EVOLUTION: FROM IDEA TO PRODUCT

$\int d\Omega$

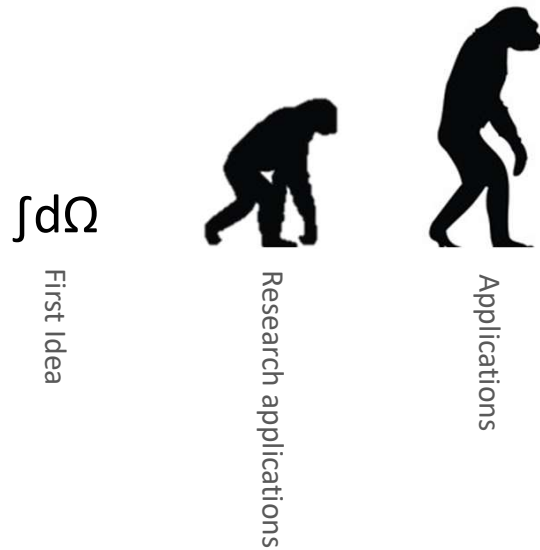
First Idea



Research applications

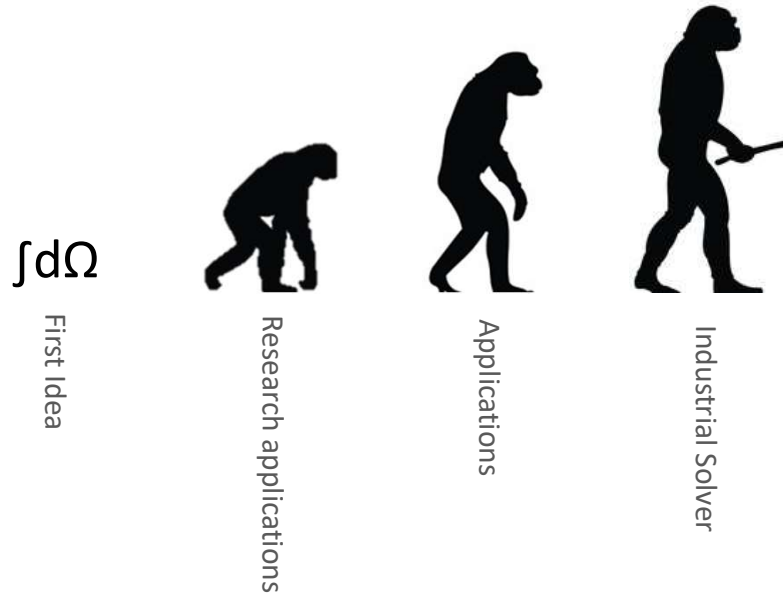
- Some experimental development
- A working prototype with less associated risk
- Suitable for a small test problem

EVOLUTION: FROM IDEA TO PRODUCT



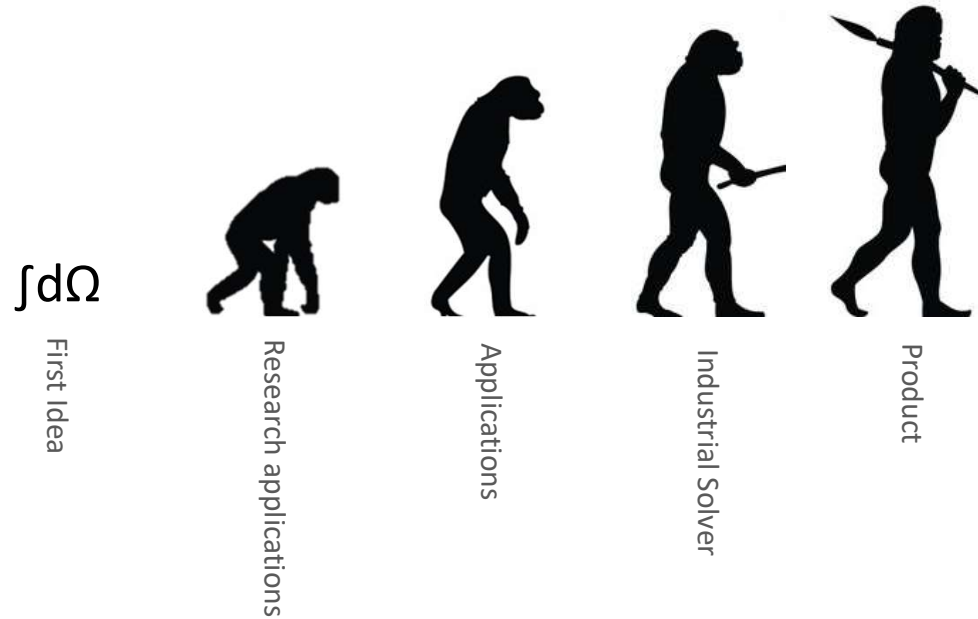
- Rewriting the prototype in more efficient way (Cpp/Fortran)
- Making it more generic
- Avoiding common bottlenecks
- Increasing the code quality
- Adding QA

EVOLUTION: FROM IDEA TO PRODUCT



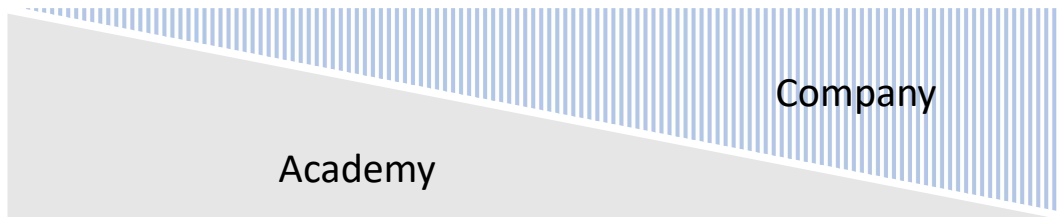
- More physics and calibration
- Improving the robustness
- More QA
- Improving the code quality

EVOLUTION: FROM IDEA TO PRODUCT

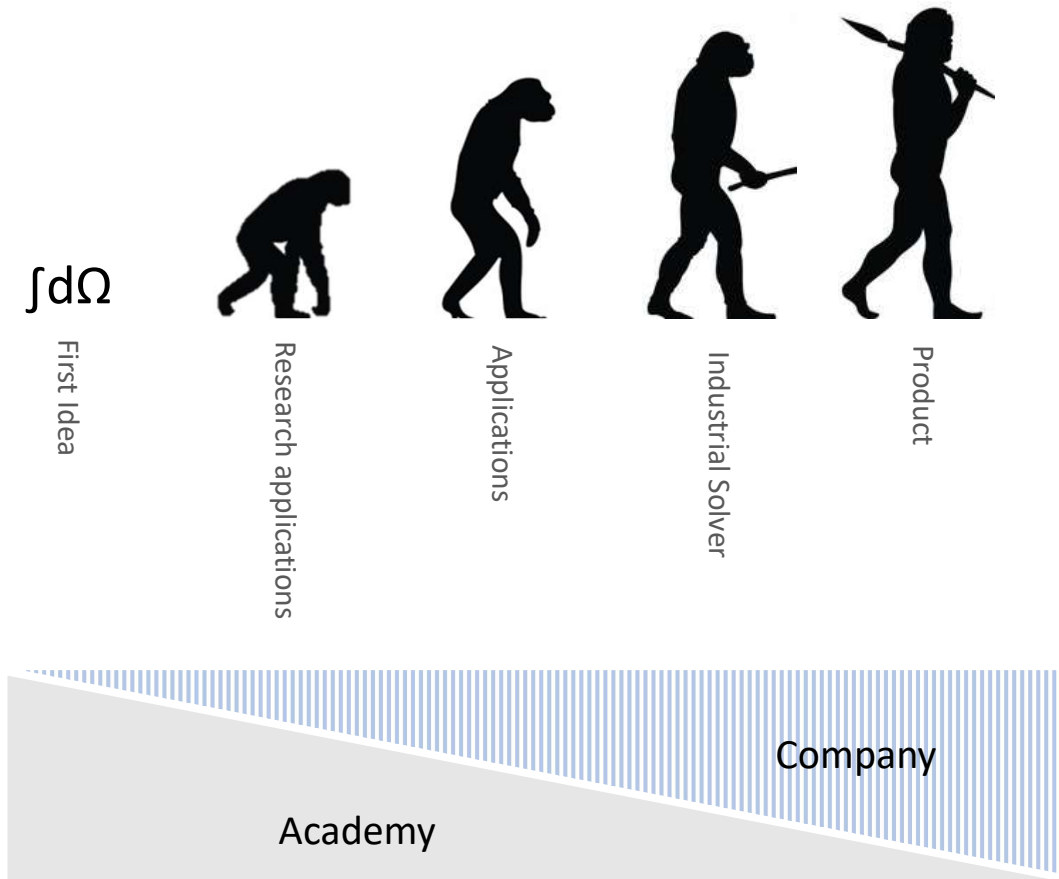


The Solver +

- GUI
- Validation
- Automatization
- Final tuning
- Maintenance
- Documentation
- Distribution
- Support



EVOLUTION: FROM IDEA TO PRODUCT



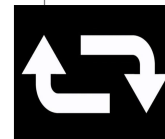
Slow evolution

- Large implementation times
- Ignoring scalability and efficiency



Difficulties in passing to industry

- Reimplementing the idea
- Problems in knowledge transfer

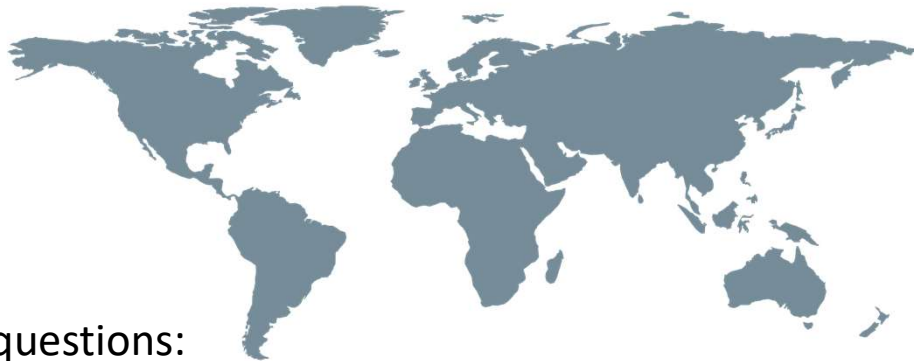


Repeating again and again...

- Starting from scratch
- Reimplementing others' works to test

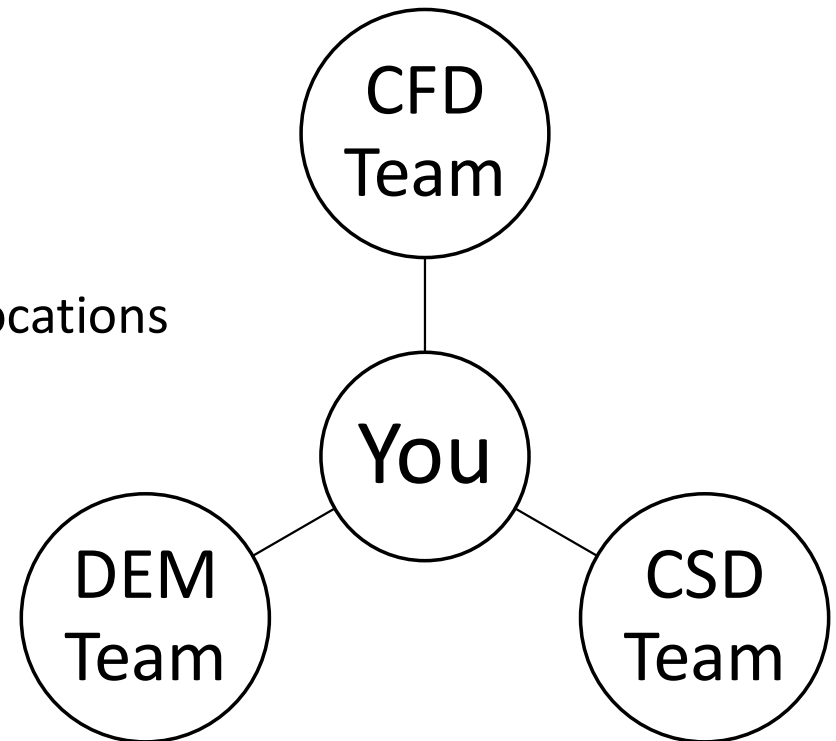
Multi Disciplinary Implementation

- Team Working is an issue in multi-Disciplinary implementation
- Even more complicated with teams at different locations



Common questions:

- HOW to find innovative solutions?
- WHOM to collaborate with?
- HOW to collaborate?



Objective

Creating a monster?

~ 120 developers

~ 1 million line of code



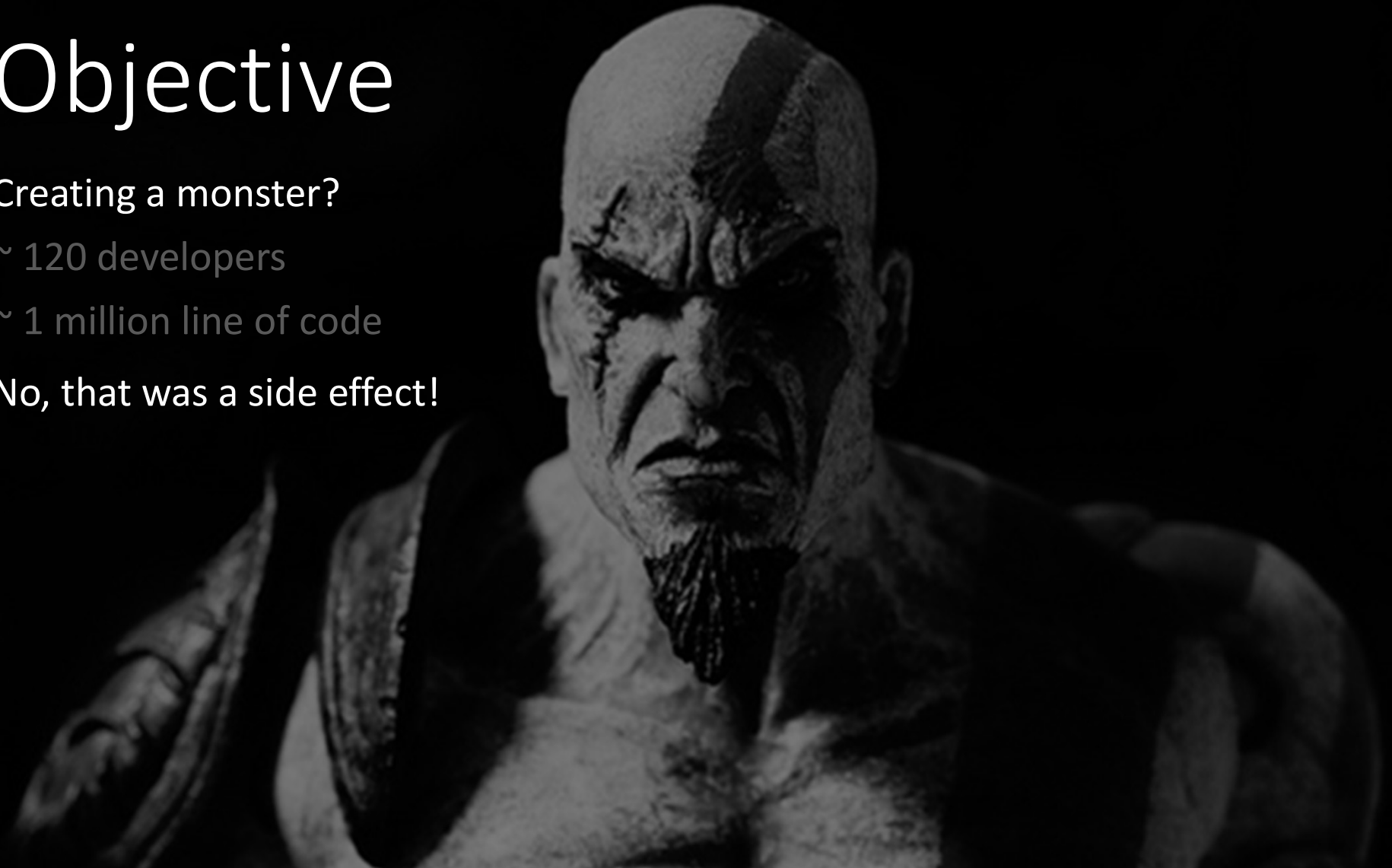
Objective

Creating a monster?

~ 120 developers

~ 1 million line of code

No, that was a side effect!

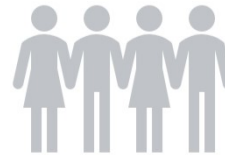




Accelerate
the evolution



Bridge for
technology
transfer



Enable Multi-
Disciplinary
team working



An Open
repository of
knowledge

Design Criteria

Adding a new physic/solution should be easy

Flexibility and **Extensibility** are the keywords

Design Criteria

Multi Disciplinary problems can become very large

Large scale calculation and **scalability** are other keywords

Design Criteria

Multi Disciplinary problems involve teams with different expertise

Collaborative environment and **community** are further keywords

They should also **share the code** so being **Open-Source** is a key



A bit of history

Kratos Timeline

Everything started in C4 office where **Xavi Royo** and I decided to join in development of an adaptive multi-physics solver



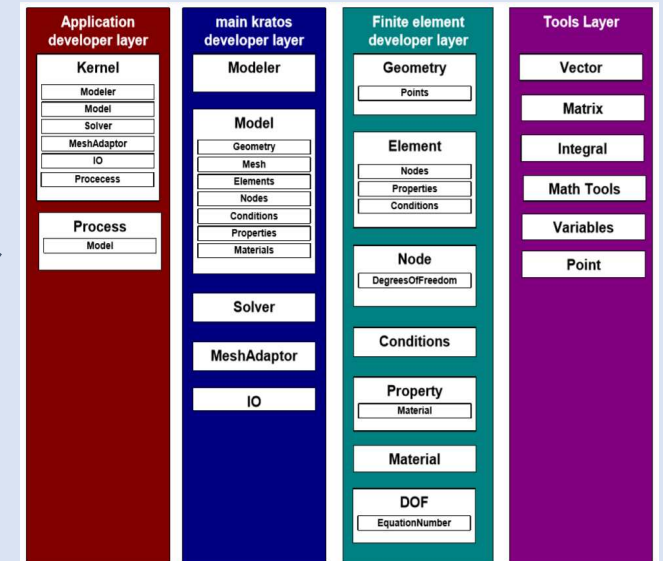
Most of the design has been done during pilgrimage to Santiago with **Xavi**



Javier Mora
First who joined



First design



LGPL License

Unfortunately, Xavi passed away some months later leaving me alone on this way...

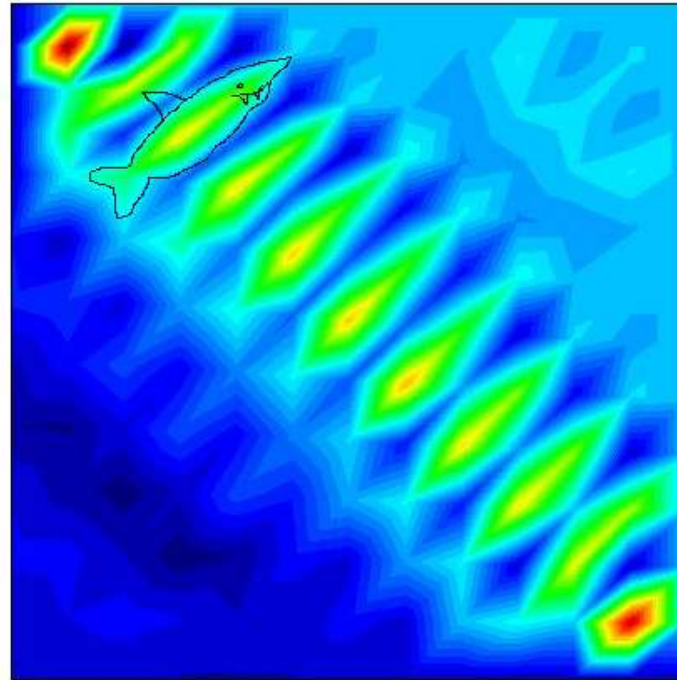
2001

2002

Riccardo joined the team



Convection Diffusion solution still with
stabilization problem!



Too fast shark in the sunset

The first Kratos art!

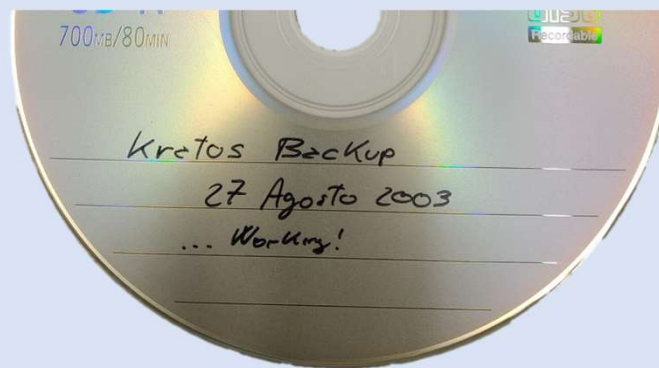
2003

All star team

- Riccardo Rossi
- Santiago Badia
- Roberto Flores
- Ruben Otin

- Structural
- Fluid
- Electro magnetics
- Thermal

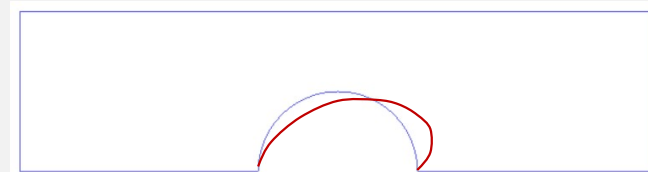
Working!



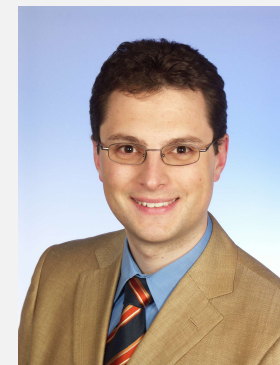
but the code was too complex and slow

Riccardo is back!

First FSI: a 2D bubble

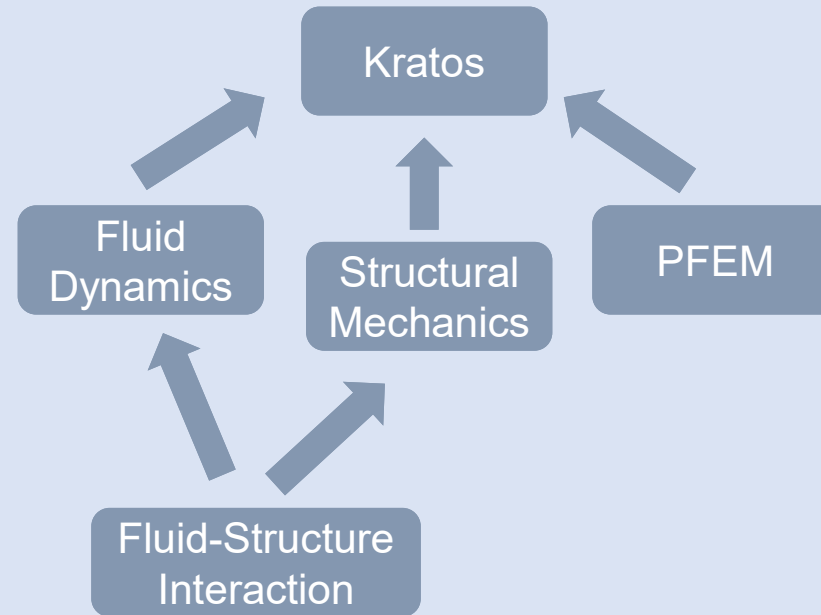


Catching Roland's attention!



Reimplementing the code from scratch base on the original design but taking into account the version 0 experience

- Reducing flexibility
- Increasing the performance
- Separating applications form core
- Reducing implementation complexities
- Still very flexible data structure
- Changing our own scripting input with Python



2004

Meanwhile all developers **except Riccardo** changed to other codes...

2005

Institute for Structural
Mechanics joined the project



giving an important push to
structural application

Prof. Idelsohn support
for PFEM in Kratos



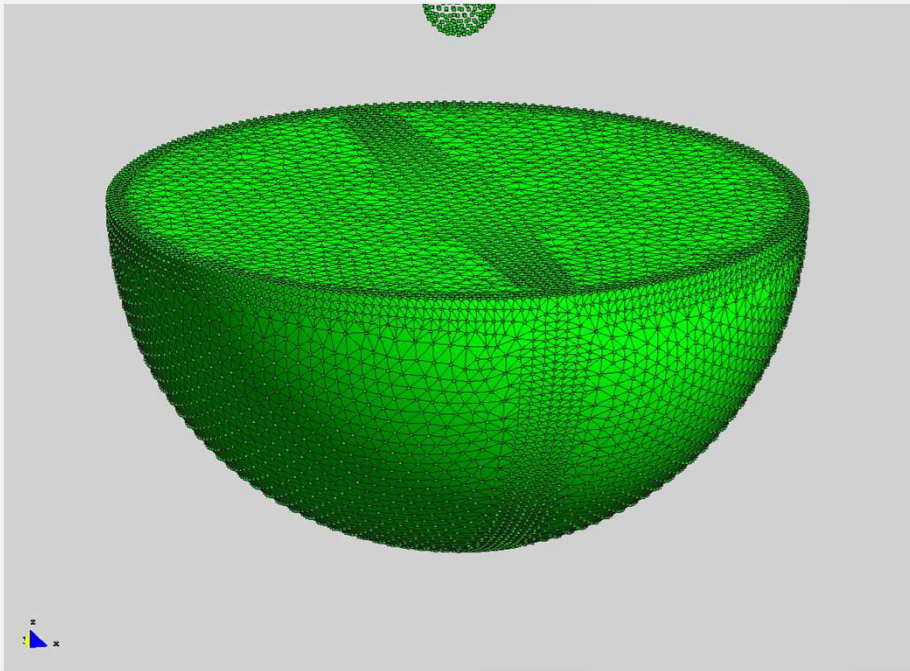
Pavel Ryzhakov
started with PFEM



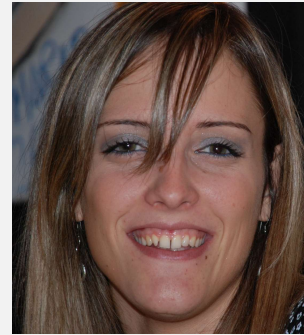
Janosch Stascheit
started with
tunneling process



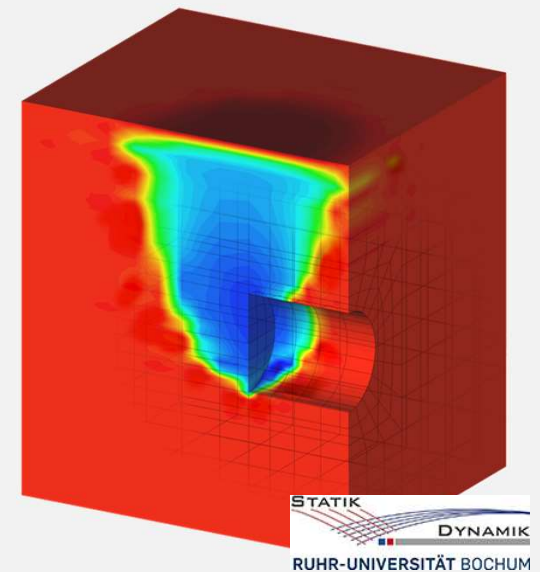
PFEM implementation in Kratos improves the remeshing capability and robustness of fluid solver



Antonia Joined!

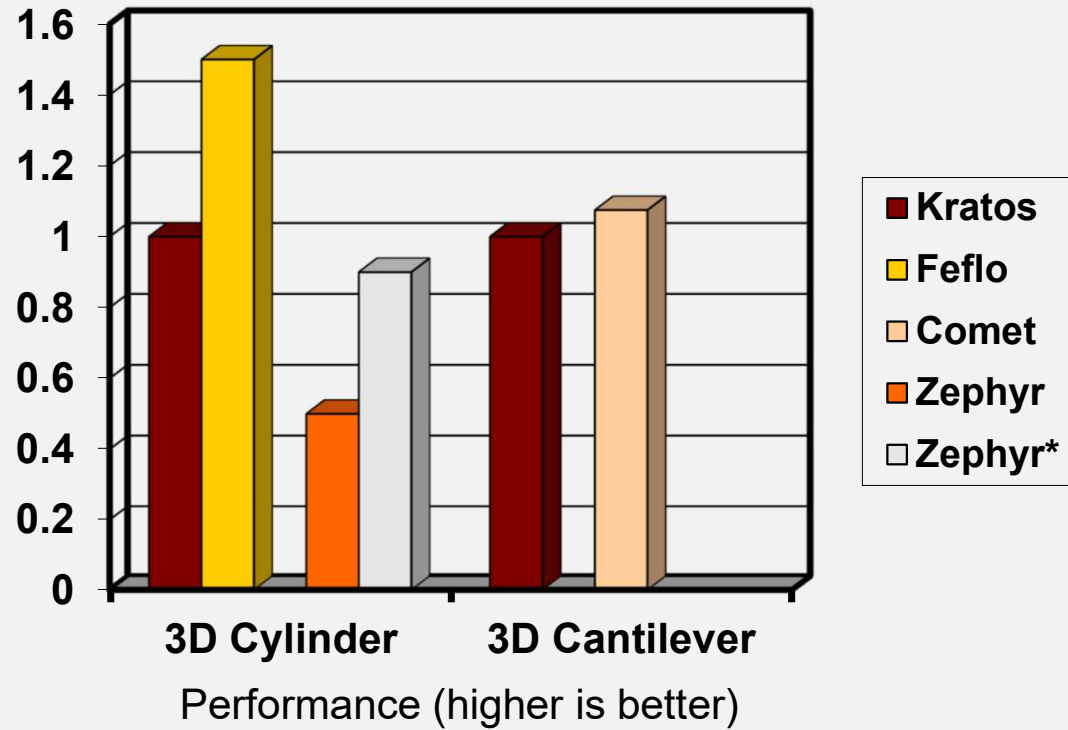


Tunneling process
courtesy of Janosch Stascheit



2006

Kratos starting to be comparable to single purpose codes in performance



Starting with MPI parallelization

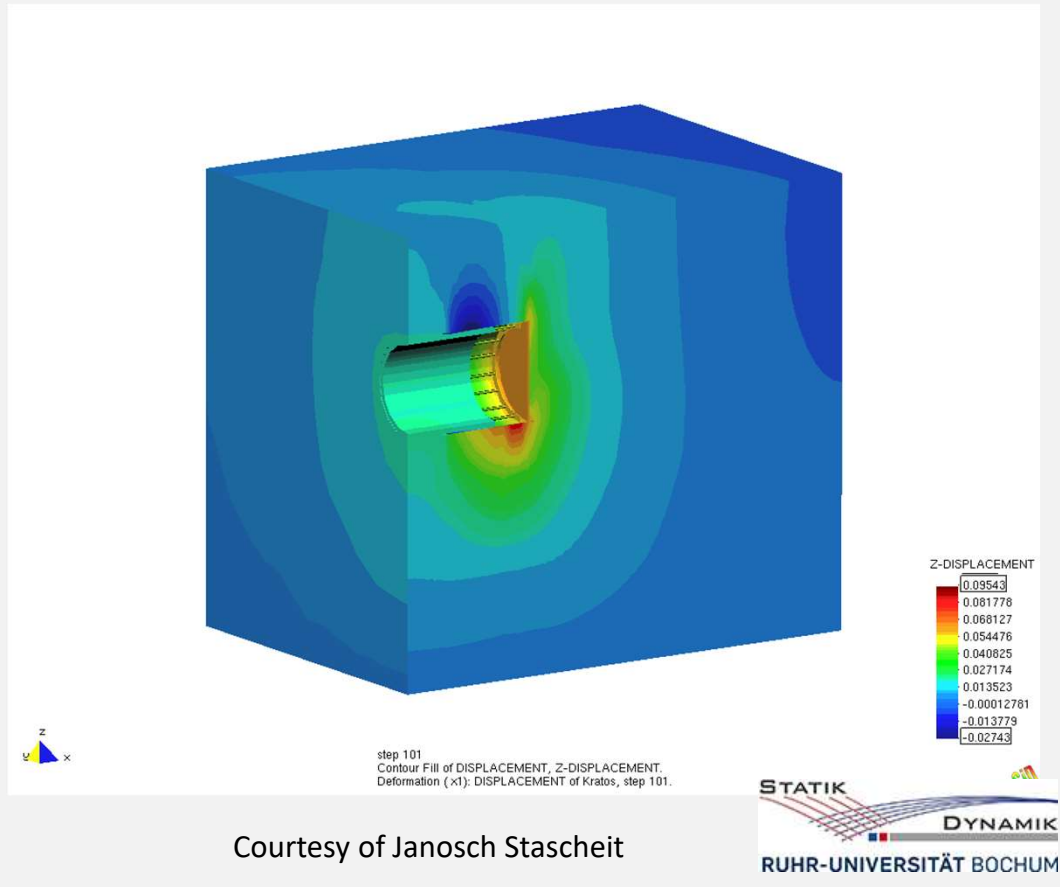


First Industrial use as composite shell solver inside a company inhouse solution



2007

MPI parallel tunneling process



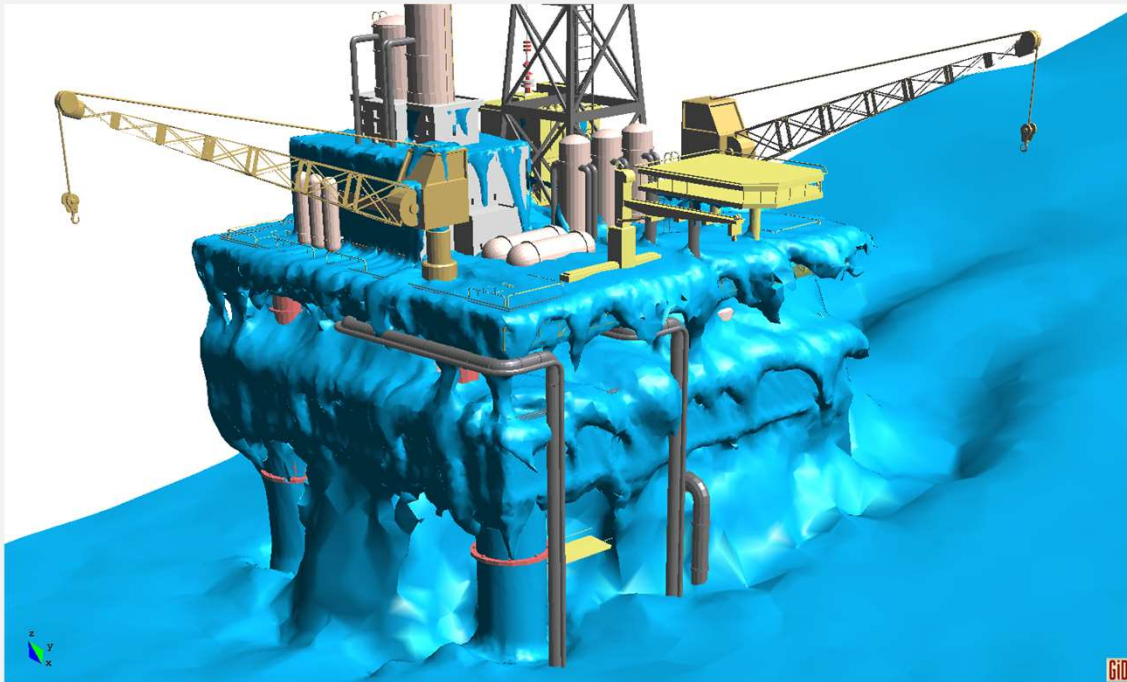
Courtesy of Janosch Stascheit

Jordi Cotela joins



2008

Level set free surface tracking



Team growing!



2009

Click2Cast solver prototype

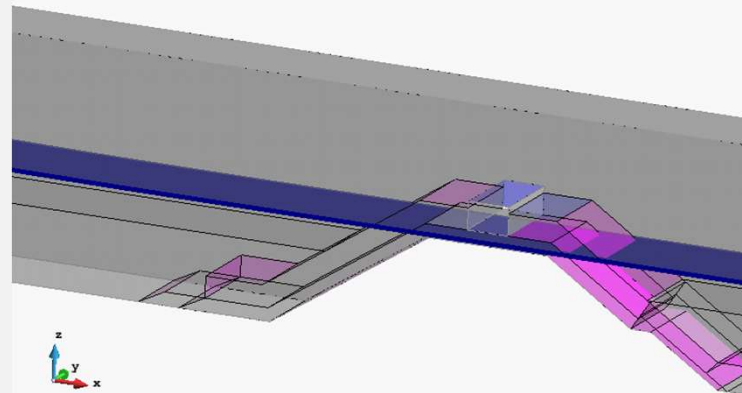


Static chair on board!

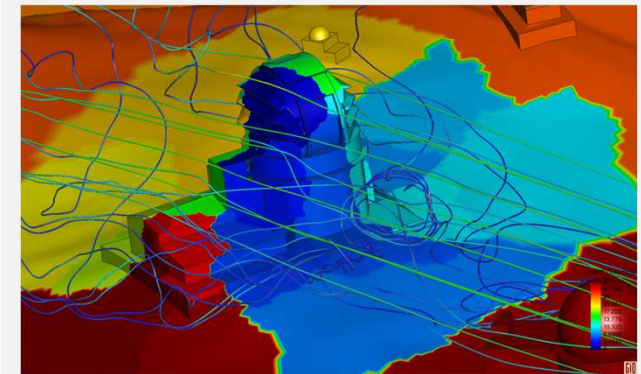


Beginning of a long and very close collaboration with strong impact on many parts of Kratos!

Fluid in porous media in virtual wave channel

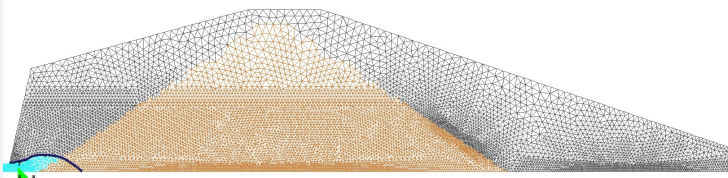


Reaching 10M elements



2010

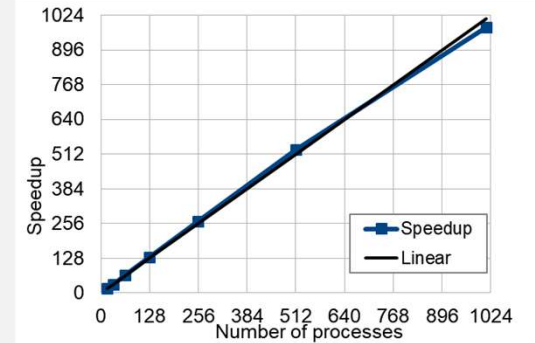
Porosity with damage prediction



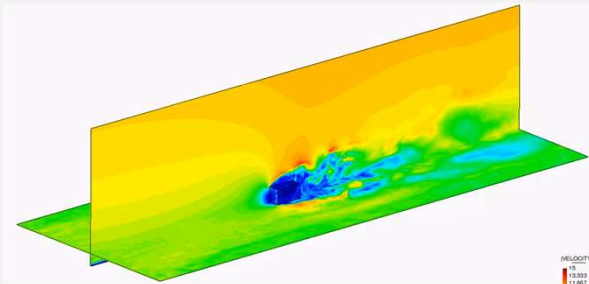
Carlos Roig Joined!



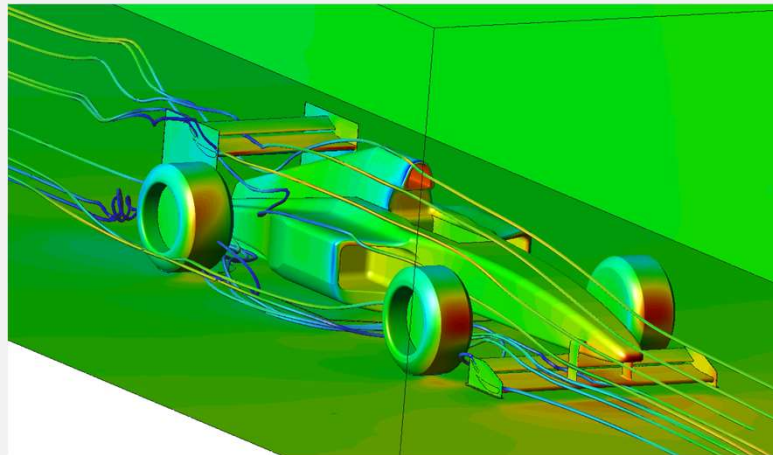
Reaching 1000 cores scalability



Silsoe cube the first collaborative master thesis TUM-CIMNE



Reaching 100M elements



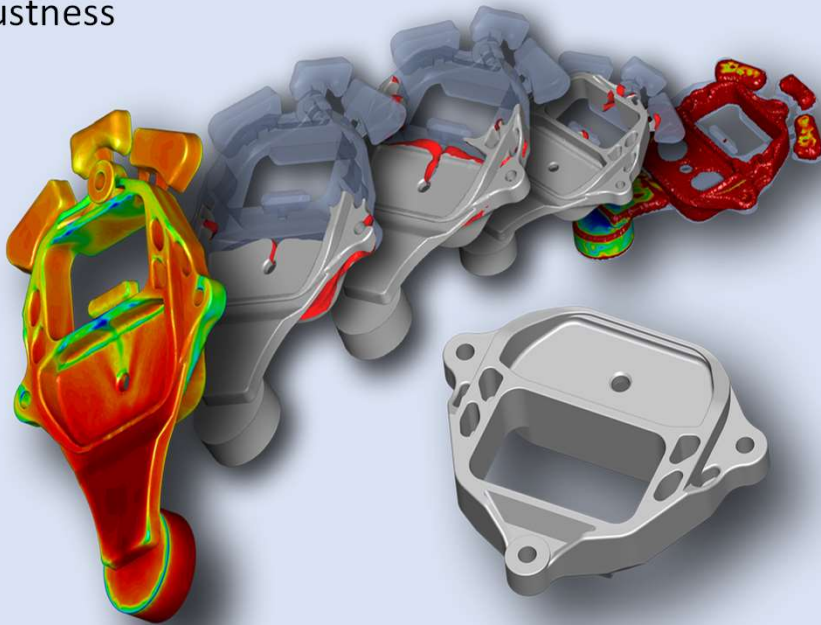
Pablo Becker starting PFEM2



2011

Click2Cast V1.0 release

Click2Cast development pushes considerably the level set and two fluid solutions among increasing the fluid solver robustness



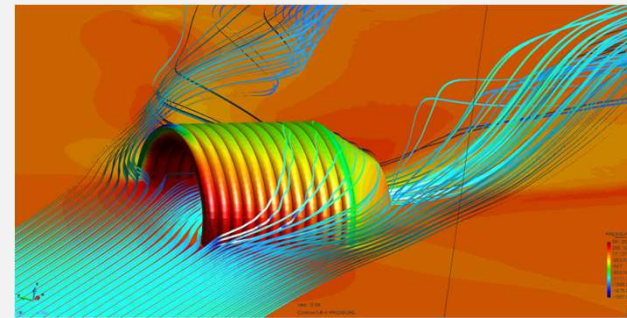
Kratos FSI course

First Kratos as **platform for teaching** at TUM
A collaborative course between TUM and UPC

BSD License

An important move toward more collaborative framework

Wind simulation of inflatable hangar



Miguel Angel
starting DEM

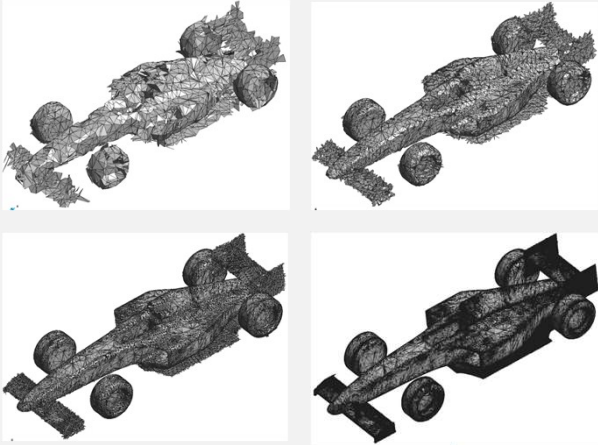


Josep M. Joined

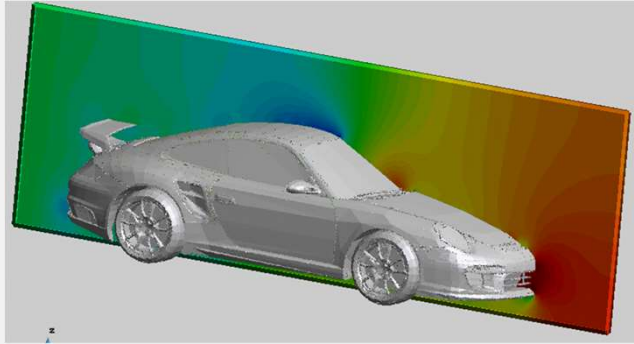


2012

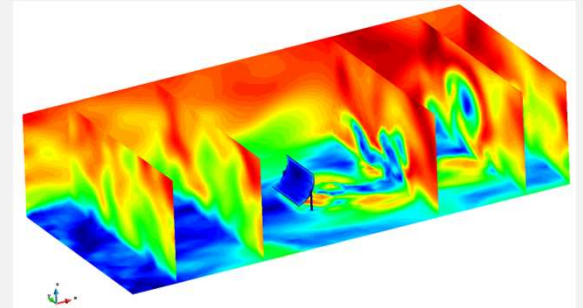
Implicit geometry description



Embedded Fluid Prototype



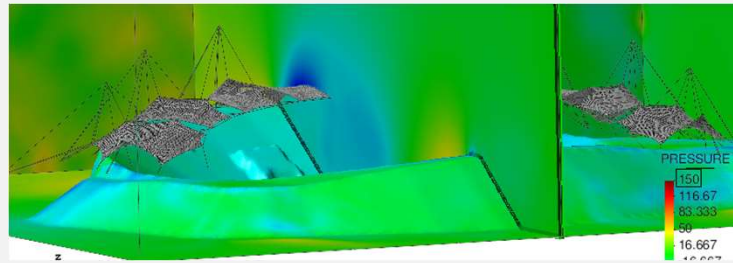
ABENGOA supporting Kratos



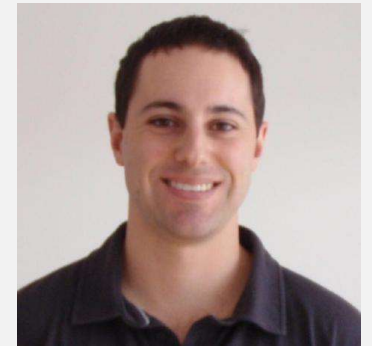
Daniel Baumgärtner
joined



Computational Wind Engineering

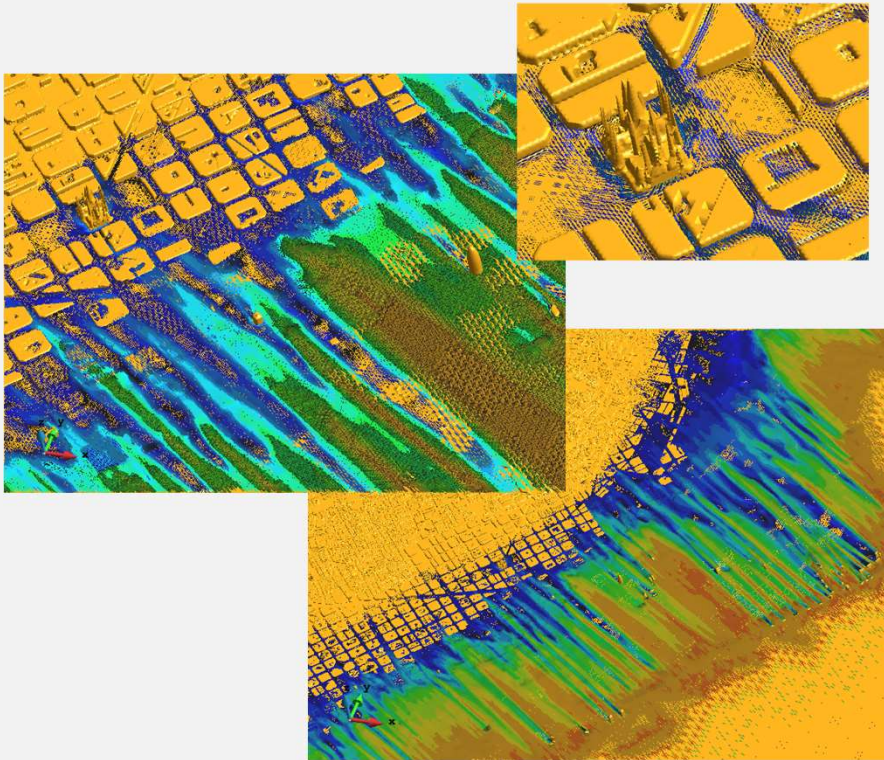


Michael Andre joined



2013

Wind simulation over entire Barcelona city with 4m resolution reaching 400 millions of elements

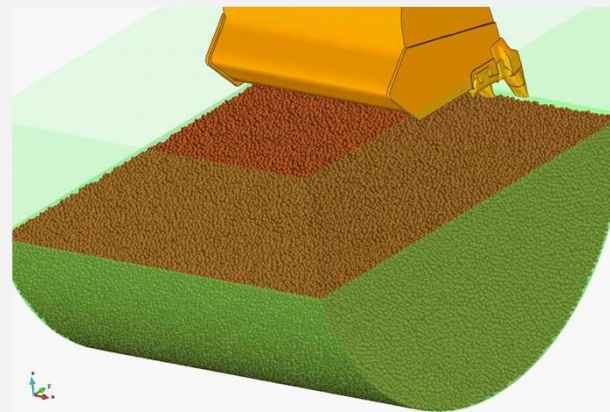


Robust mesh updating strategy



Andreas Winterstein
TUM

DEM Spherical particles

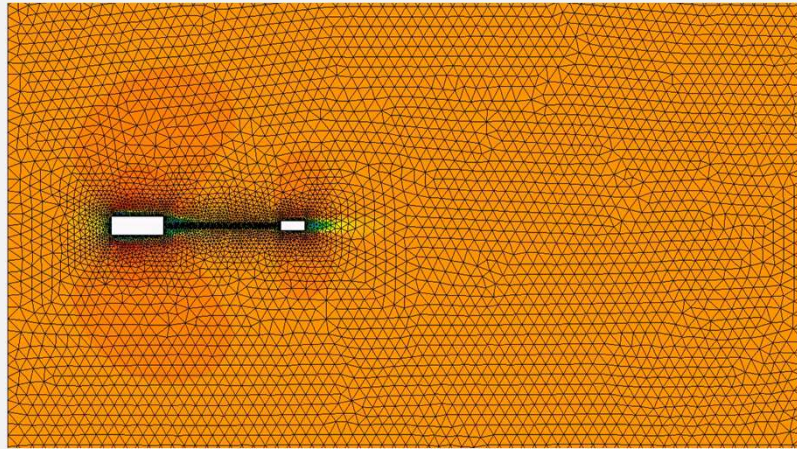


Co-simulation started
by Aditya Ghantasala

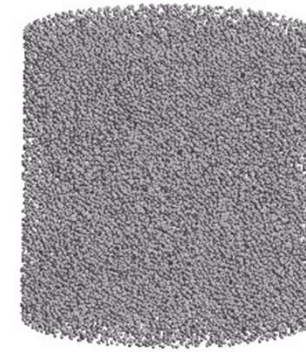


2014

FSI becomes robust Andreas Winterstein@TUM



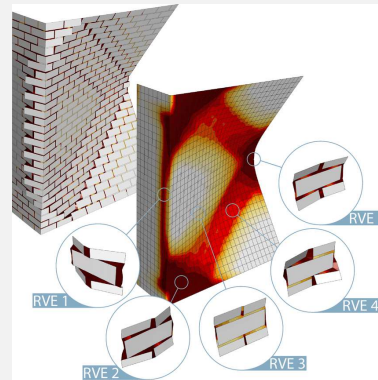
MPM application



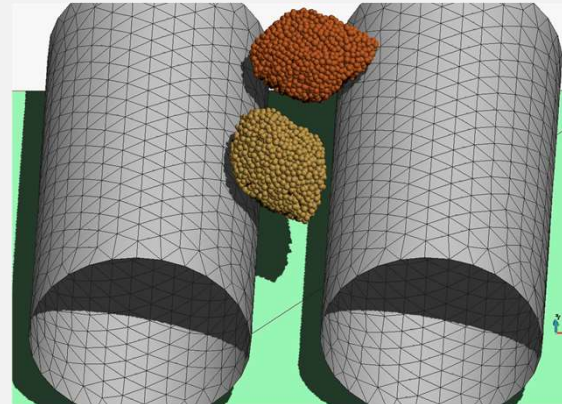
Altair acquired Clic2Cast and became aware of Kratos



Multi scale simulation



Multi spheres DEM

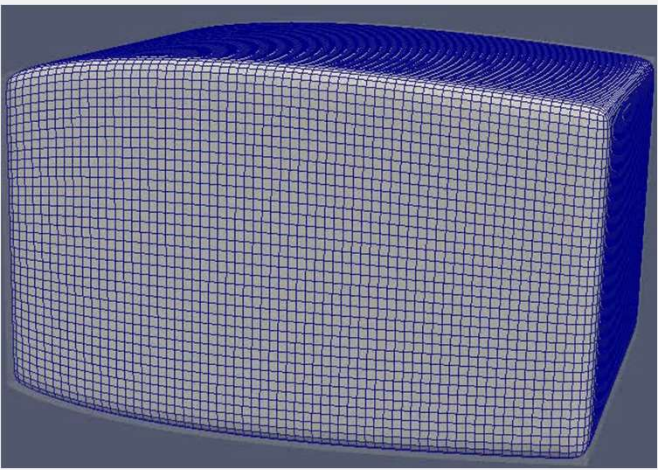


Vicente Mataix start committing!



2015

Optimization



by Daniel Baumgärtner @ 

Altair started new projects based on Kratos  Altair


Siemens on board! Their continuous support since then was an **important push** in Kratos as open R+D framework

Swimming dem



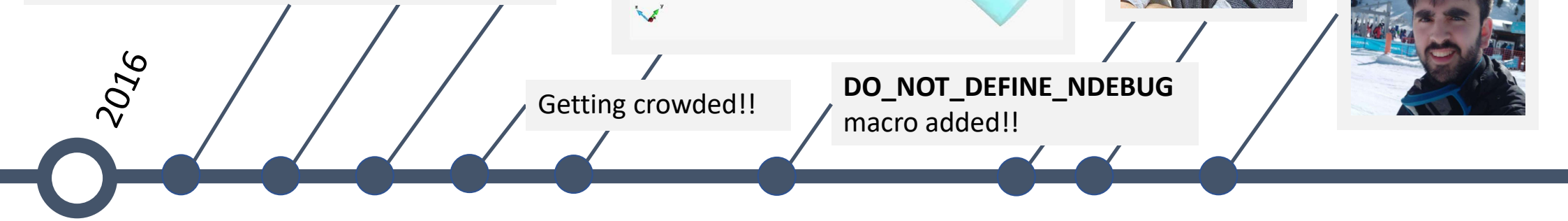
Philipp joined!



Ruben Joined!



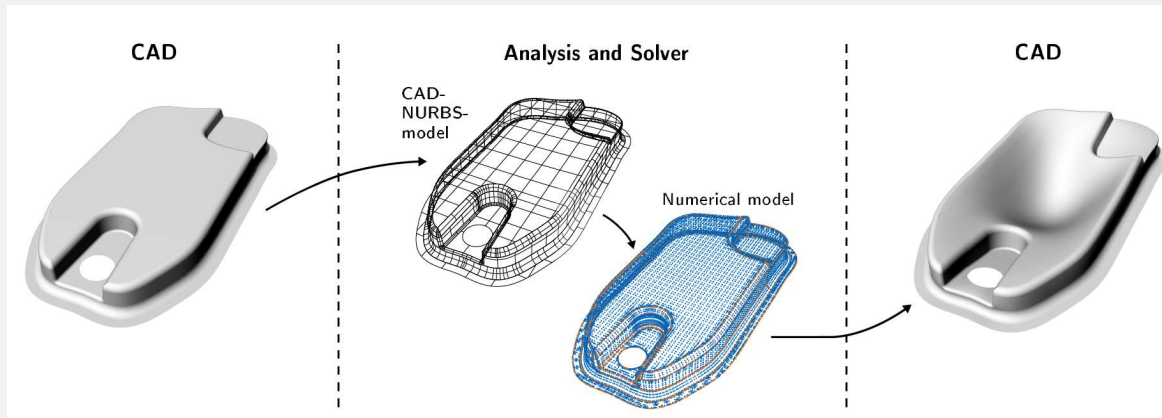
Dani Joined!



Getting crowded!!

`DO_NOT_DEFINE_NDEBUG` macro added!!

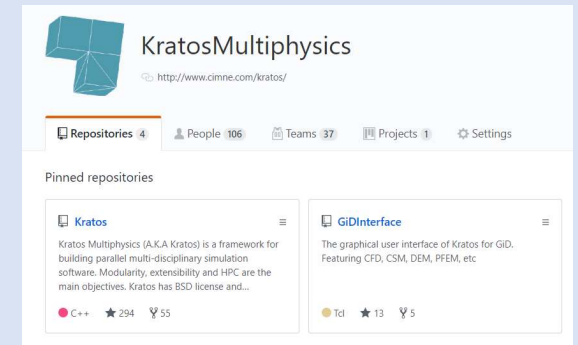
IGA application



Tobias Teschemacher @ **TUM**

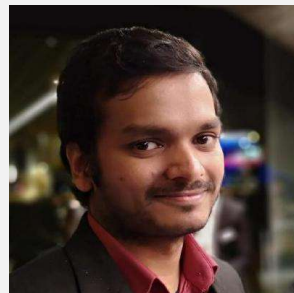
Moving to GitHub

One of the most important moves toward larger community



Airbus joined **AIRBUS**
Important collaborative push in research in Kratos framework

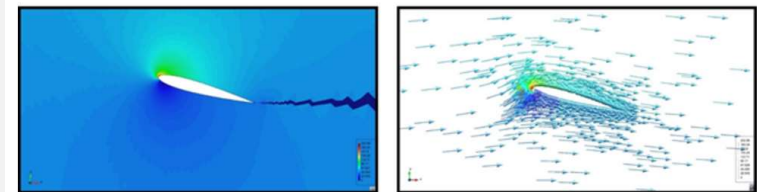
Suneth Joined!



Alejandro Joined!



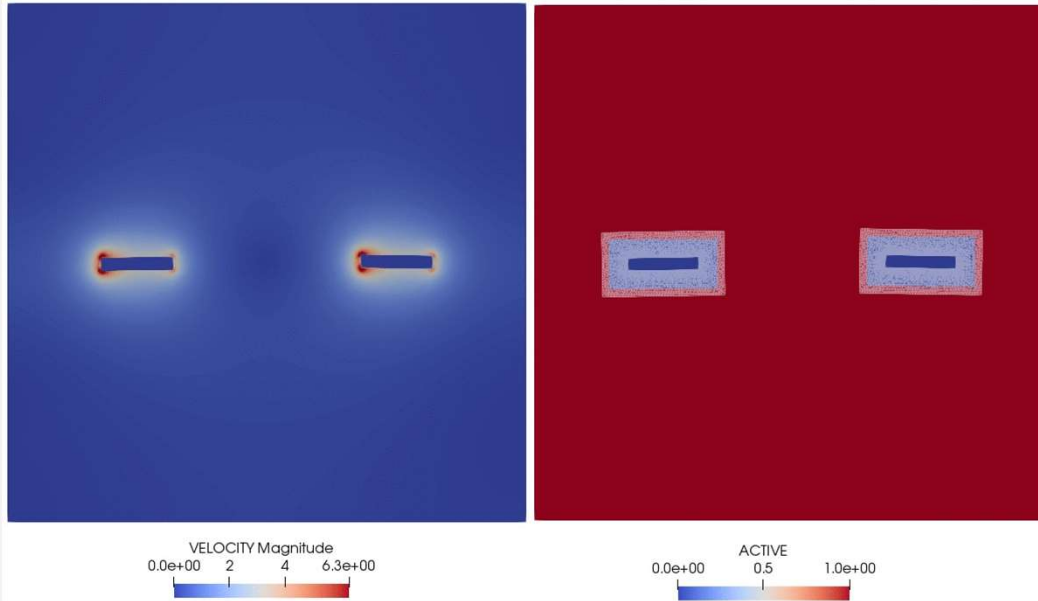
Potential flow for aero-space applications



Not matching grid mapping

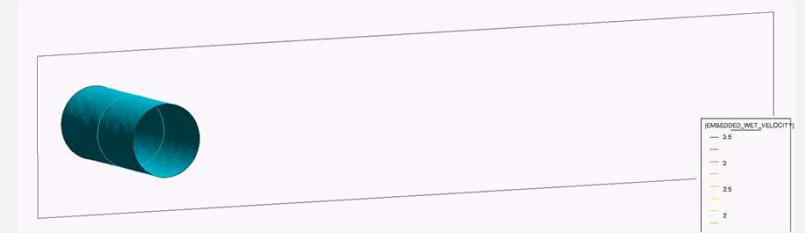
2017

Chimera

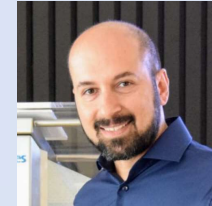


Aditya Ghantasala @ 

FSI with embedded



Catching attention of Vahid Galavi
from  **Deltares**
Enabling Delta Life



Opening to other
codes...



MPC Added to the core

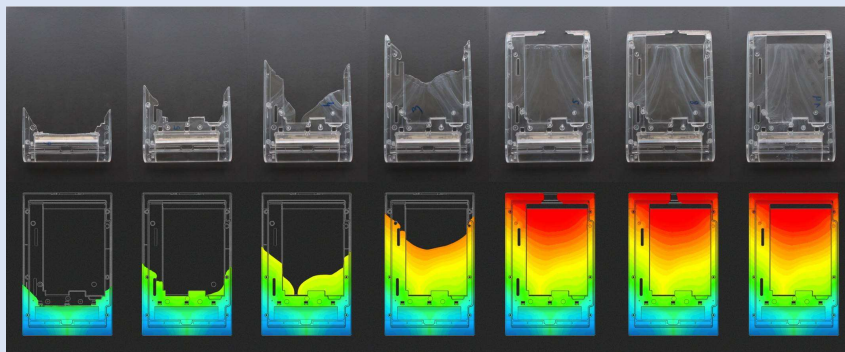
Pizza for each release!

2018

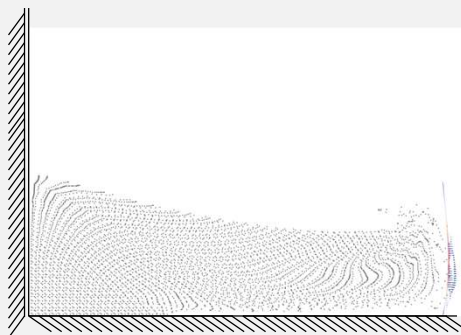
MPI Core

- Easier development in MPI
- Seamless data transfer
- Processes working correctly in serial and parallel

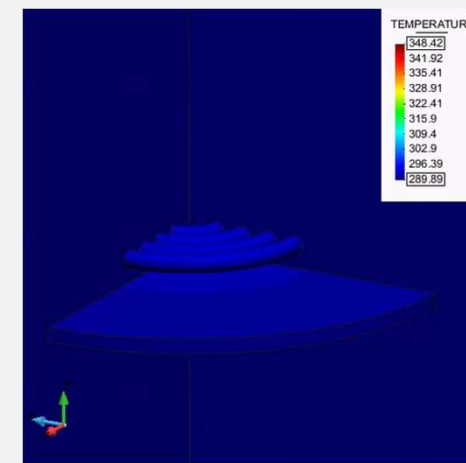
Inspire Mold based on Kratos



Coupled MPM-FEM

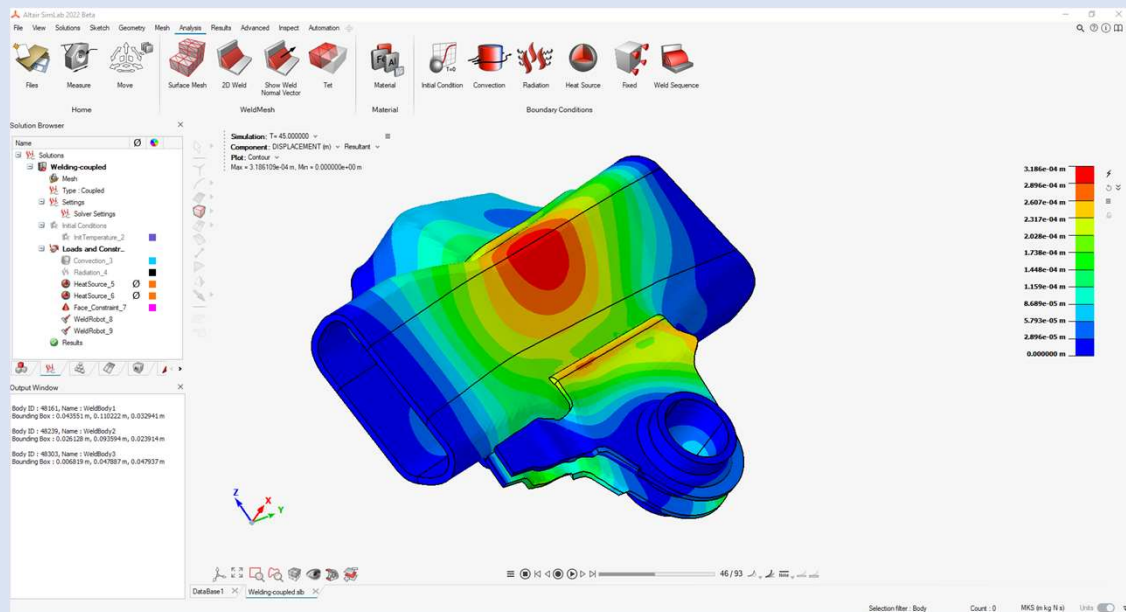


Conjugate heat transfer



2019

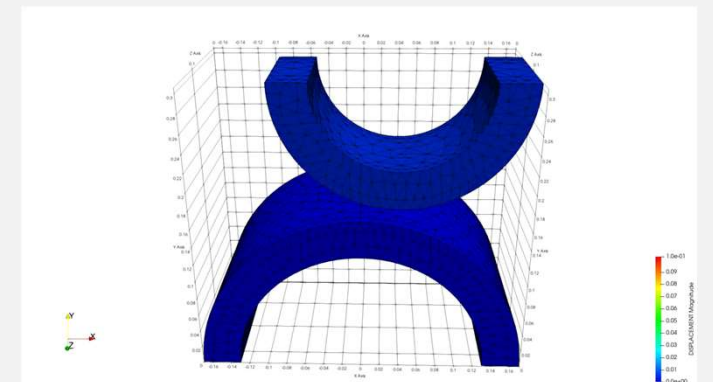
AMS welding solver based on Kratos



MPM-DEM Coupling

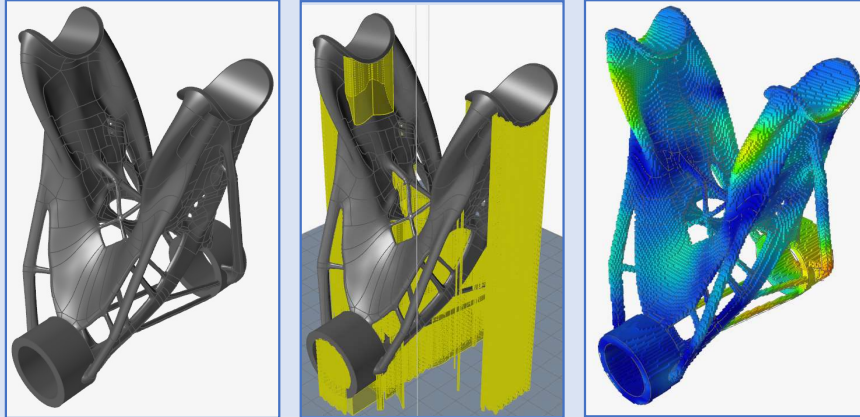


Contact Mechanics

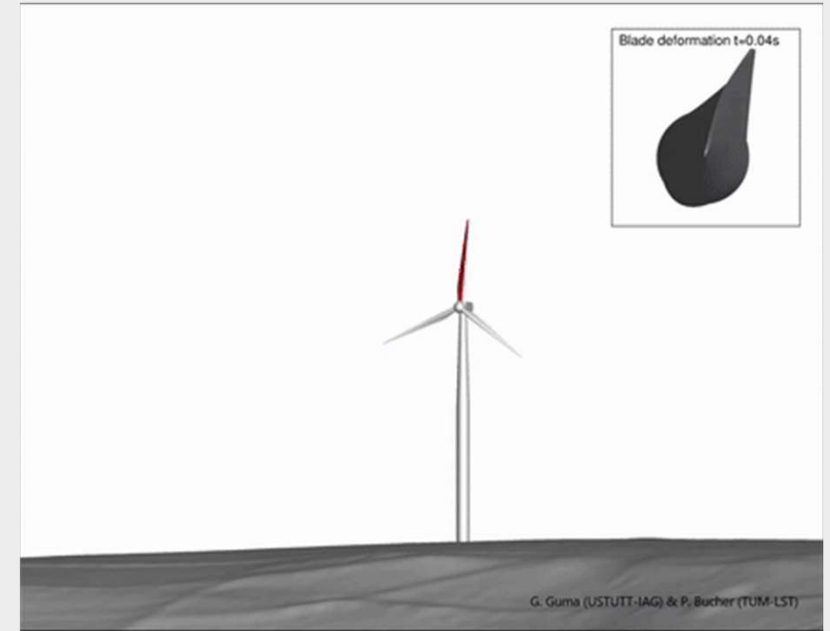


2020

New Inspire 3DPrinting inherent strain solution



Large scale co-simulation Philip Bucher, Giorgia Guma @TUM



ISD chair Joined!

Institut für Statik
und Dynamik



Technische
Universität
Braunschweig

Deltares on board

Deltares

Enabling Delta Life



A new push to geomechanics

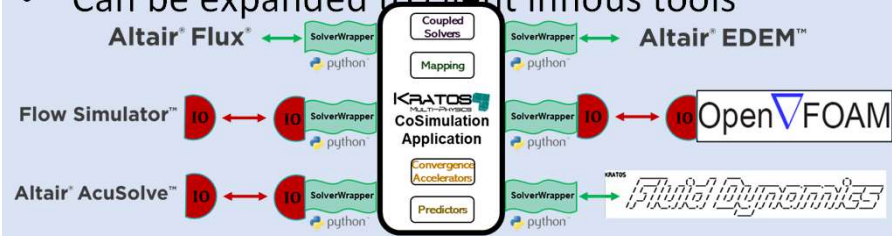
Monte Carlo and beyond



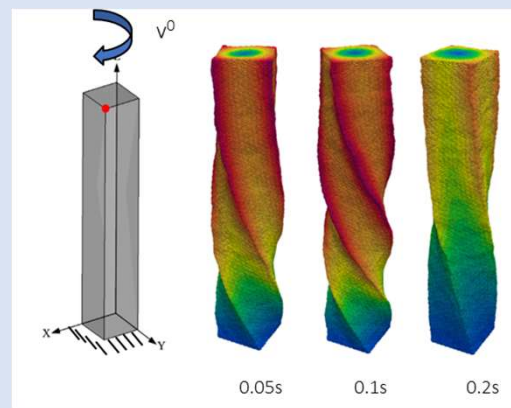
2021

CoSimulation between solvers:

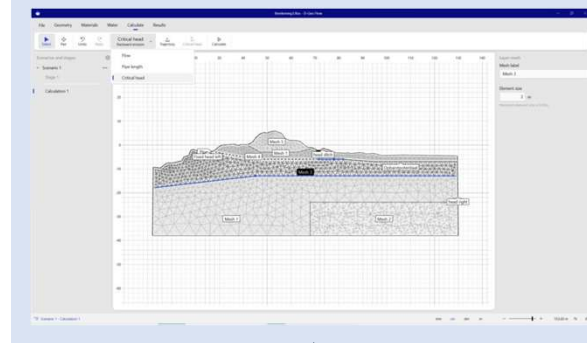
- Connecting to several Altair solutions
- Can be expanded to client inhouse tools



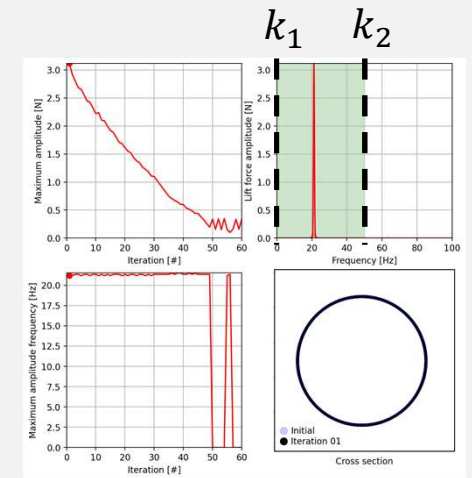
MPM MIXED (u-p) Formulation



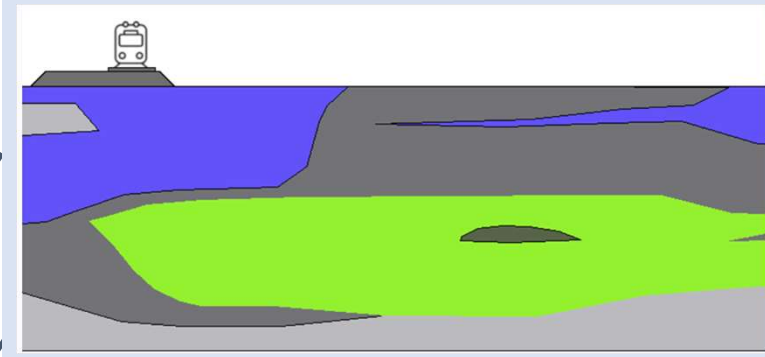
First DGeoFlow based on Kratos



Transient adjoint analysis

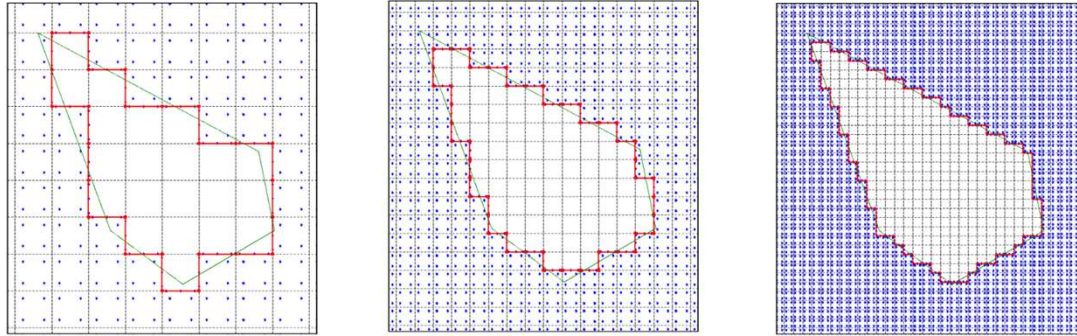


First version of STEM

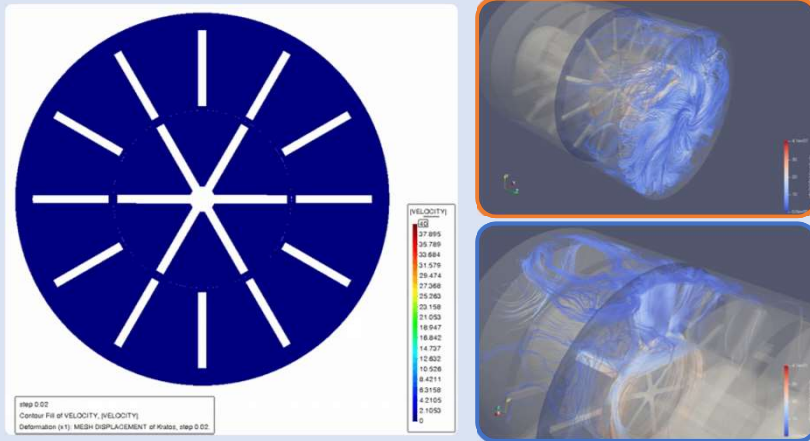


2022

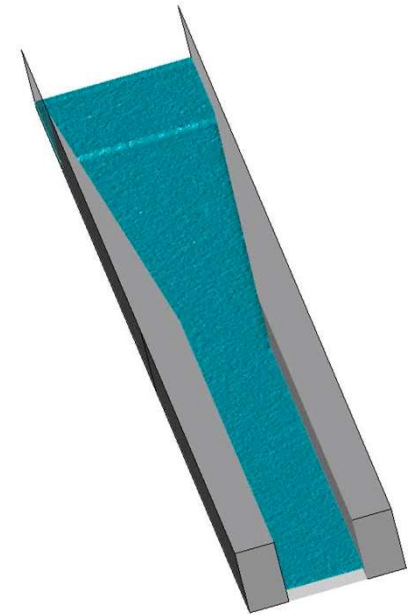
The Shifted Boundary Method in IsoGeometric Analysis



ROM over e-motors



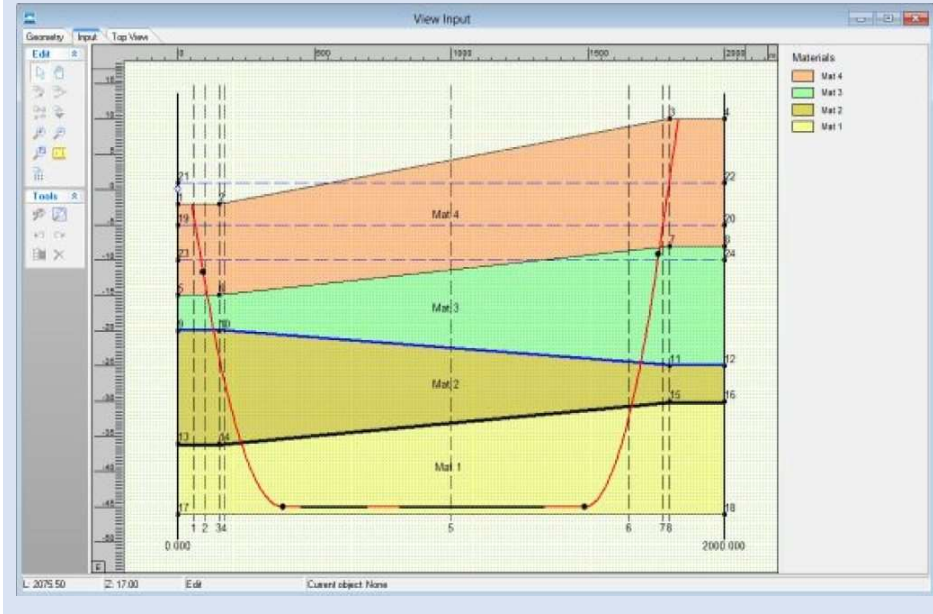
Debris Flows



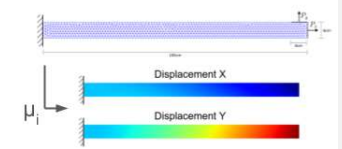
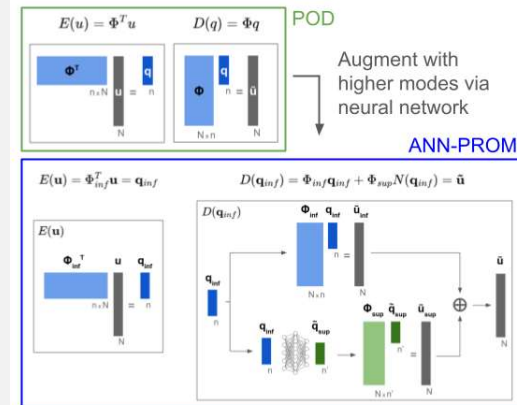
step 0.02227679
Iso Surfaces of DISTANCE

2023

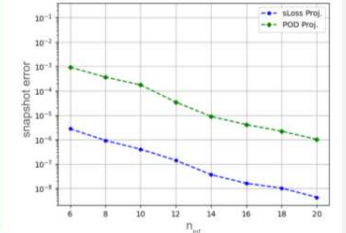
DGeoPipeline based on Kratos



Non-linear ROM (ANN-PROM)

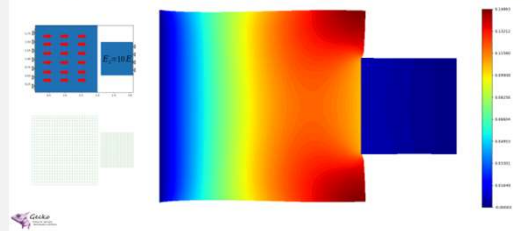


Fewer modes needed for the same accuracy as POD:



Now

Contact in IGA



Kratos Multiphysics

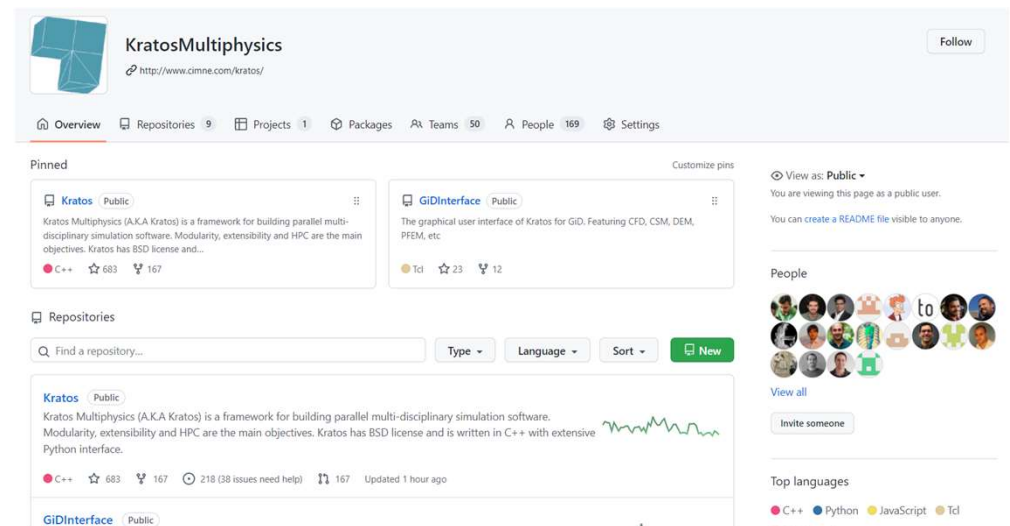
Framework



Kratos Multiphysics

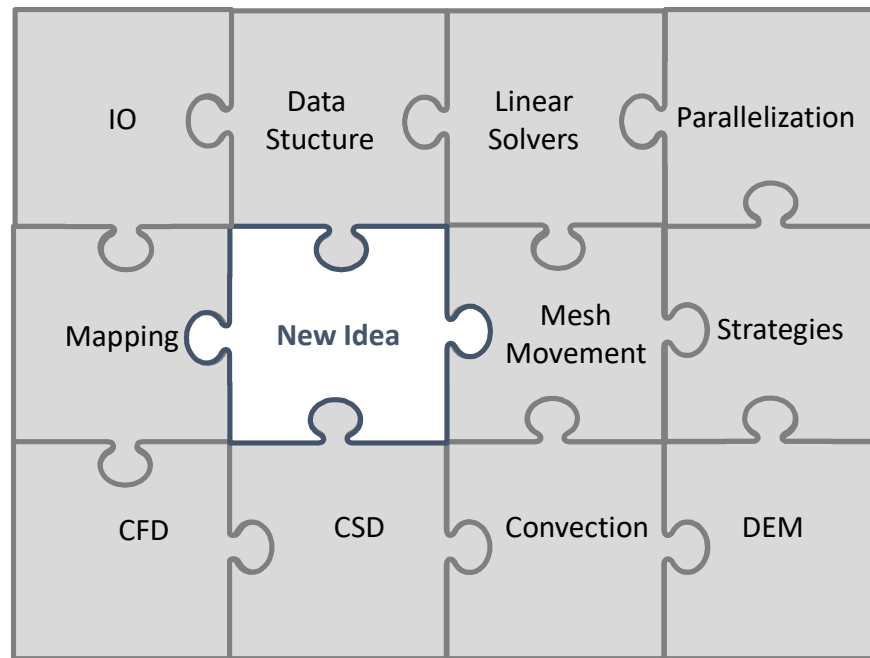
An open framework for parallel Multi-physics programs development

- Since 2001
- ~ 1 million lines of code
- ~ 100 developers
- 20 years of development
- Very modular
- High performance
- Object Oriented C++
- Extensive Python interface
- Open Source and Free (BSD Licence)

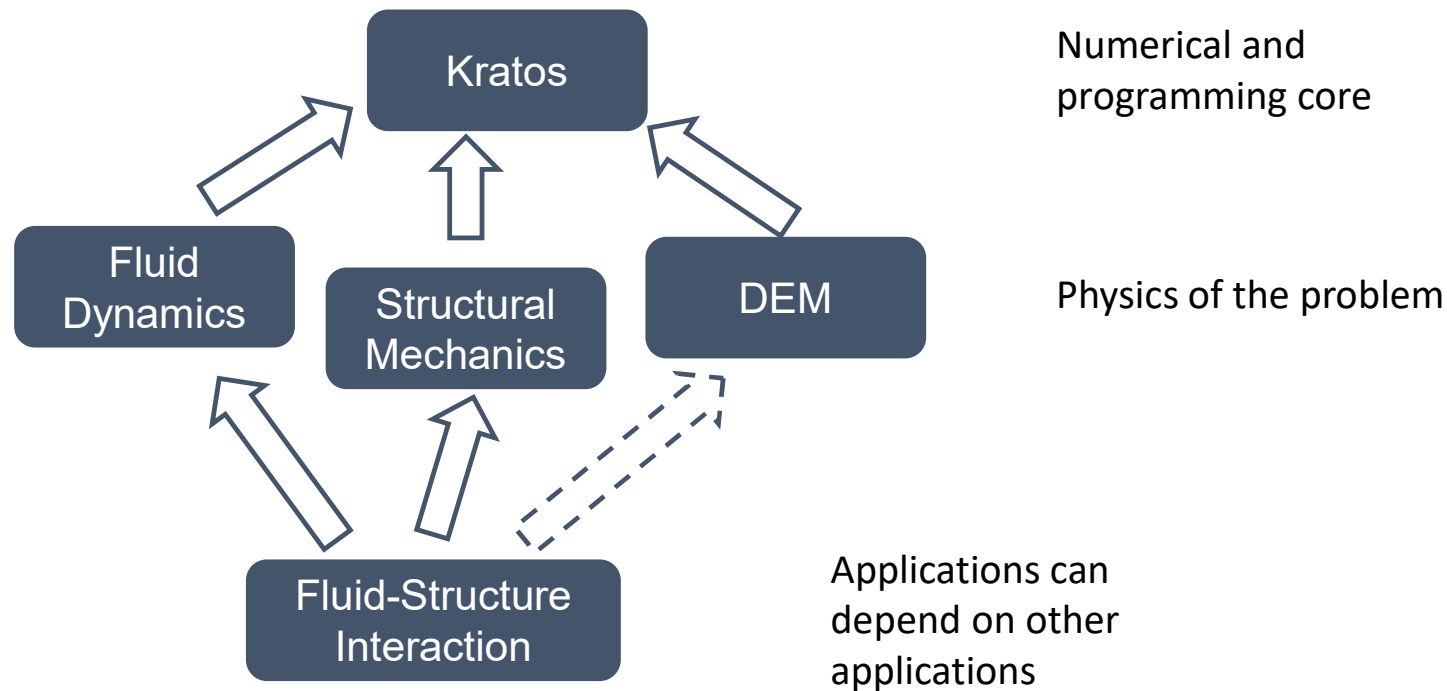


<https://github.com/KratosMultiphysics>

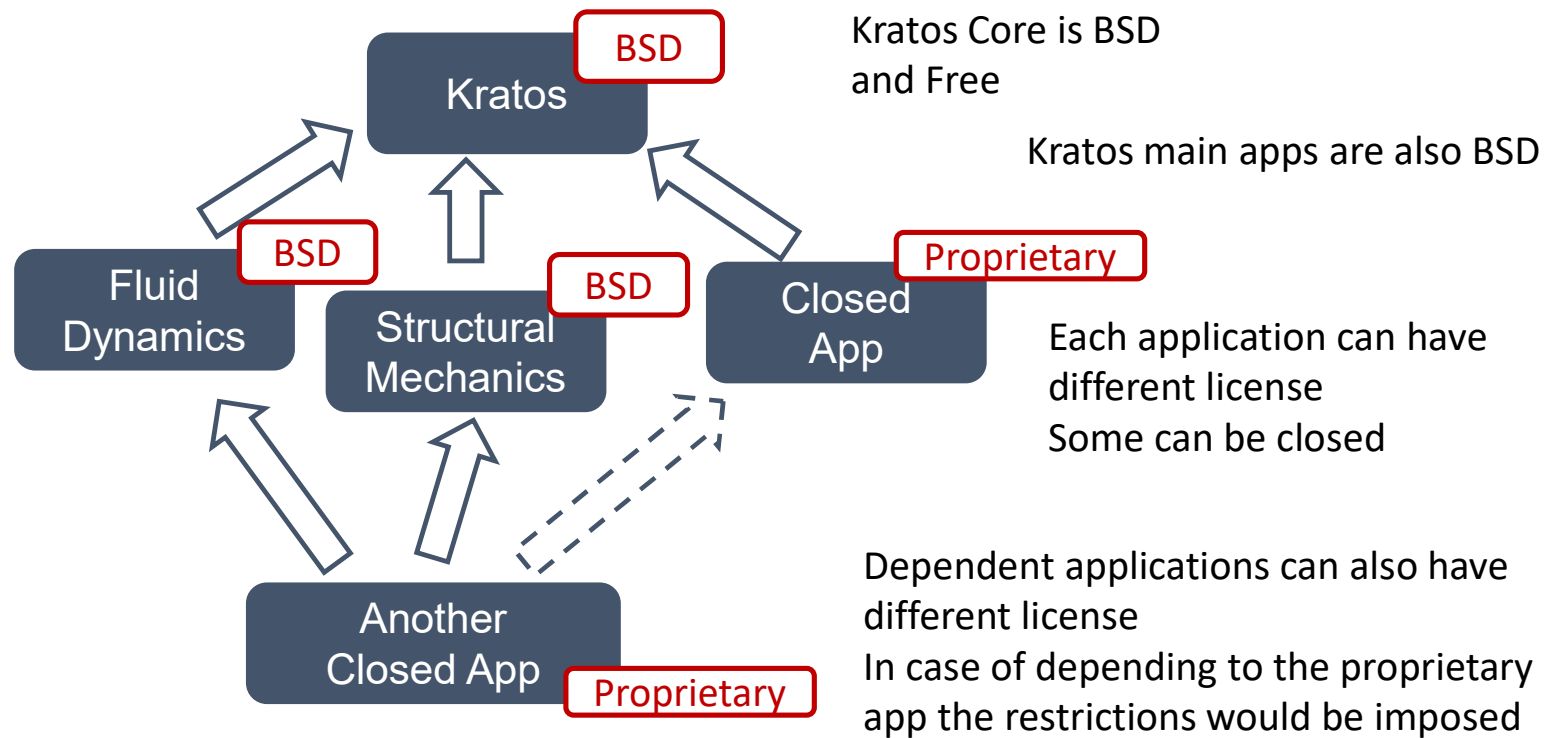
Kratos Multiphysics



Highly Modular Design & Multi-disciplinarity



Flexible License



Clear Interfaces

- Interfaces are generic but intuitive:

```
node.GetValue(PRESSURE)
node.GetValue(VELOCITY)
node.GetValue(VELOCITY_X) // Equivalent to node.GetValue(VELOCITY)[0]
```

- C++ interfaces impose type-safety

```
// The following code will not compile due to the type mismatch.
// because VELOCITY is array_1d and not a std::vector!
std::vector<double>& velocity = node.GetValue(VELOCITY)
```

- Python interface is relatively easy

Python Interface

```
from KratosMultiphysics import *

from KratosMultiphysics.StructuralApplication import *

from KratosMultiphysics.FluidDynamicApplication import *

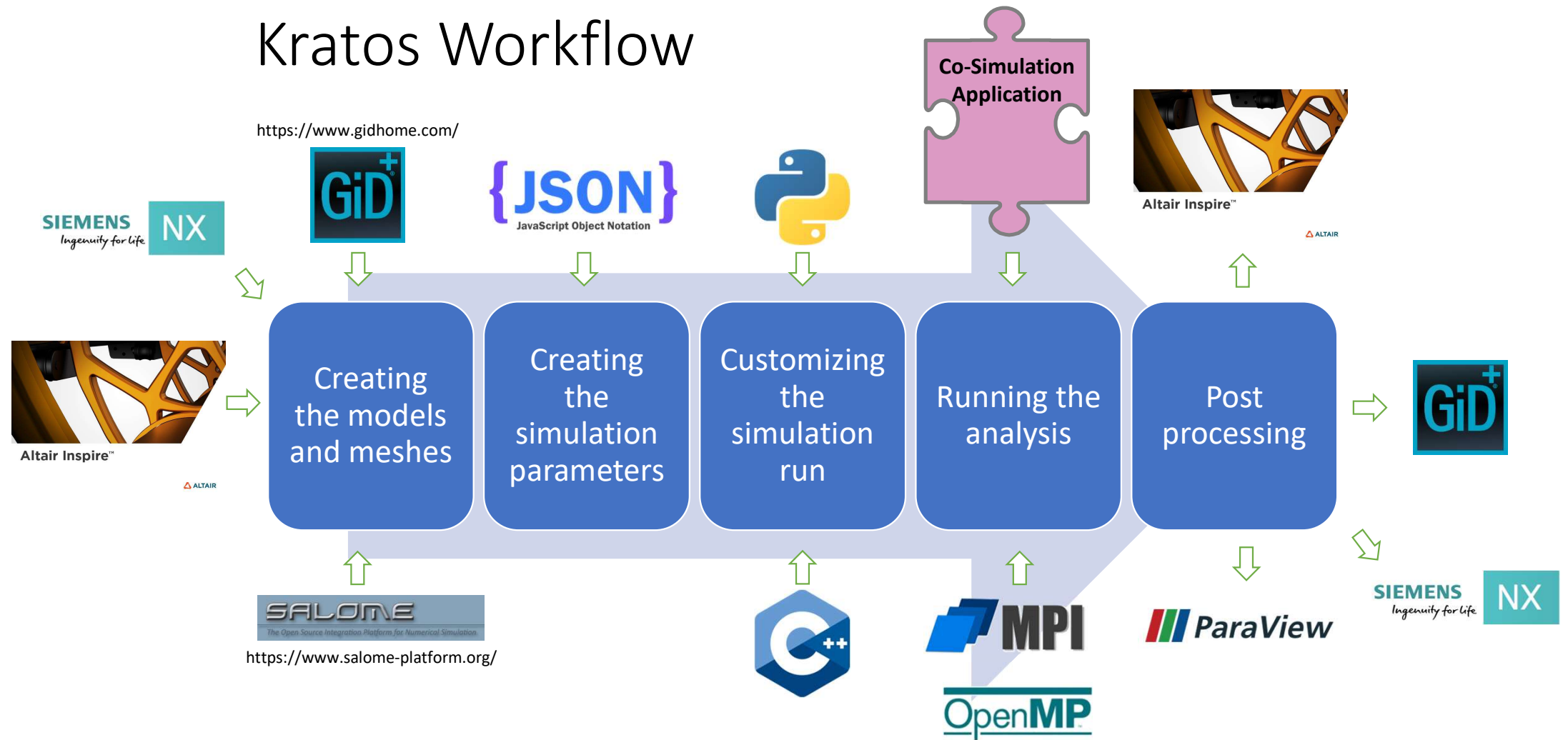
from KratosMultiphysics.FSIApplication import *


class ExplicitCoupling :

    def Solve(self):
        ## solve the structure(prediction)
        (self.structural_solver).Solve()
        ## map displacements to the structure
        (self.mapper).Map(DISPLACEMENT, DISPLACEMENT)
        ## move the mesh
        (self.mesh_solver).Solve()

        ## set the fluid velocity at the interface to
        ## be equal to the corresponding mesh velocity
        self.CopyVectorVar(MESH_VELOCITY, VELOCITY, self.interface_fluid_nodes)
        ## solve the fluid
        (self.fluid_solver).Solve()
        ## map pressure to the structure
        (self.mapper).InverseMap(PRESSURE , POSITIVE_FACE_PRESSURE)
        ## solve the structure(correction)
        (self.structural_solver).Solve()
```


Kratos Workflow

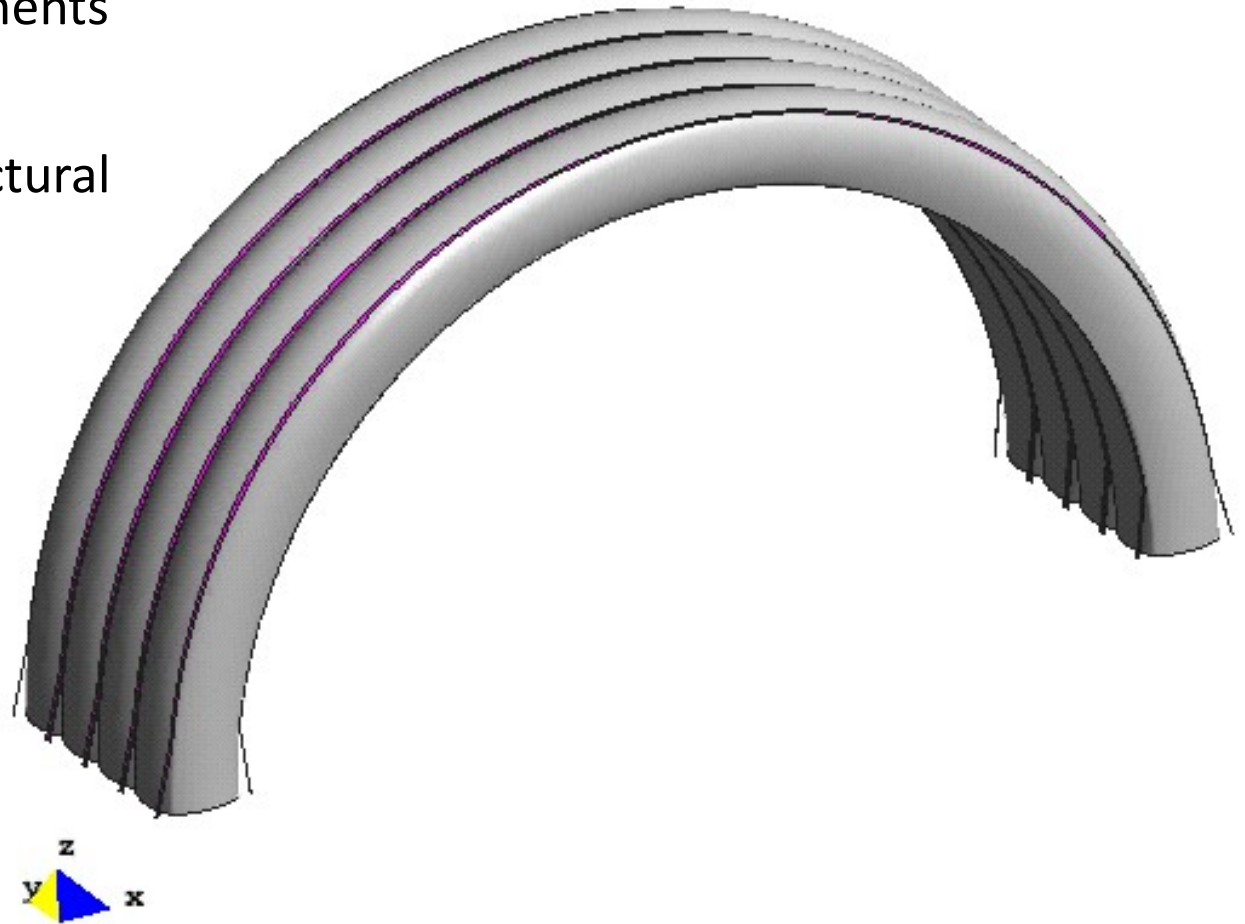


Kratos Applications

Matured and well maintained

STRUCTURAL MECHANICS

- 2D, 3D, and axisymmetric solid elements
- Beams, shells and membranes structural elements



STRUCTURAL MECHANICS

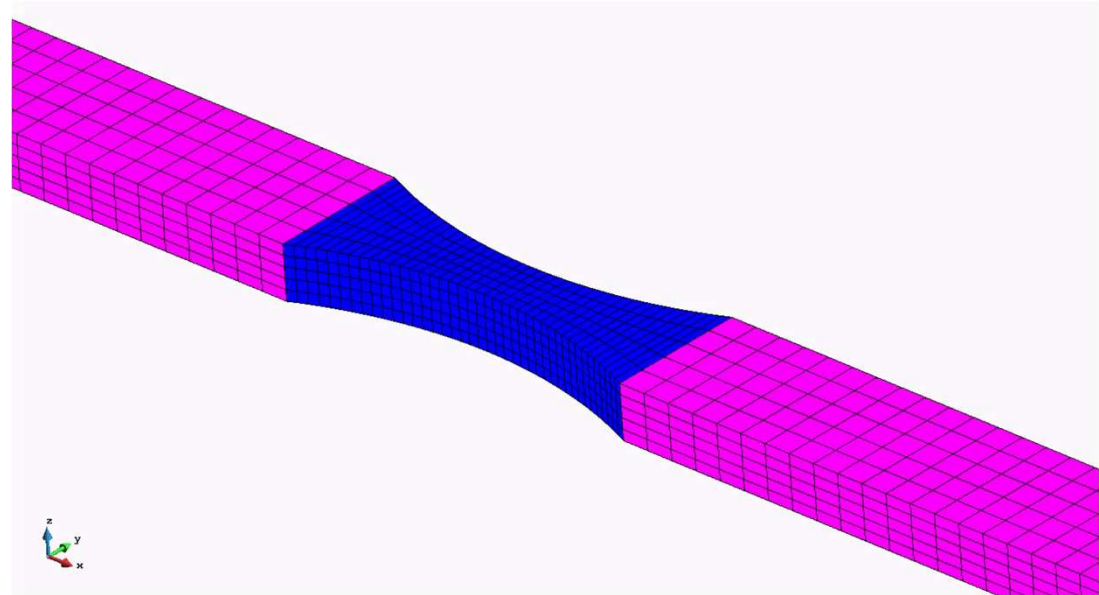
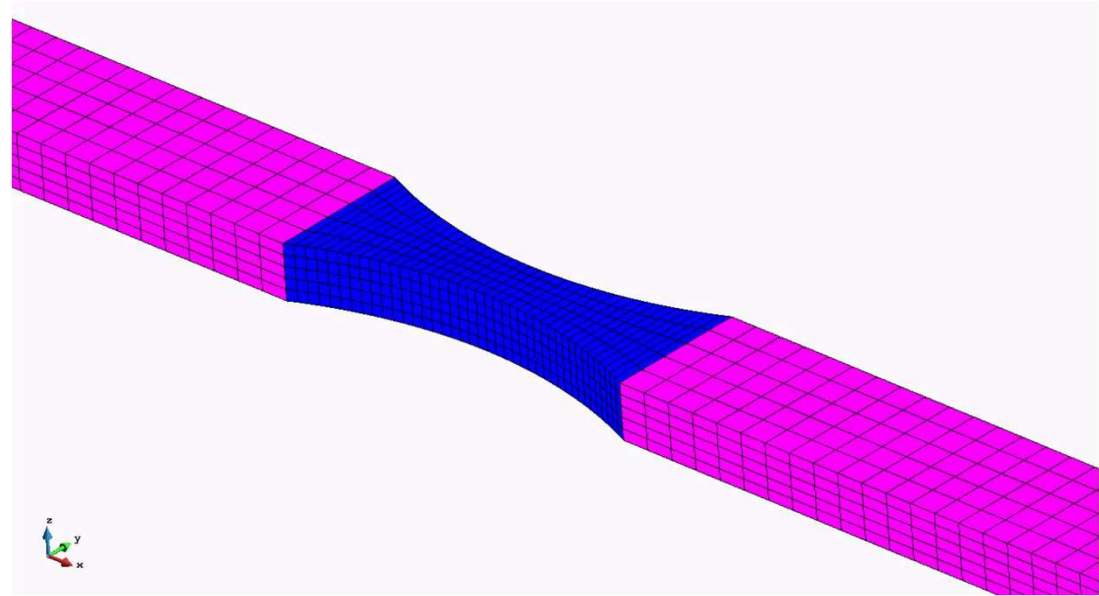
- 2D, 3D, and axisymmetric solid elements
- Beams, shells and membranes structural elements
- Linear and non-linear analysis
- Several constitutive laws

14:20 - 14:40

*Structural mechanics capabilities, FE technologies and Constitutive Law
modeling in Kratos*

Alejandro Cornejo (CIMNE Barcelona / Polytechnic University of Catalonia)

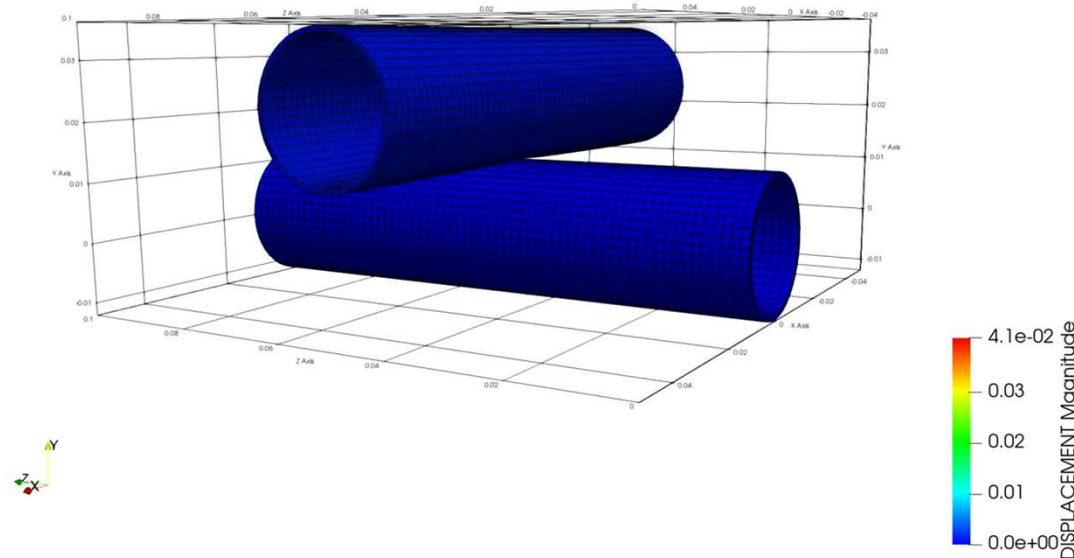
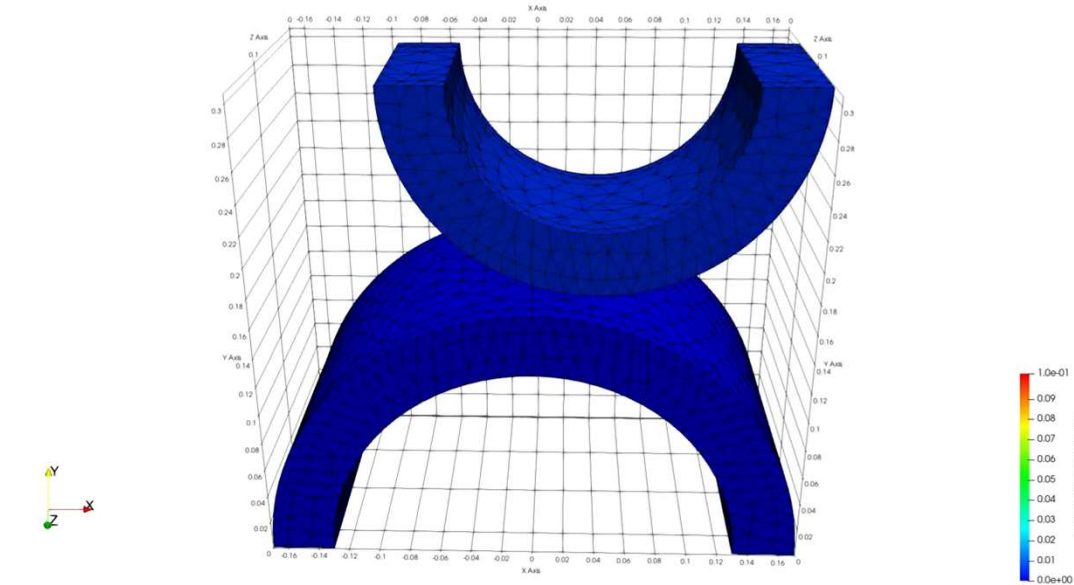
By: Alejandro Cornejo



STRUCTURAL MECHANICS

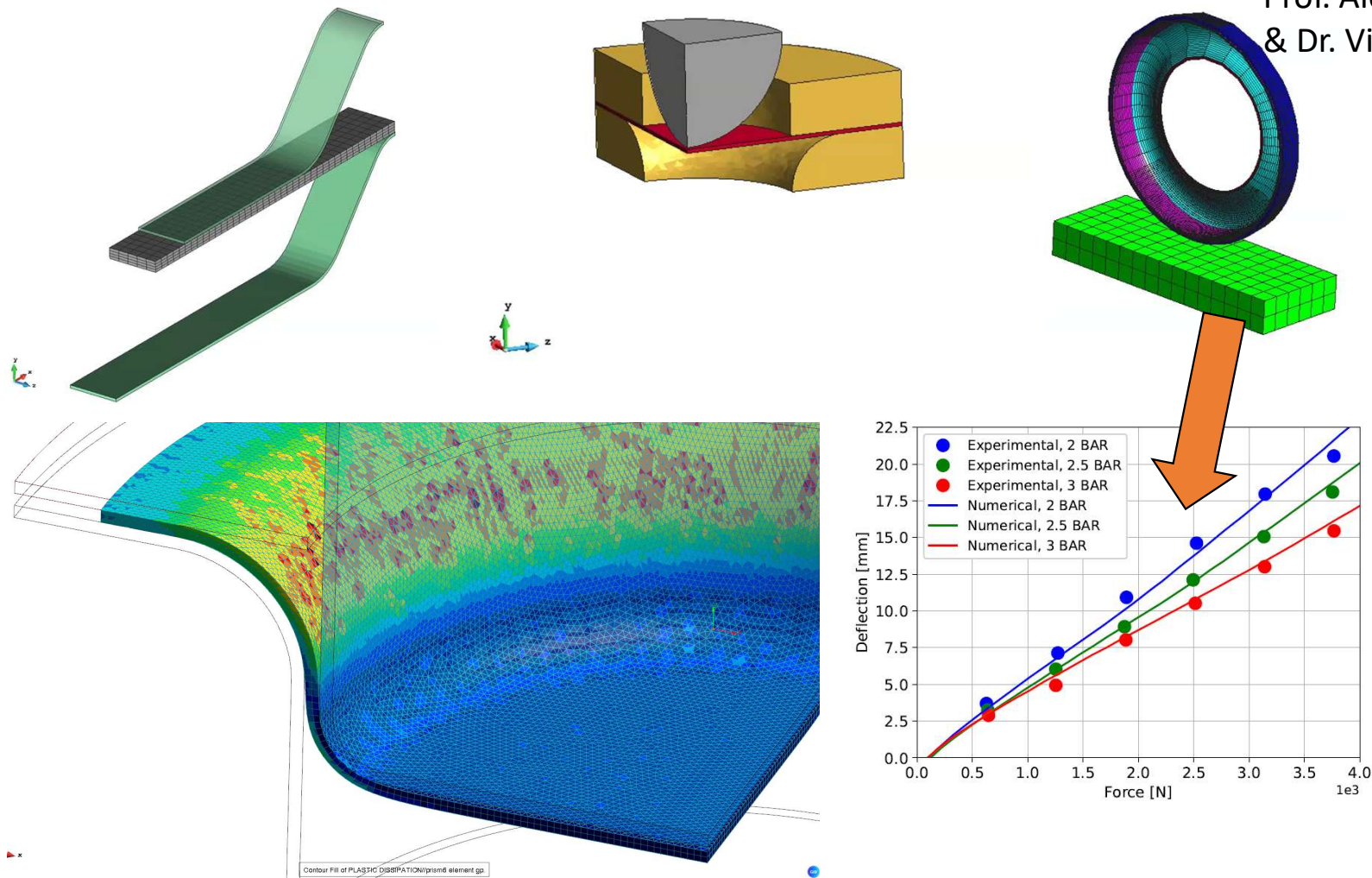
- 2D, 3D, and axisymmetric solid elements
- Beams, shells and membranes structural elements
- Linear and non-linear analysis
- Several constitutive laws
- Contact

By: Vicente Mataix

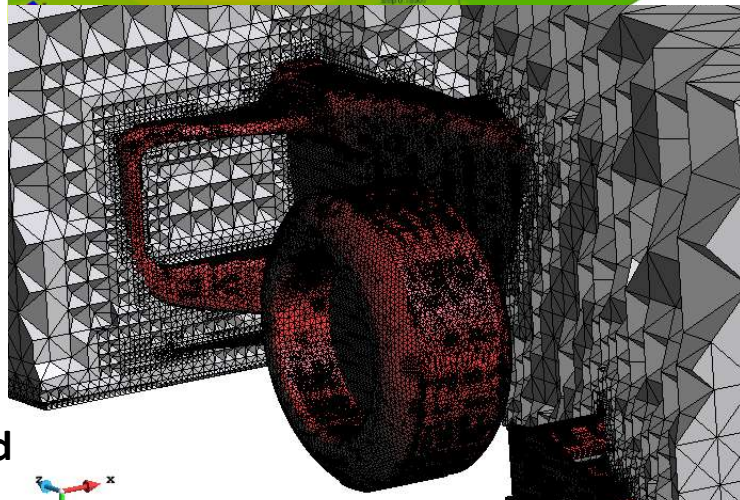
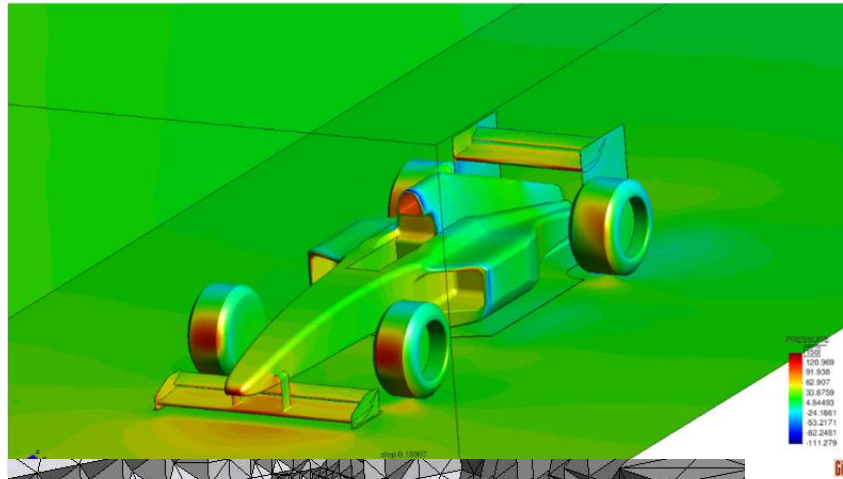


Computational Contact Mechanics

Prof. Alessandro Cornejo
& Dr. Vicente Mataix



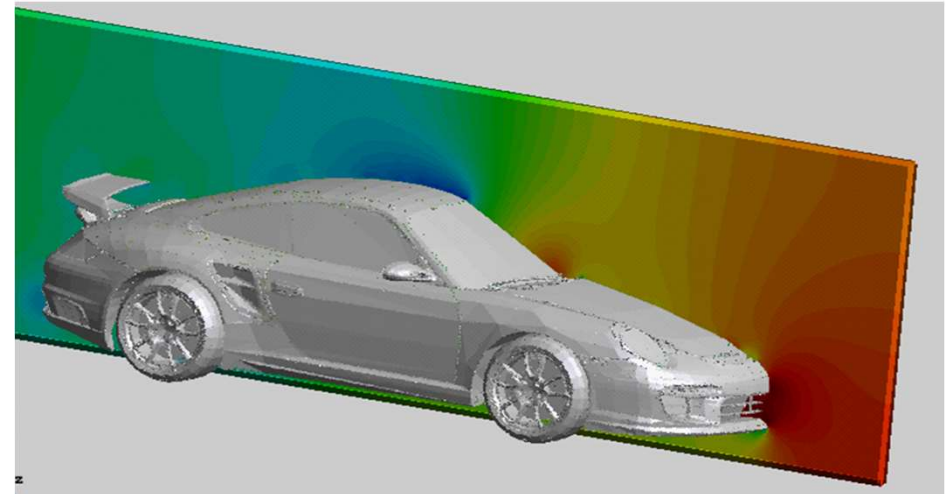
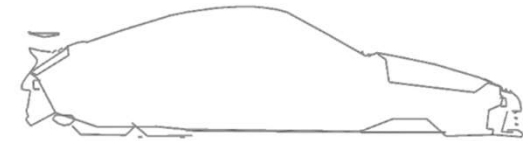
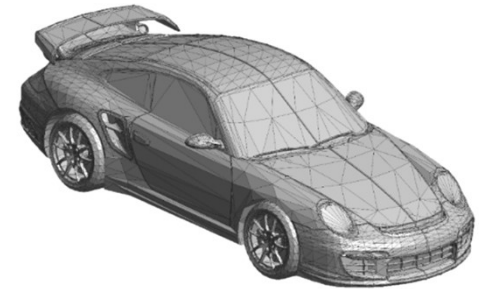
Fluid Dynamics Application



Body fitted

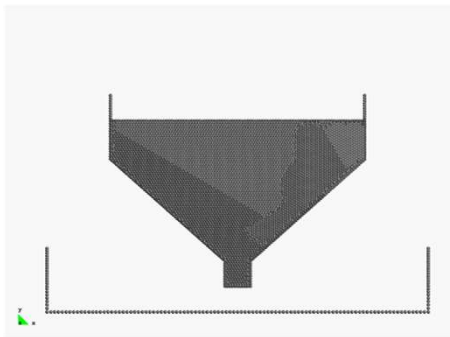


Embedded

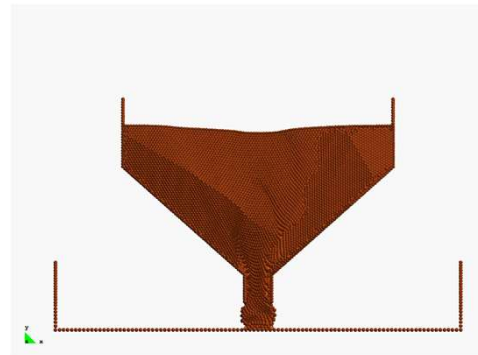


Incompressible Flow Solver

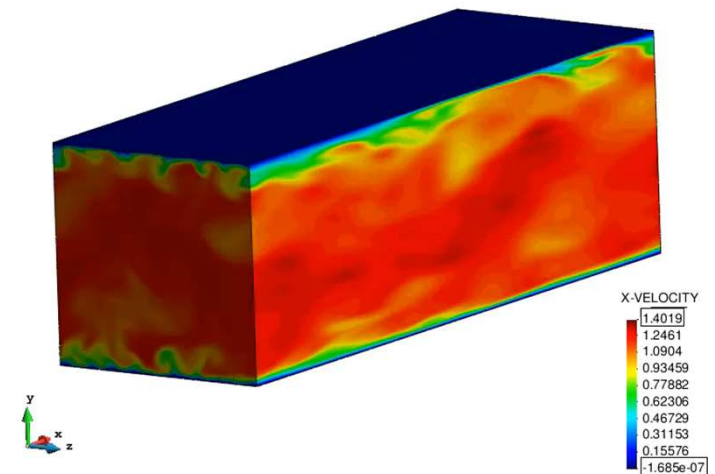
- Navier-Stokes Equations
 - Stabilized finite element formulation (GLS, Finite Integral Calculus, SUPG)
- Newtonian and viscoplastic fluids (Bingham, Hershel-Bulkley).
- RANS and LES turbulence models based on an eddy viscosity (Currently available: Spalart-Almaras, Smagorinsky).



Hershel-Bulkley



Bingham Fluid

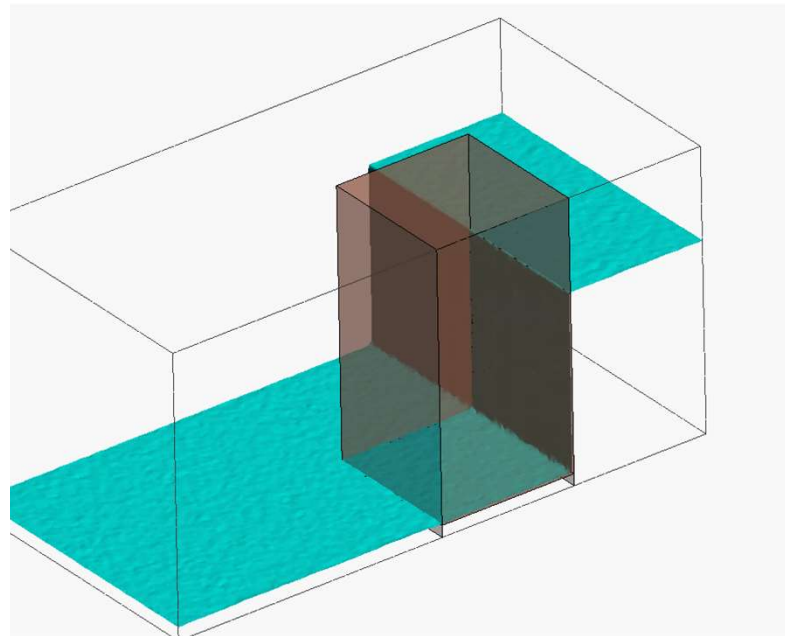


Turbulent Flow

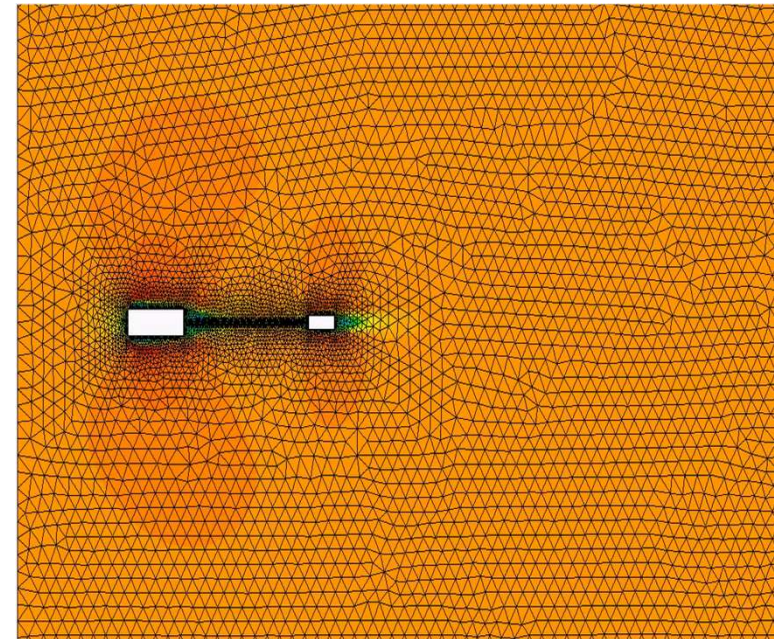
Incompressible Flow - Multiphase and FSI



Free Surface

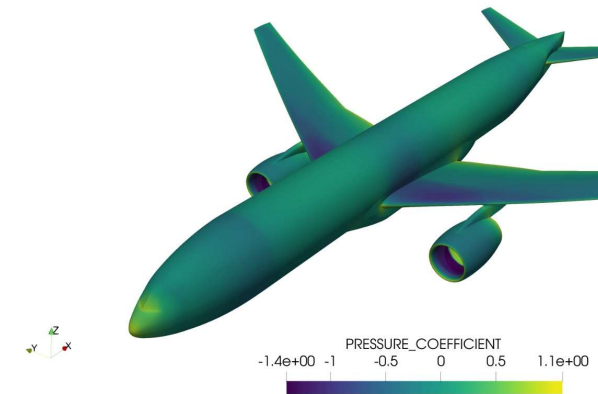
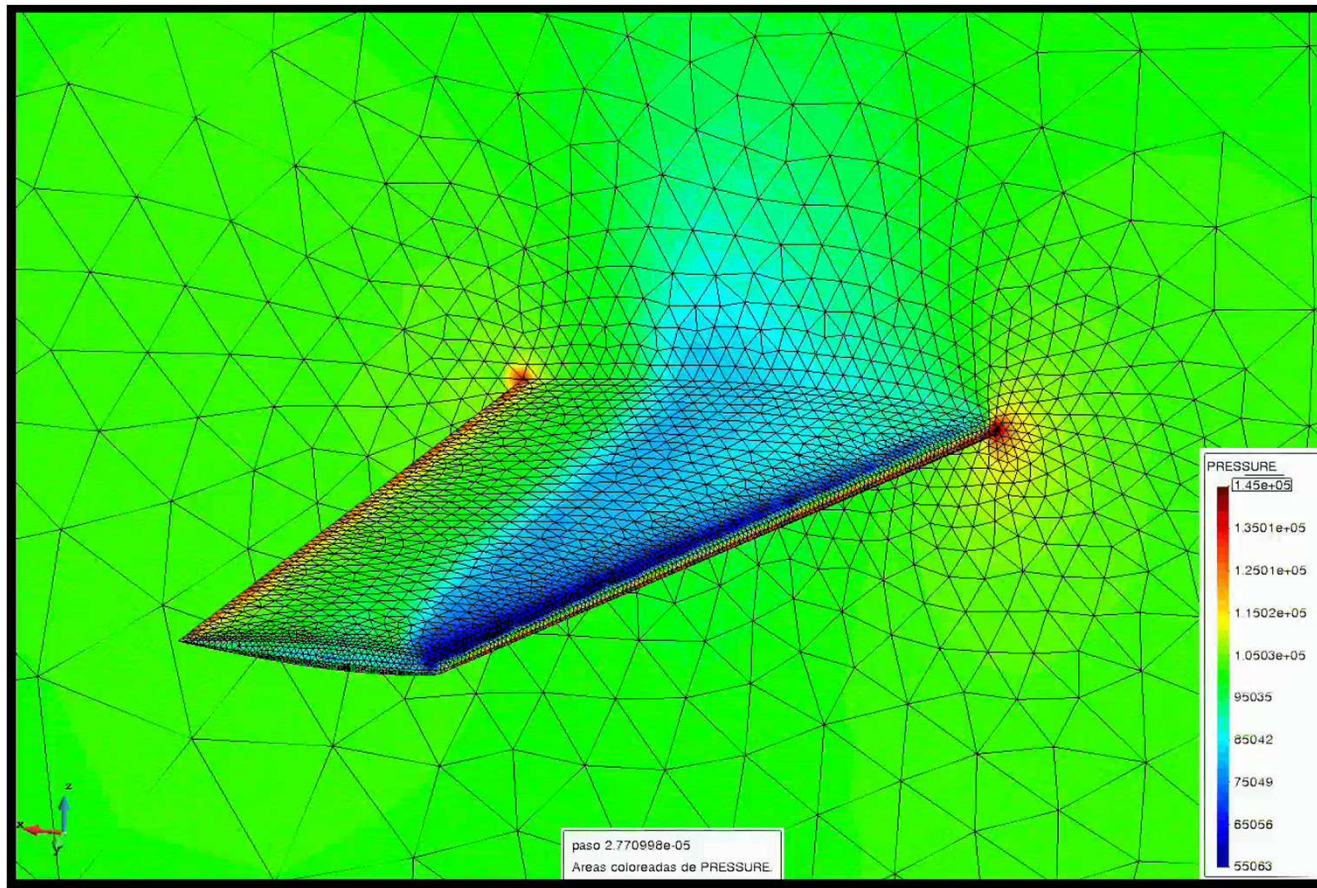


Ground Water Seepage



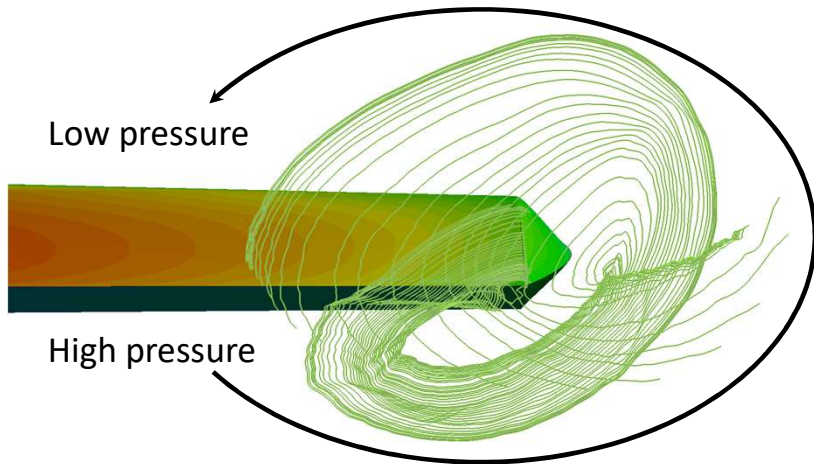
FSI and ALE

Compressible CFD (multiple levels of accuracy)



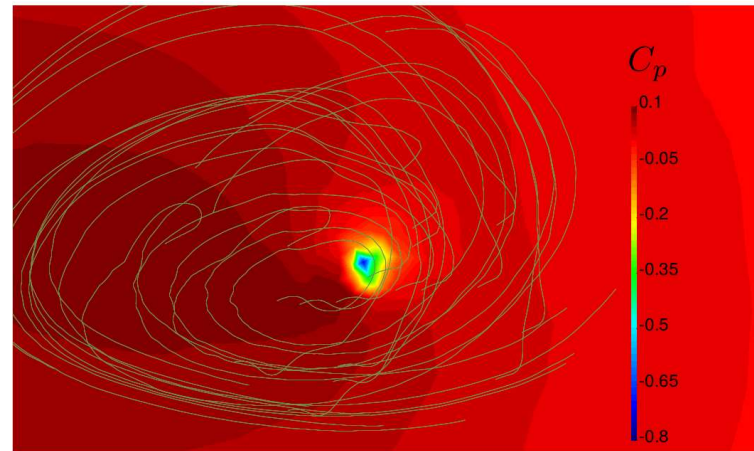
Transonic Potential Flow Solver

ONERA M6 wing

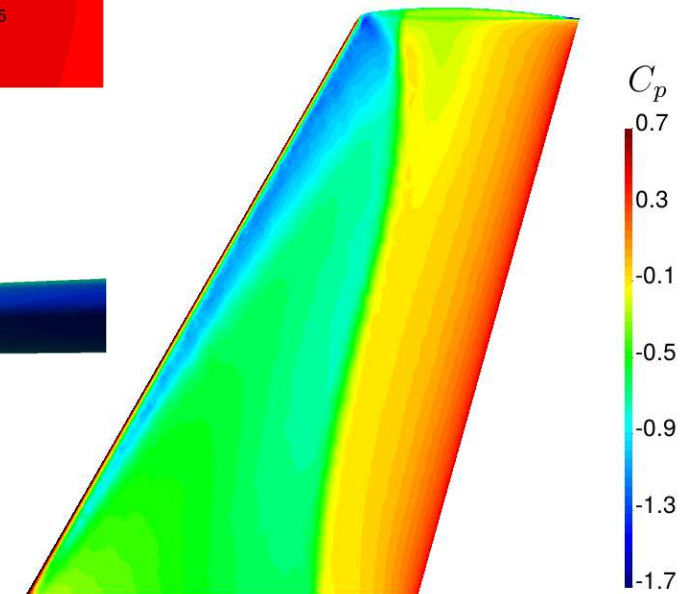
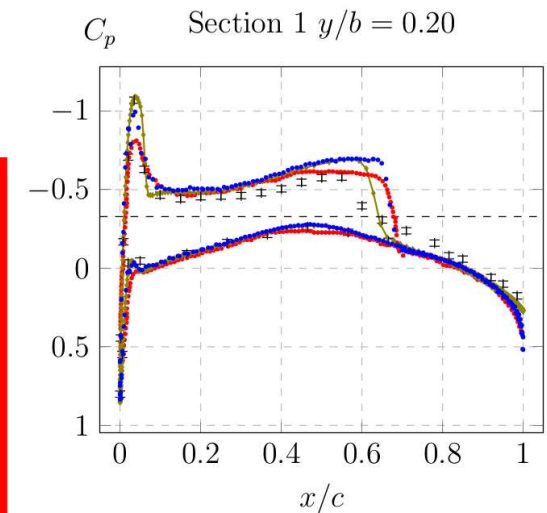


Source: NASA Langley Research Center

Iñigo Pablo López Canalejo@TUM

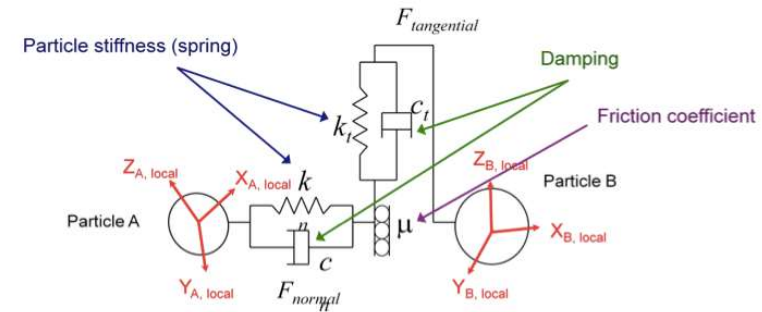


• FE Potential Solver (Kratos) — RANS
 • FV Potential Solver I Experiment

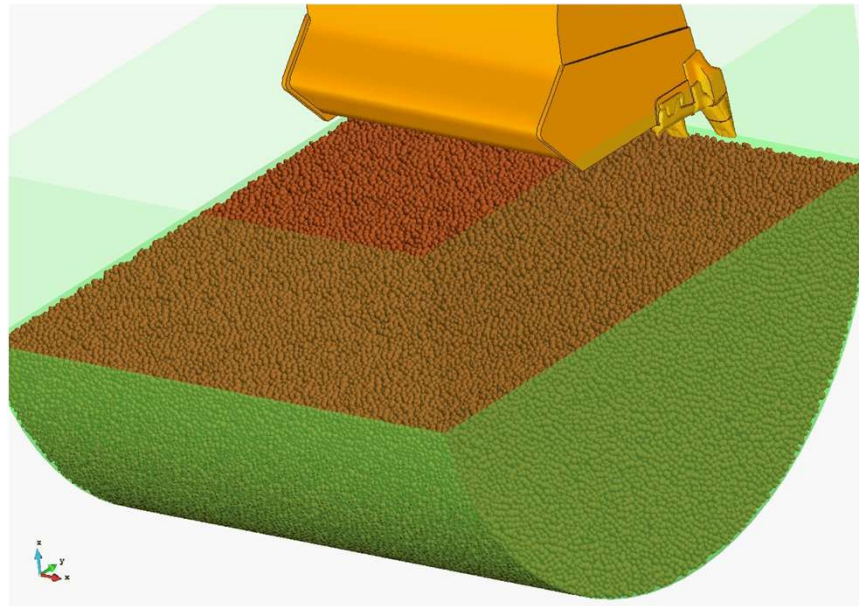


DEM

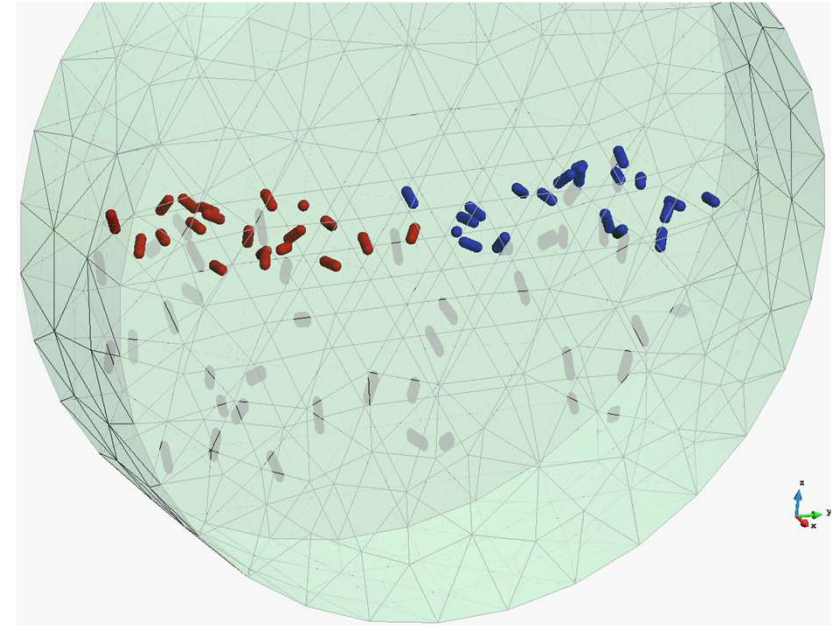
- Cohesive and non-cohesive materials
- Spherical and non-spherical particles
- Parallel computing
- Several contact laws
- Python interface
- Easy/fast coupling with other Kratos apps



Spherical particles



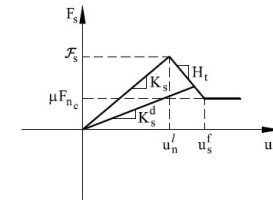
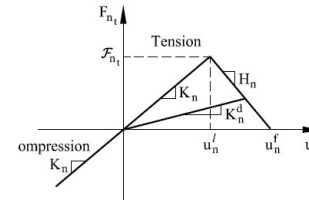
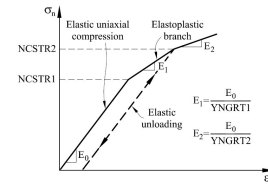
Non-spherical particles



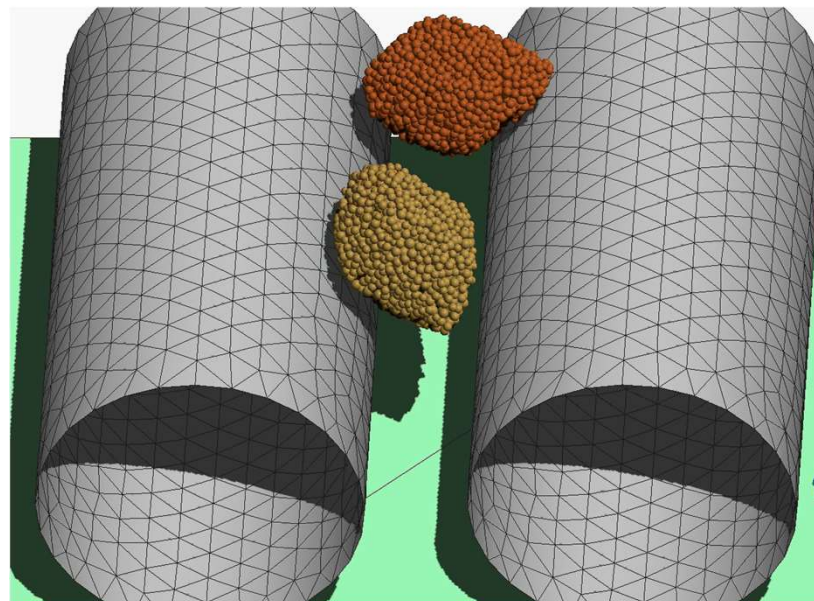
Miguel Angel Celigueta

DEM

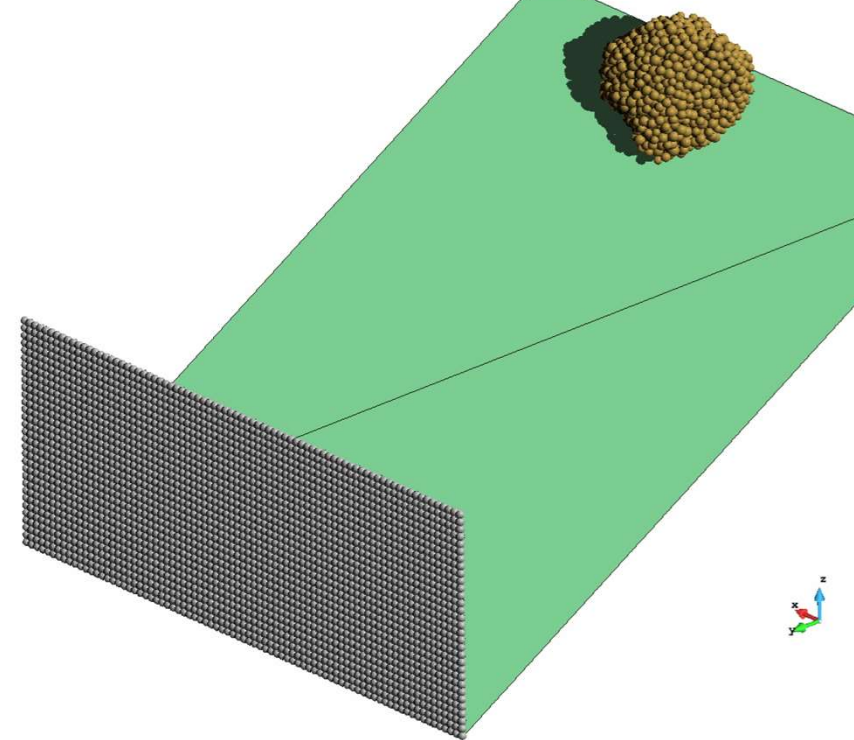
- Bonded particles
- Custom constitutive law
- Bond damage, plasticity, fracture, ...
- Sintering



Slightly cohesive



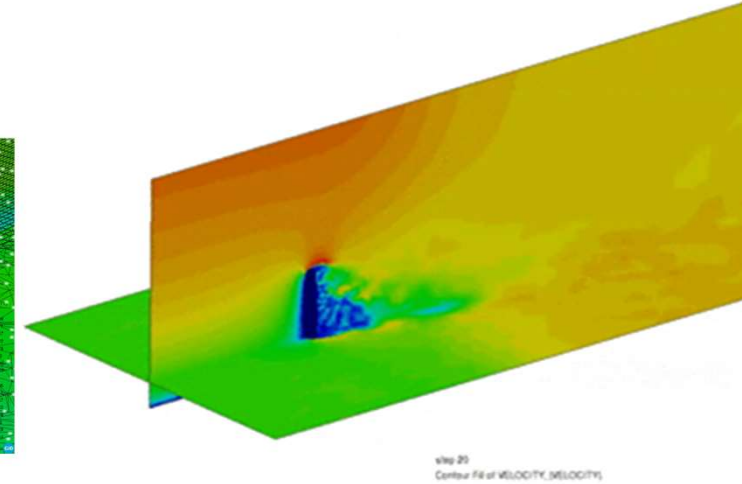
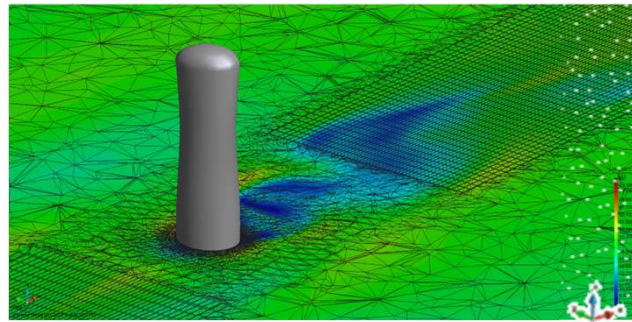
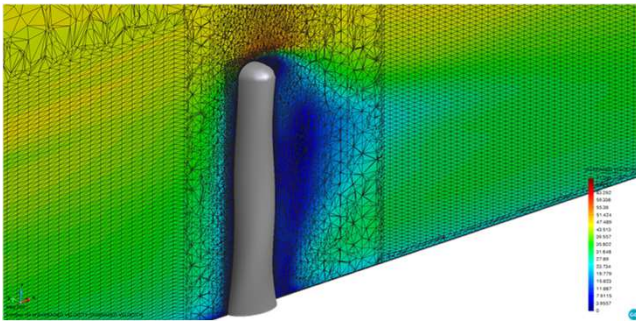
Strongly cohesive and fabrics



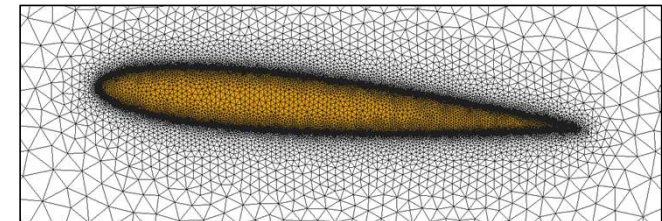
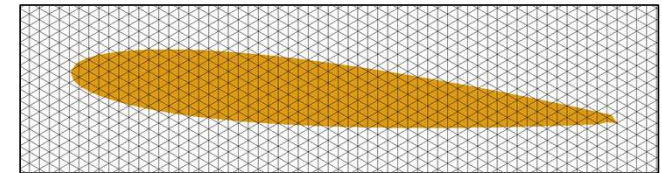
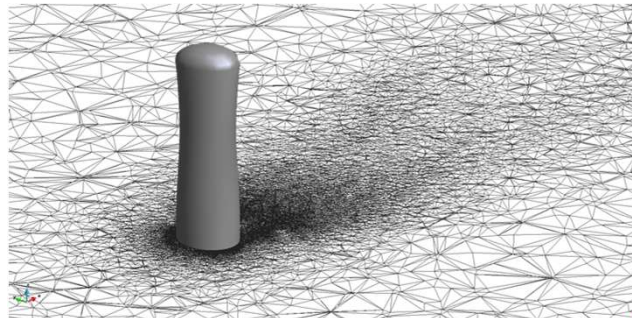
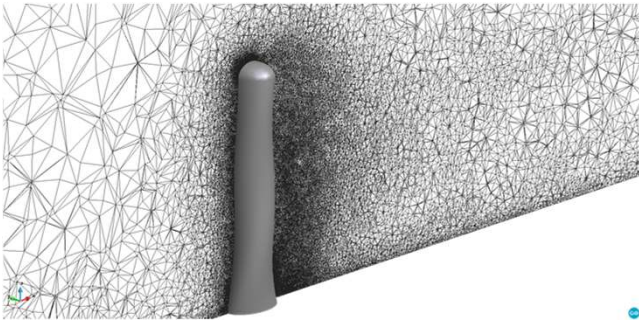
Miguel Angel Celigueta

MPI parallel remeshing

Initial mesh, solved fluid with 96 procs, 3M elements

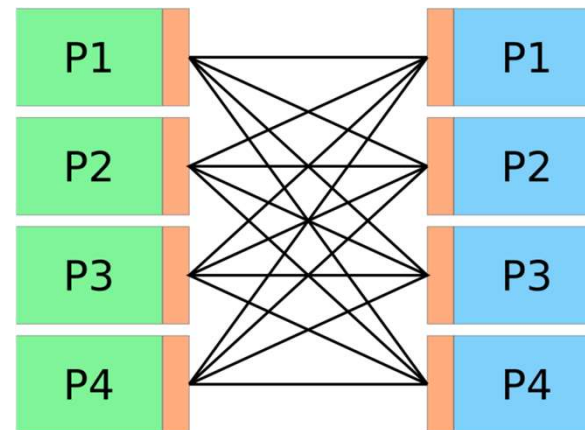
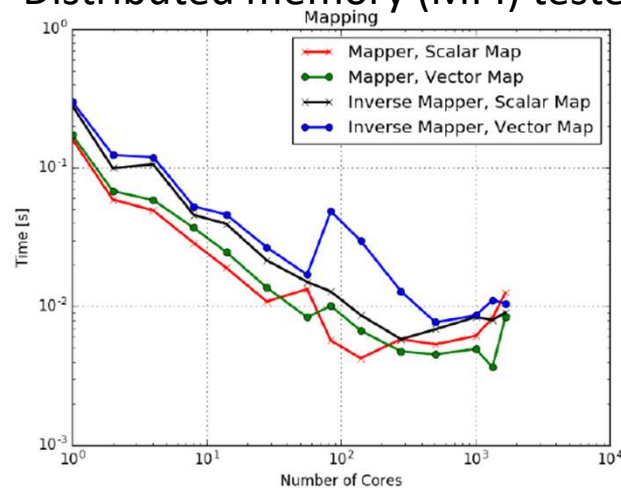


Adapted mesh with 96 procs, 5M elements



Kratos Mapping Application

- Kratos Mapper is fast and robust with following features:
- Parallelized:
 - Shared memory (OpenMP)
 - Distributed memory (MPI) tested over 1000 cores



Peer-to-peer communication, no bottleneck

Example of partitioning of wind turbine blade

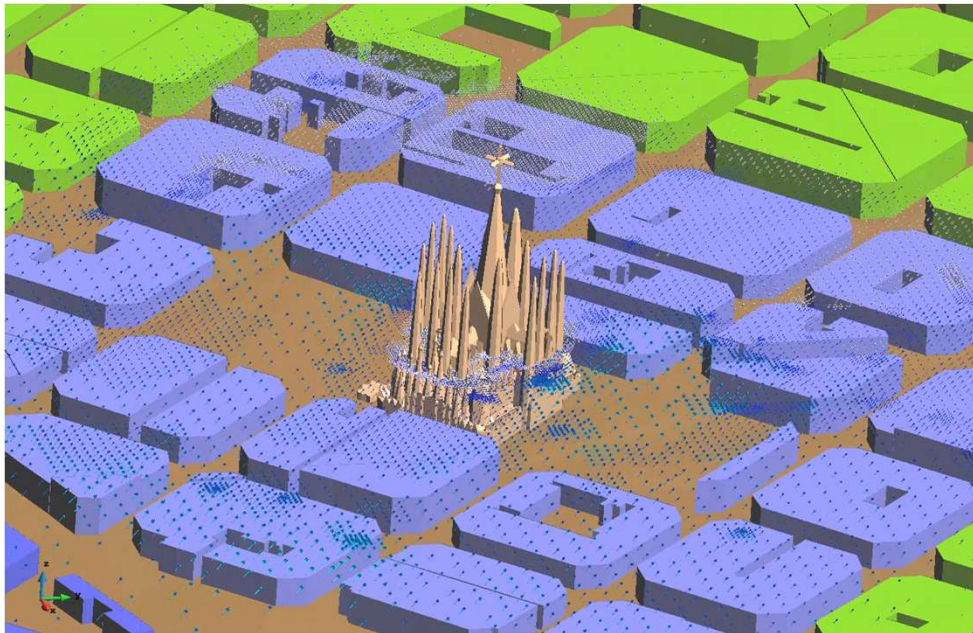
Origin: 1.153.727 nodes (2.303.452 elements)
Destination: 368.474 nodes (734.686 elements)

Philip Bucher@

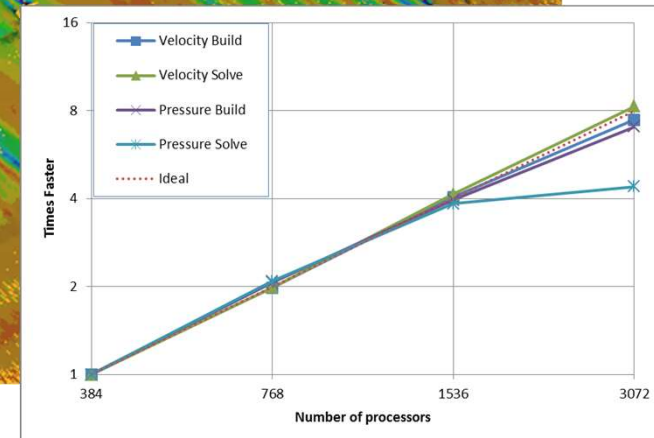
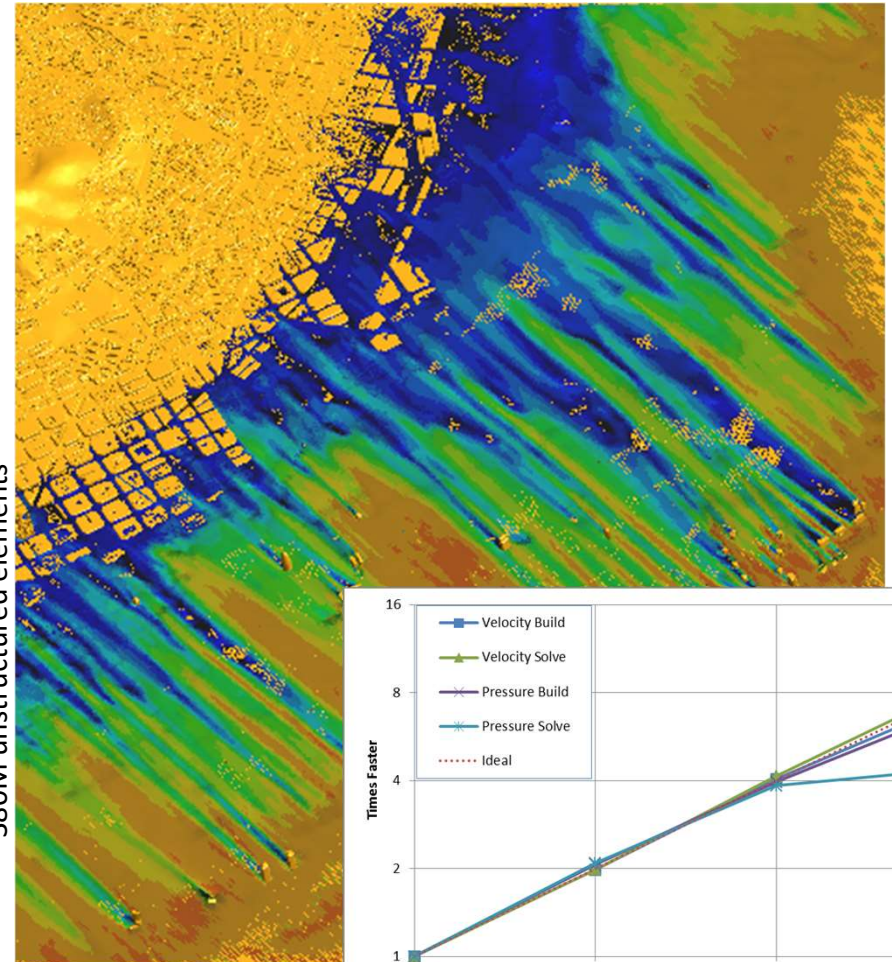


High Performance Computing

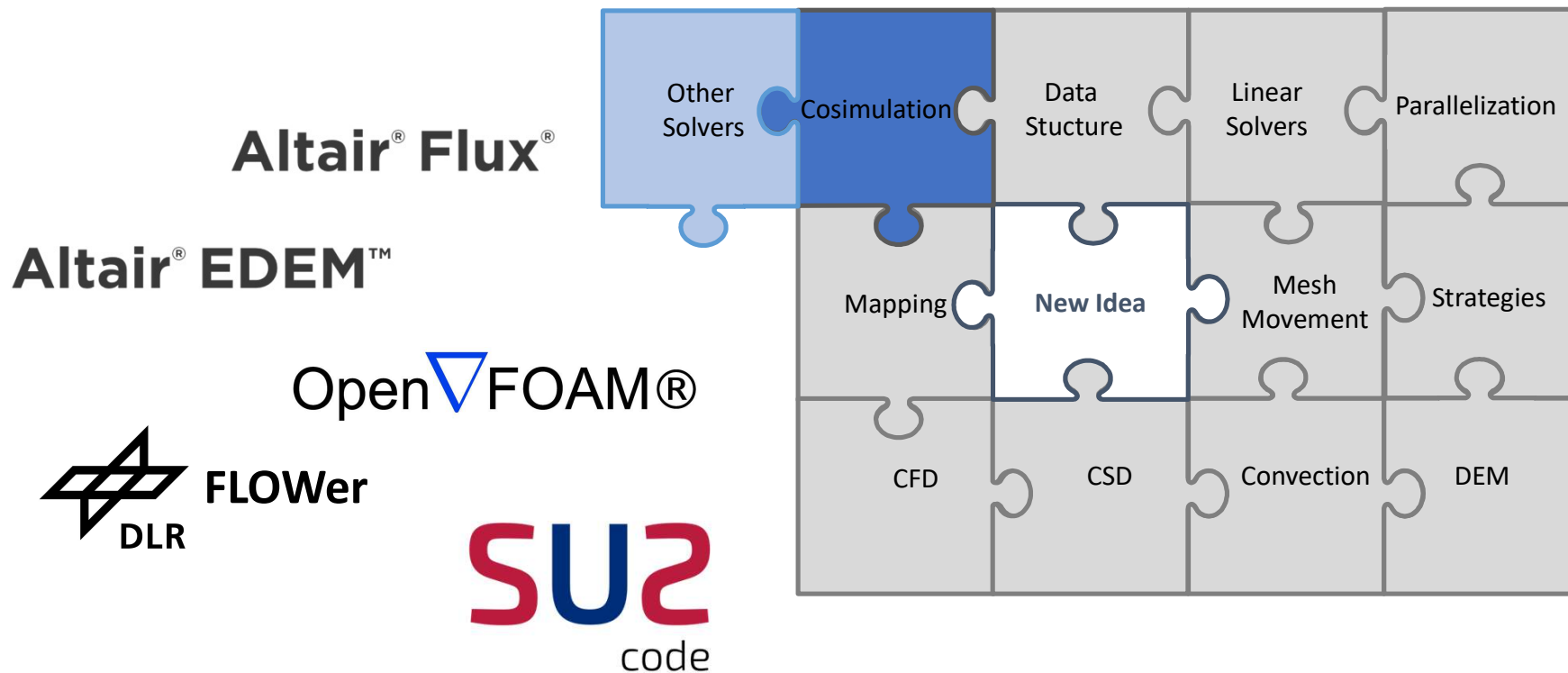
- Prepared for large scale simulations
- Shared Memory Parallelization
- Several applications are migrated to MPI



380M unstructured elements



Cosimulation Application

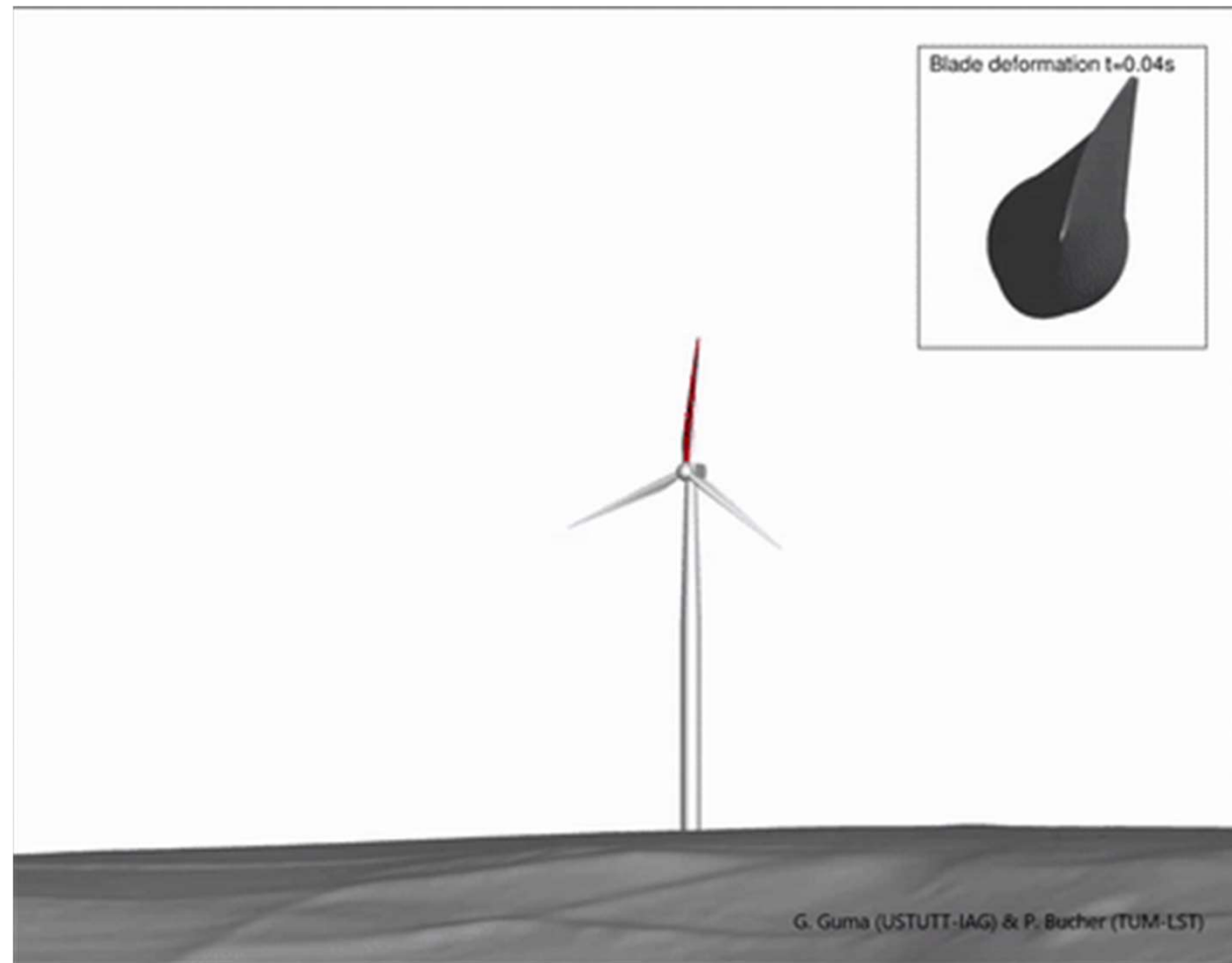


Wind Turbine FSI

FLOWer, Kratos Mapper, Kratos Structural

- FLOWer ran over 7000 cores
- Kratos Structural ran on 144 cores
- Mapping between them

Philip Bucher, Giorgia Guma @ 

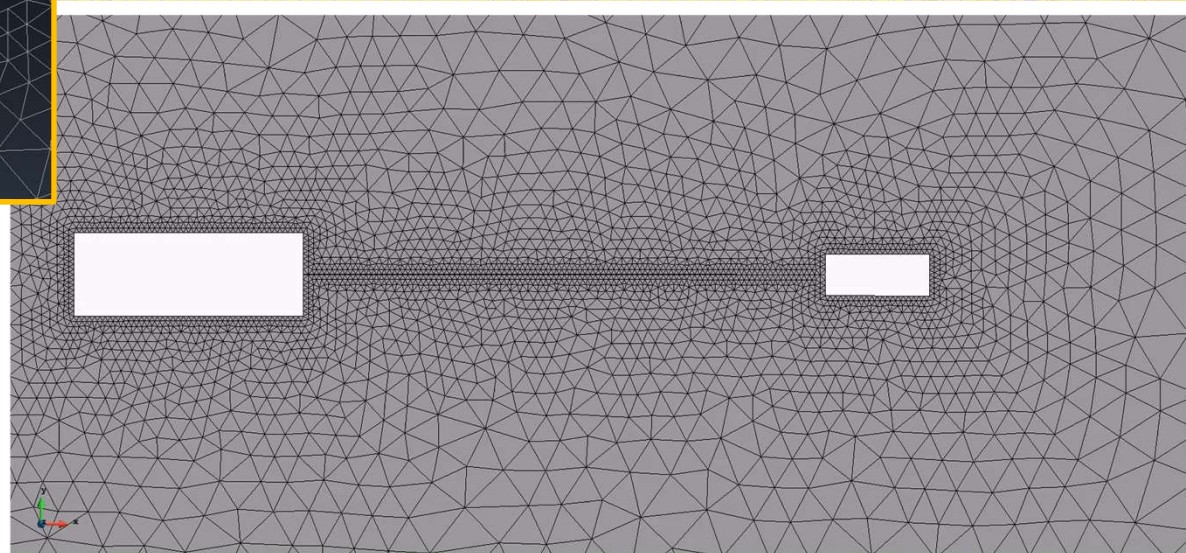
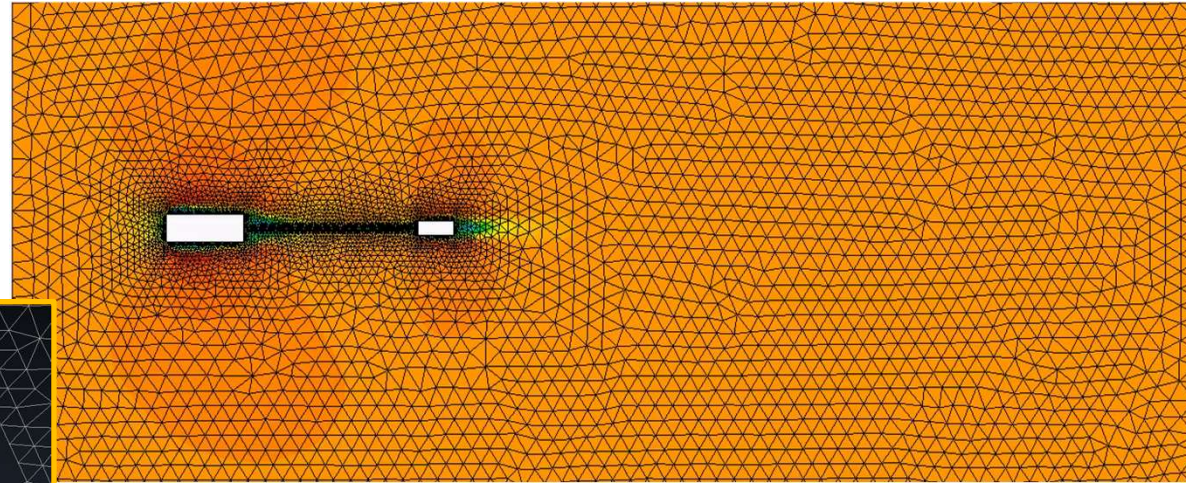
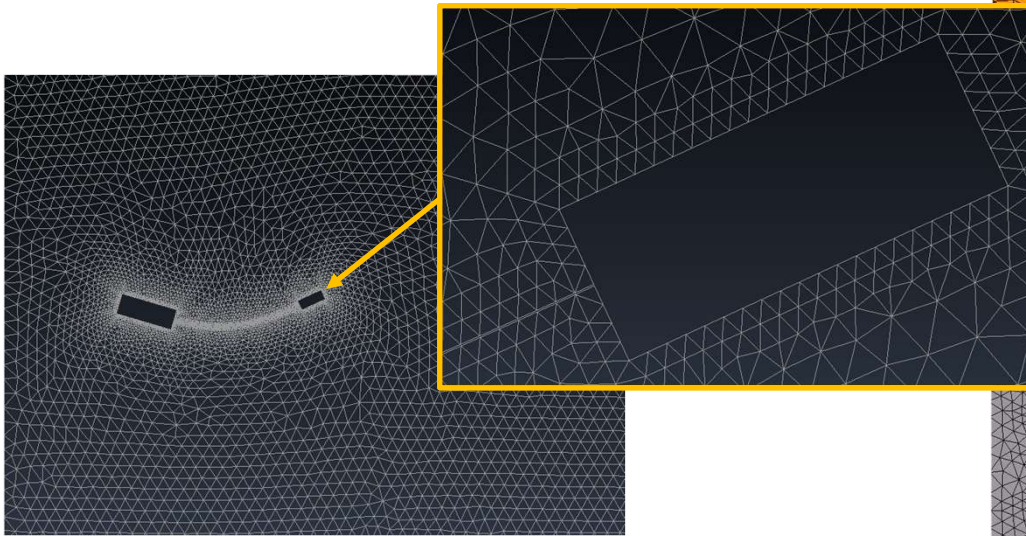


Coupling In Kratos

The combinations are growing fast

Body fitted Fluid Structure Interaction (FSI)

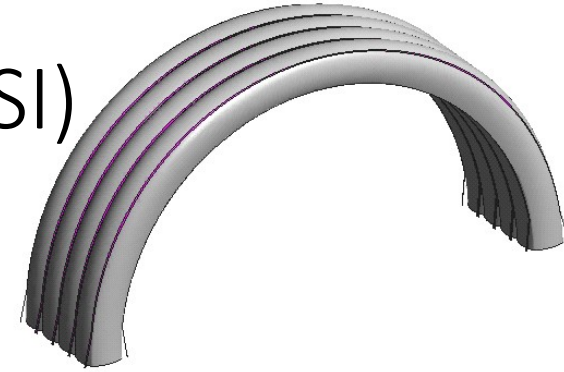
- Robust and accurate mesh movement strategies



by Andreas Mini@ 



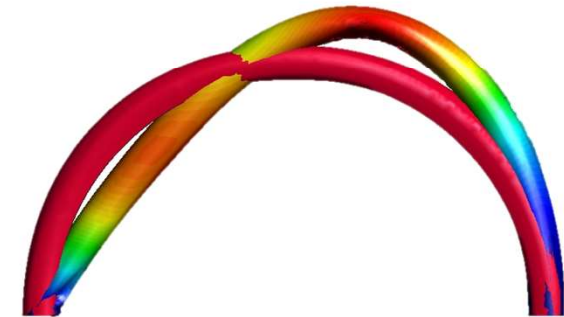
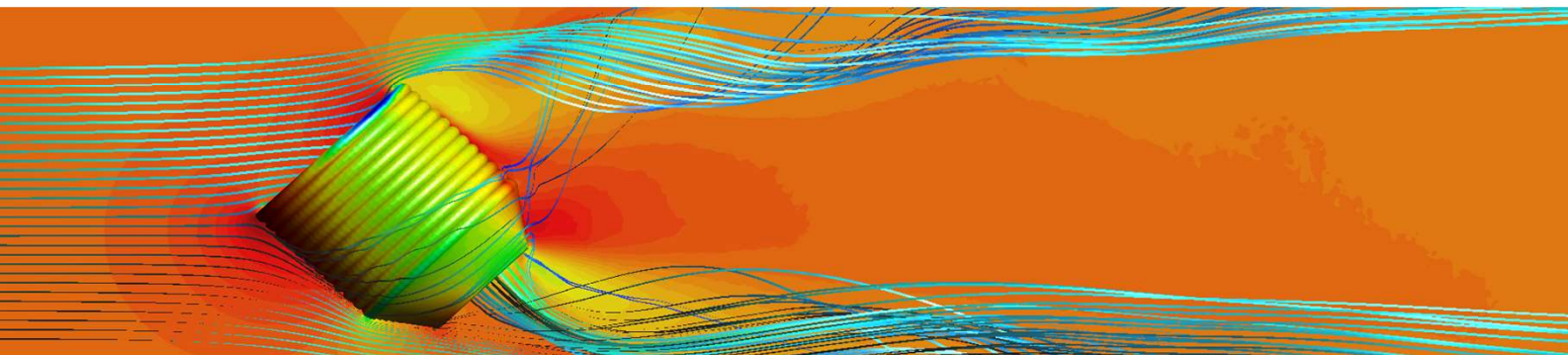
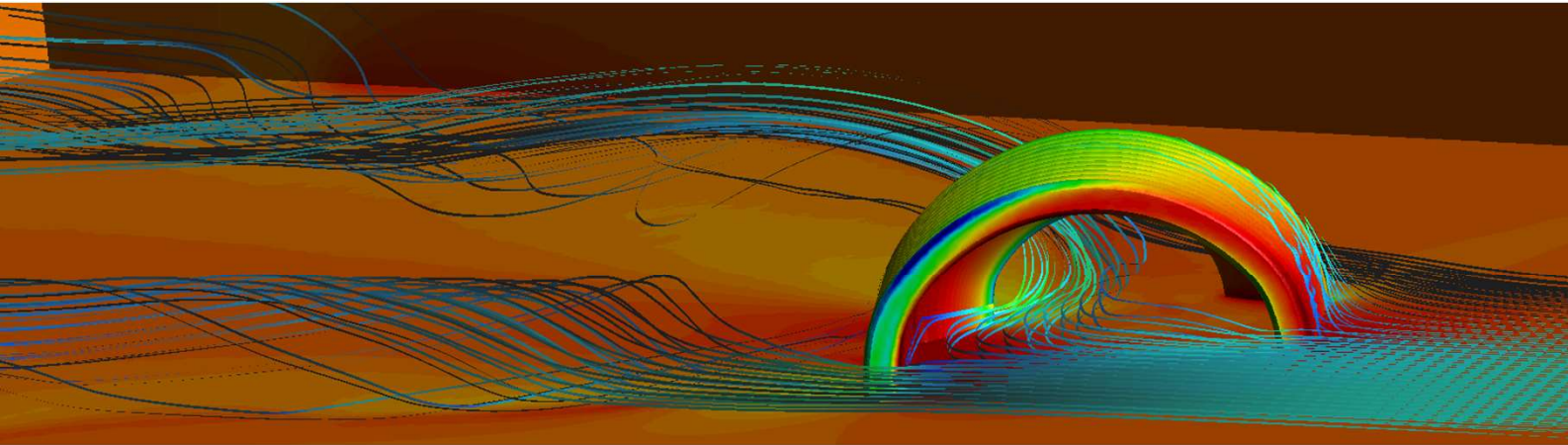
Immersed Fluid Structure Interaction (FSI)



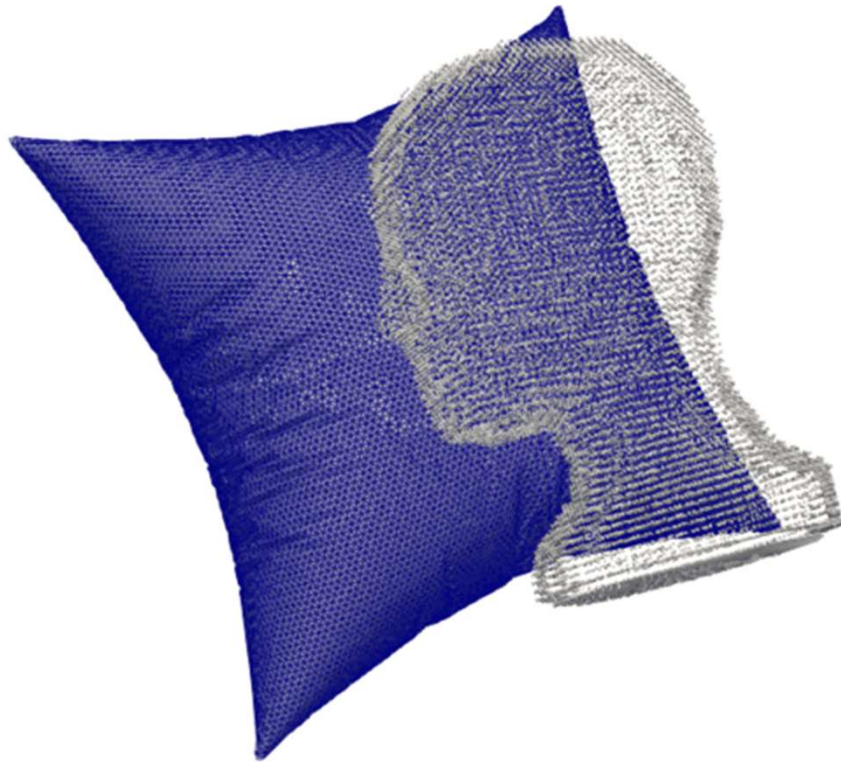
15:00 - 15:20

FE Technology, Unfitted simulation and Coupling

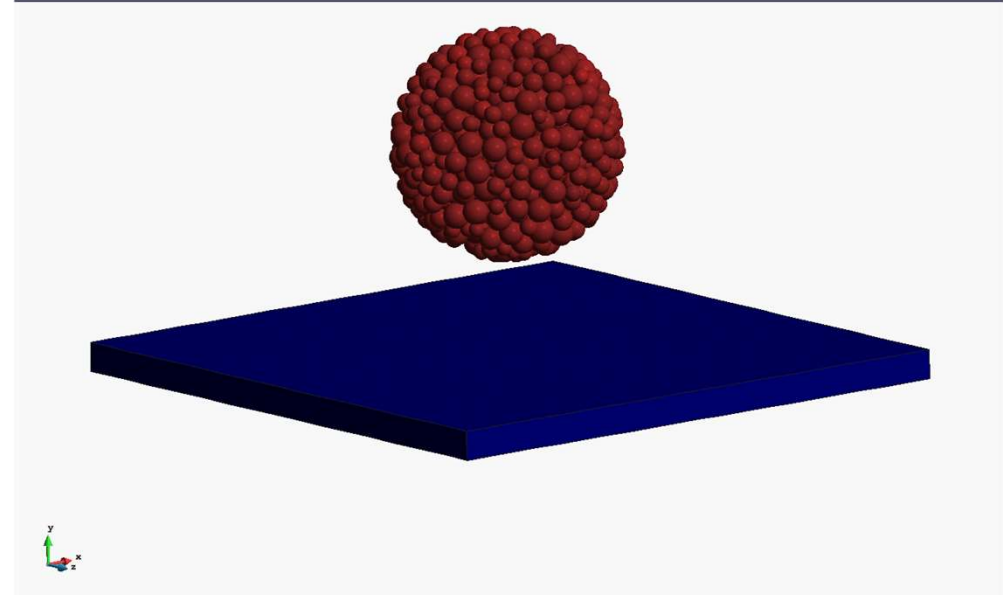
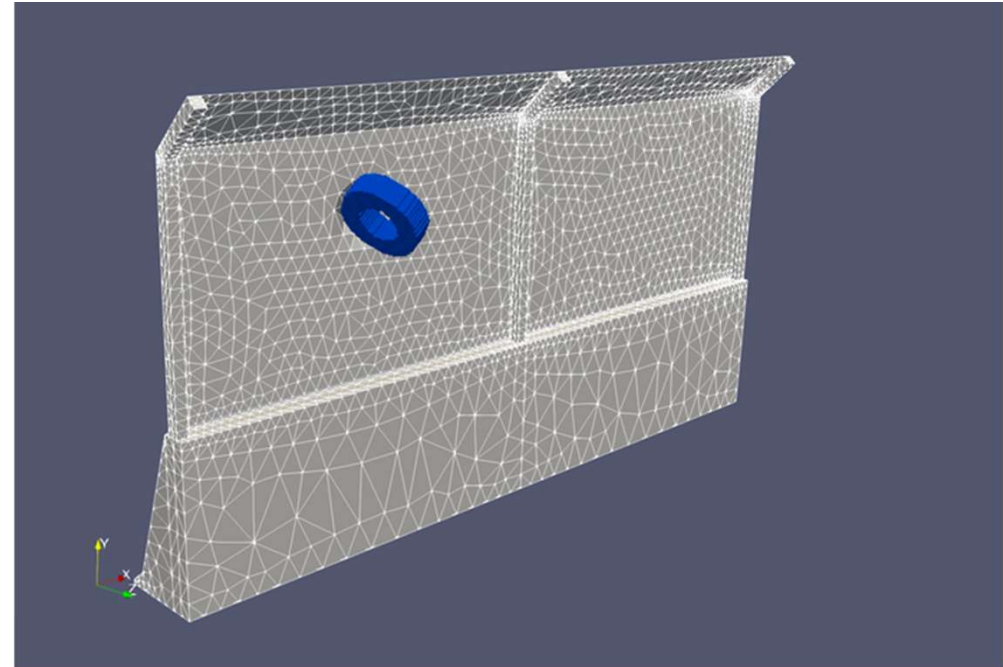
Rubén Zorrilla (CIMNE Barcelona / Polytechnic University of Catalonia)



Particles and Structure

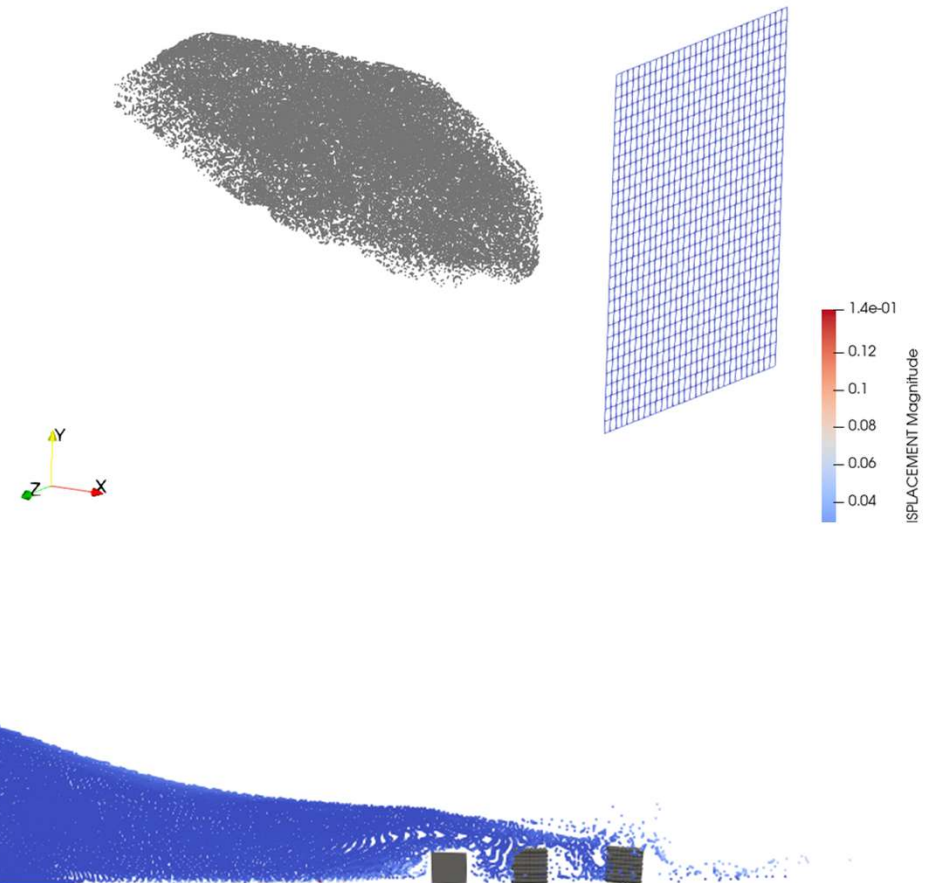
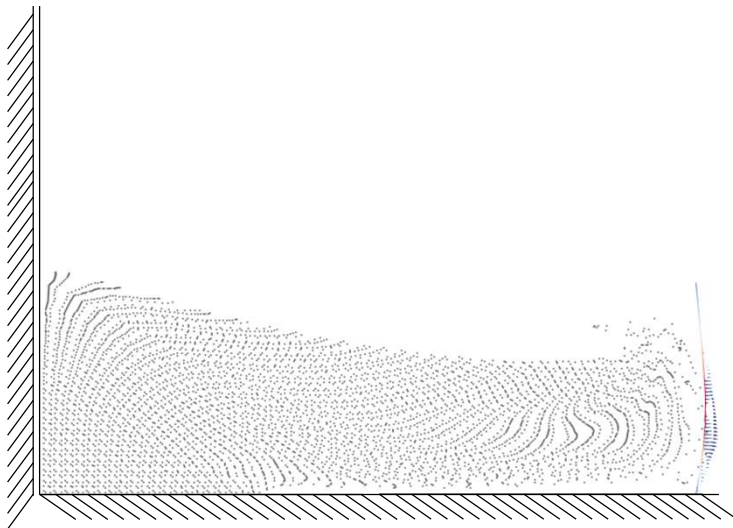


Klaus Sautter@



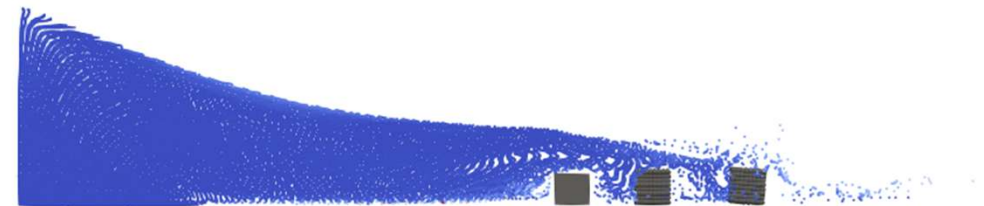
MPM-FEM-DEM coupling

MPM-FEM Granular Flow impacting flexible cable/membran



MPM DEM

Granular Flow impacting
discrete particles



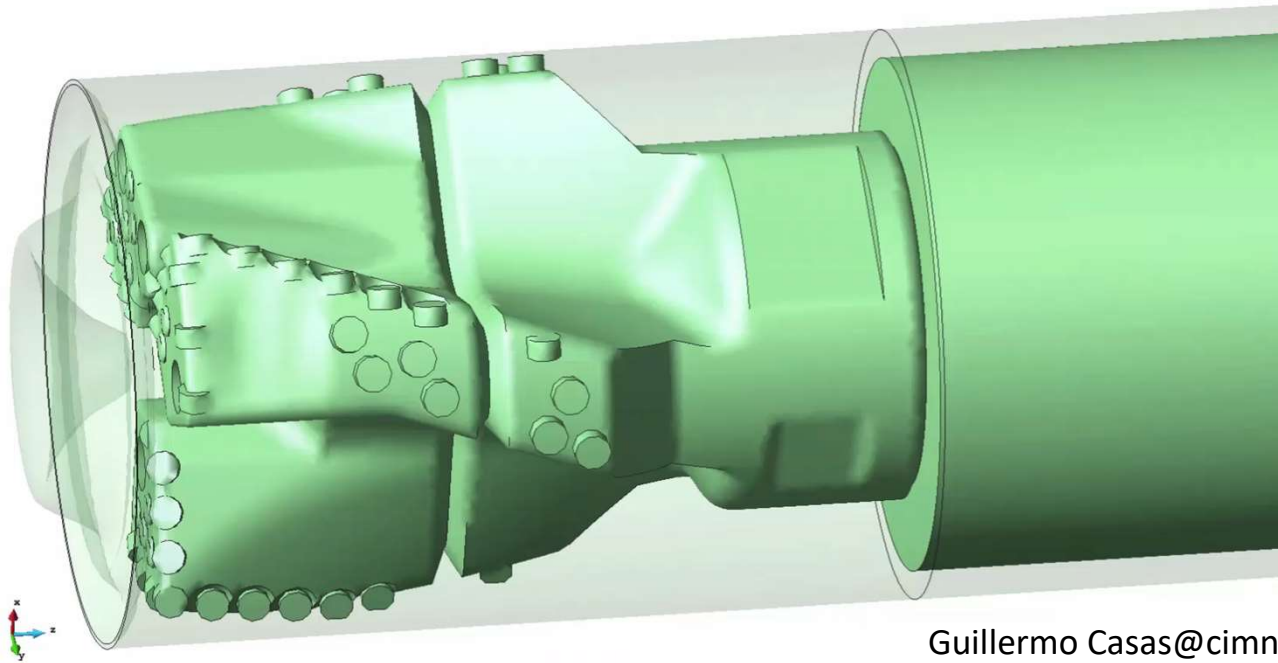
13:40 - 14:00

MPM Capabilities in Kratos
Antonia Larese (University of Padova)

Particles and Fluids

Two-way coupling:

- Fluid applies drag forces
- Particles accumulation affect the porosity
- Embedded boundary in fluid for moving parts



Guillermo Casas@cimne



Thermo-Mechanics

- Thermal dependent materials
- Thermal expansions



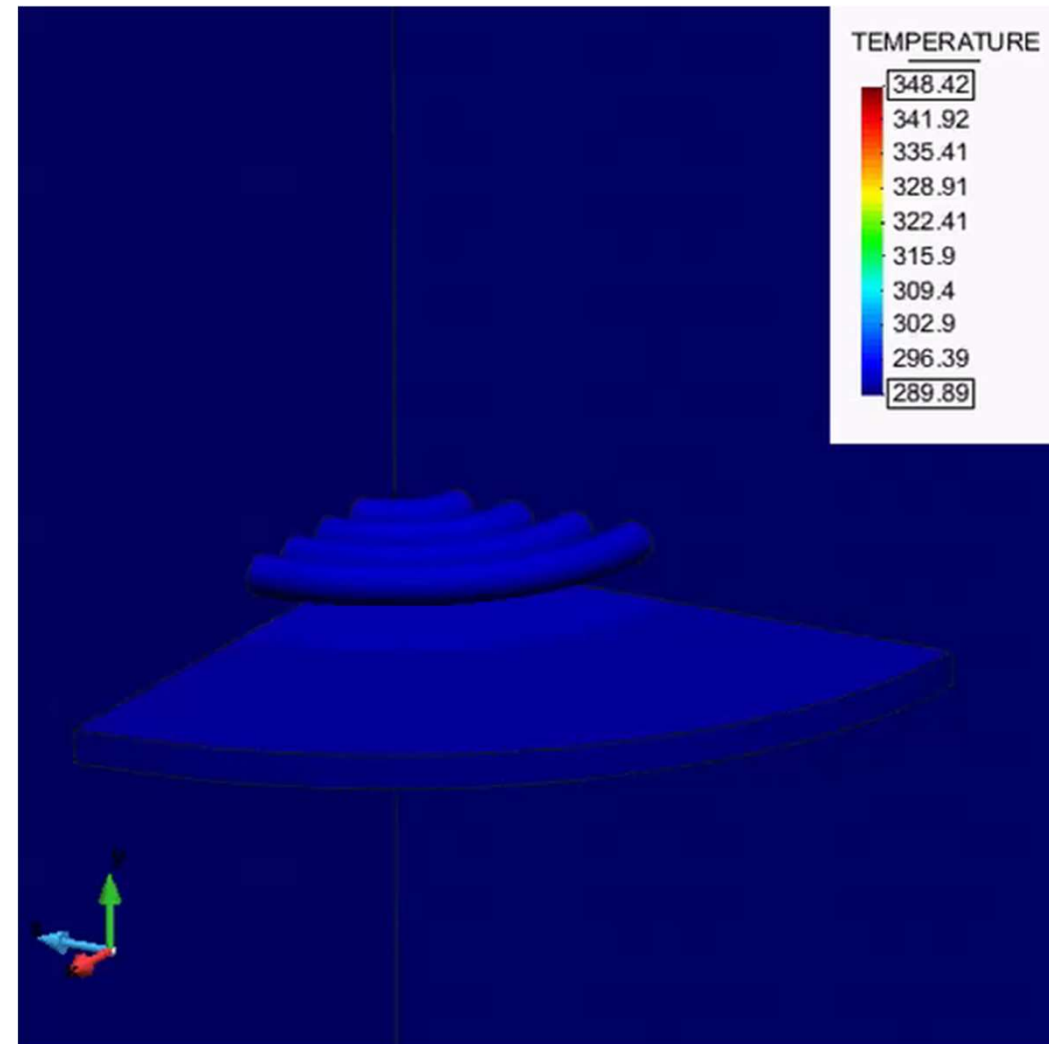
Alejandro Cornejo

step 1
Contour Fill of TEMPERATURE.
Deformation (x10): DISPLACEMENT of Kratos, step 1.



Induction with Conjugate Heat Transfer

- Electromagnetic simulation to get the heat source
- Buoyancy solution for heat transfer

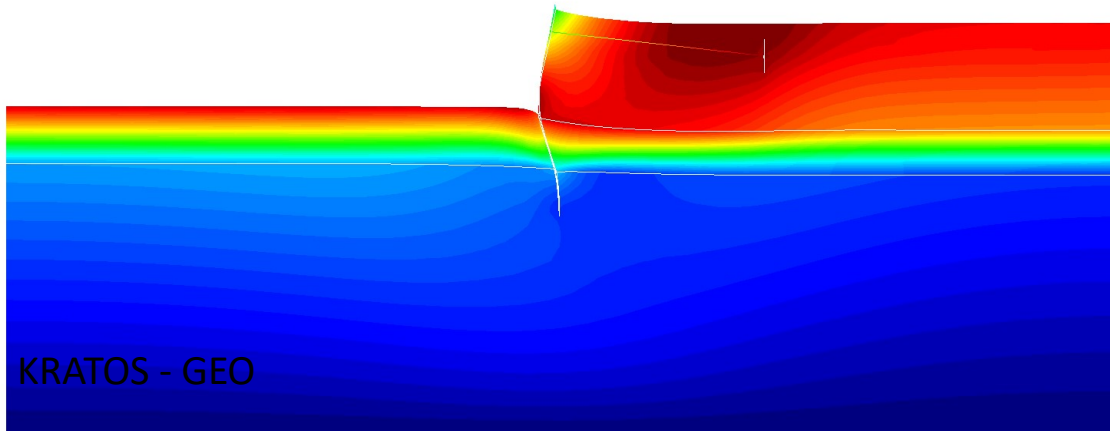


By: Jordi Cotela

Done by Kratos

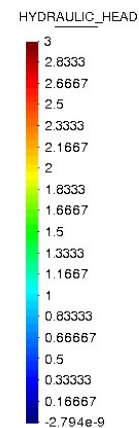
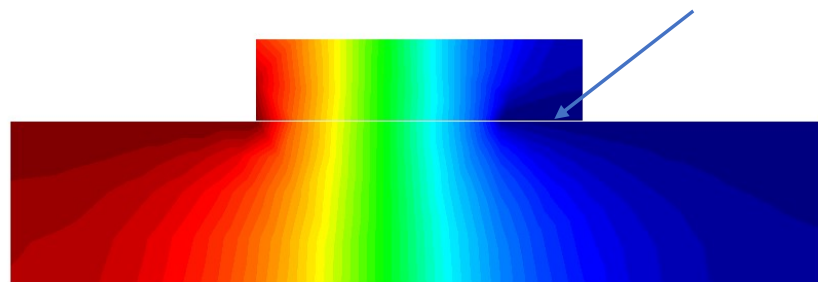
Geomechanics

Soil-structure interaction



Kratos - GeoMechanics

Pipe element



Contour Fill of HYDRAULIC HEAD/tris element ap.

11:00 - 12:30	Session 2: Computational Geo-Mechanics Chair: Vahid Galavi (Diana FEA)
	Mini-Keynote: <i>Soil-Structure Interaction in Kratos-Geomechanics</i> Vahid Galavi (Diana FEA)
	<i>Kratos Multiphysics for Dynamic Geotechnical Railway Applications</i> Aron Noordam (Deltares)
	<i>Using the Probablistic Toolkit (PTK) with Kratos</i> Rob Brinkman (Deltares)
	<i>Introduction to Deltares GeoSoftware driven by Kratos</i> Raymond van de Meij (Deltares)

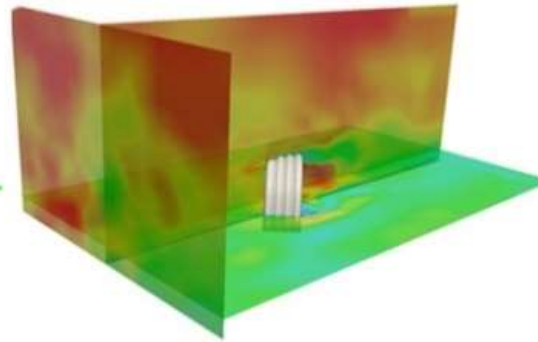
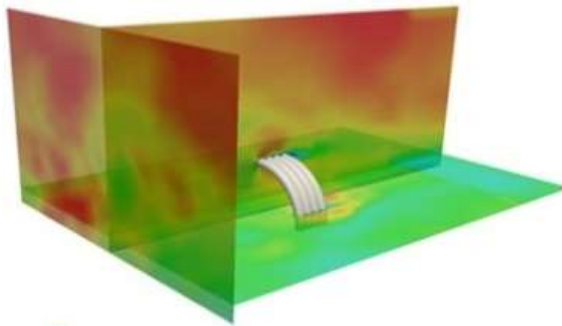
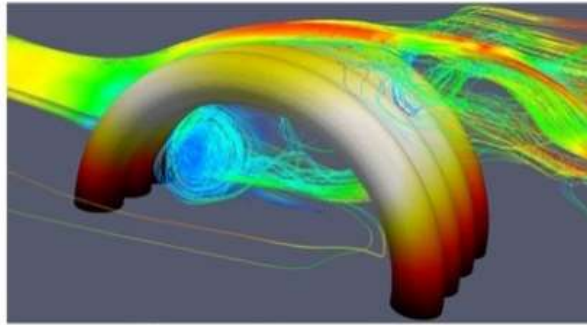
Groundwater flow and piping

By:Vahid Galavi

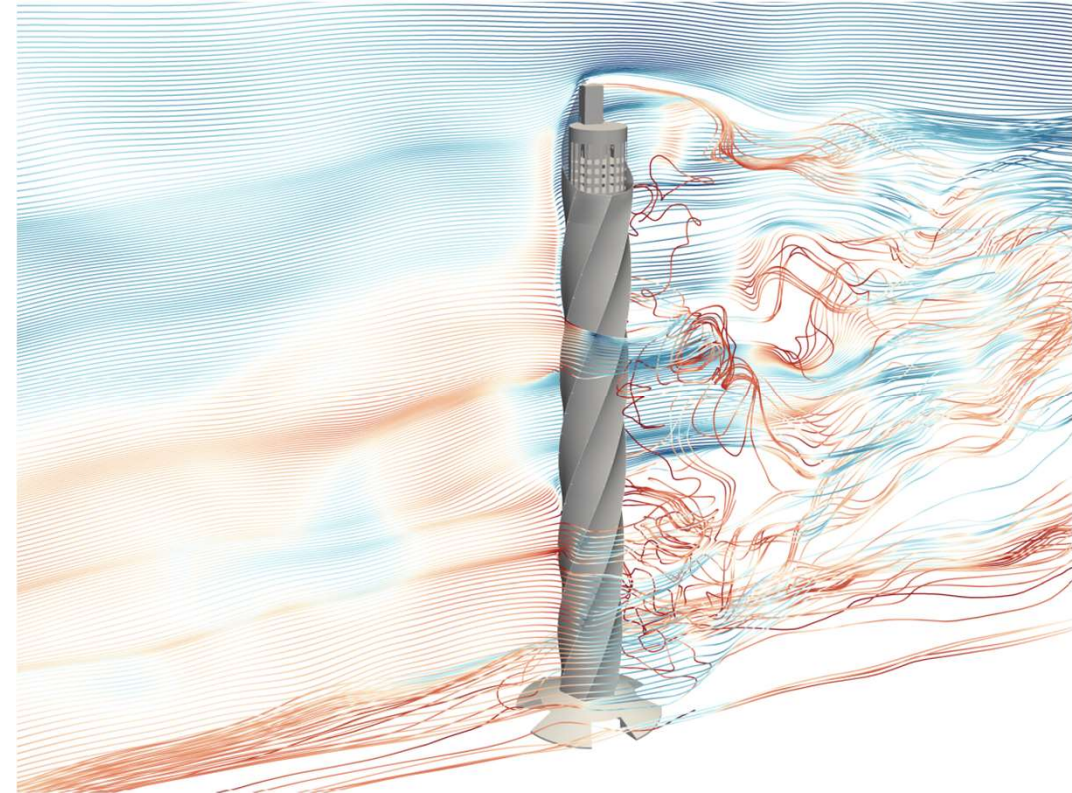
Virtual Wind Tunnel



© Buildair

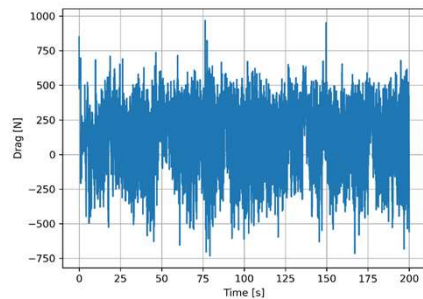


ULITES

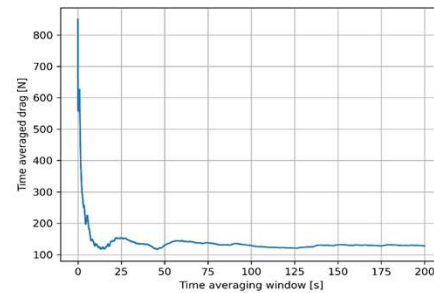


by Andreas Mini@ 

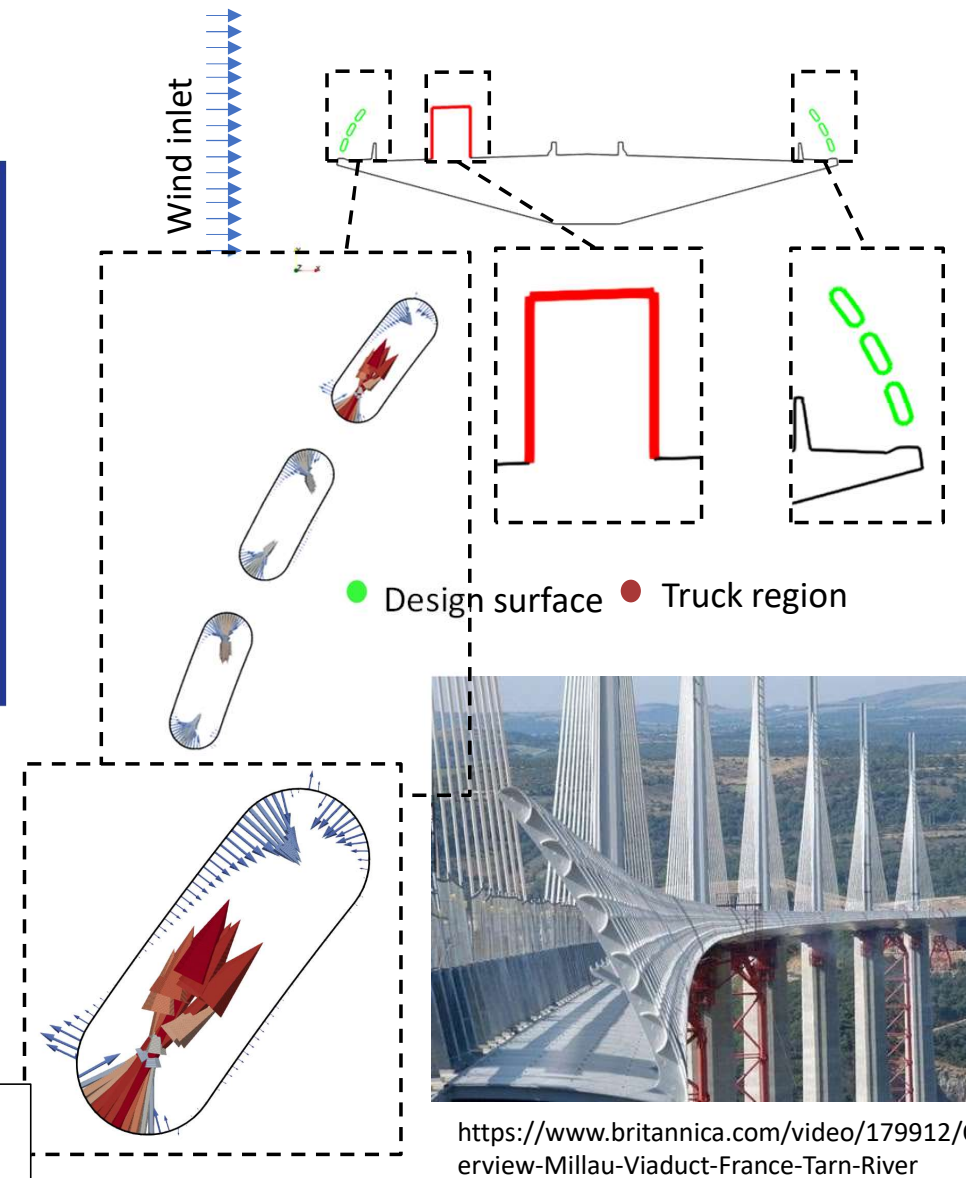
Wind Engineering



Time instantaneous drag



Time averaged drag



15:20 - 15:40

Rans and Optimization in Kratos

Suneth Warnakulasuriya (Technical University of Braunschweig)

MPM

13:40 - 14:00

MPM Capabilities in Kratos

Antonia Larese (University of Padova)

- 2D, 3D and axisymmetric formulations
- Irreducible and Mixed formulations
- Large deformation regimes
- Several constitutive laws

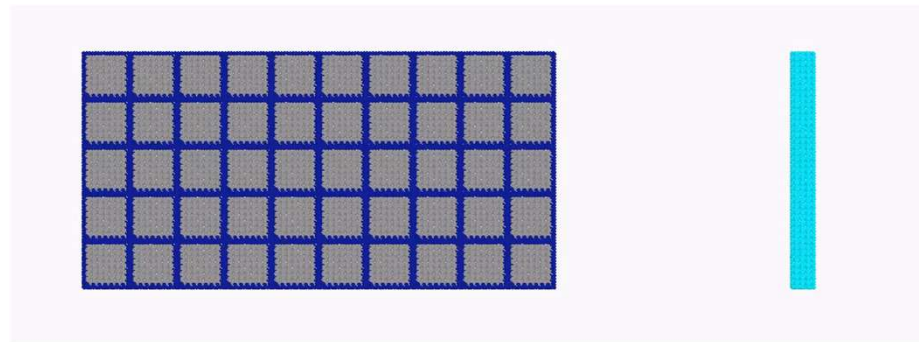


Research and
Development

GRANUFALL TEST



Rigid strip footing

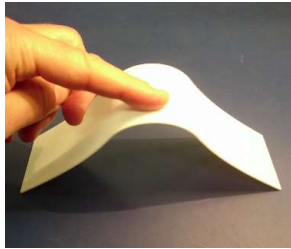


Viscoplastic dambreak on a flexible body

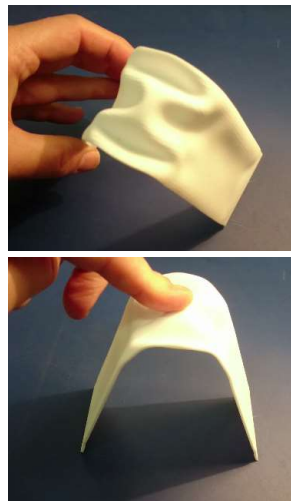


shape optimization and CAD reconstruction

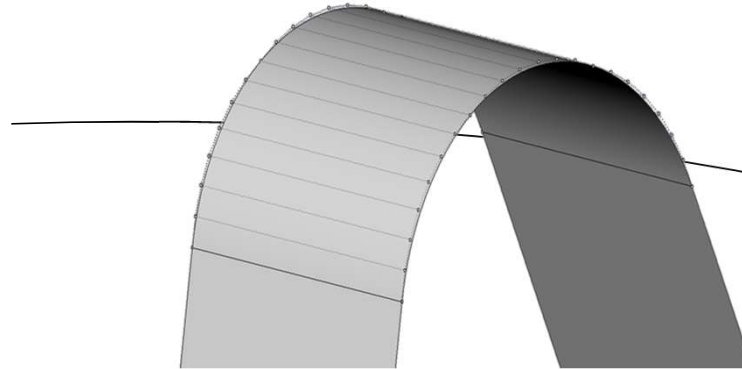
Design problem



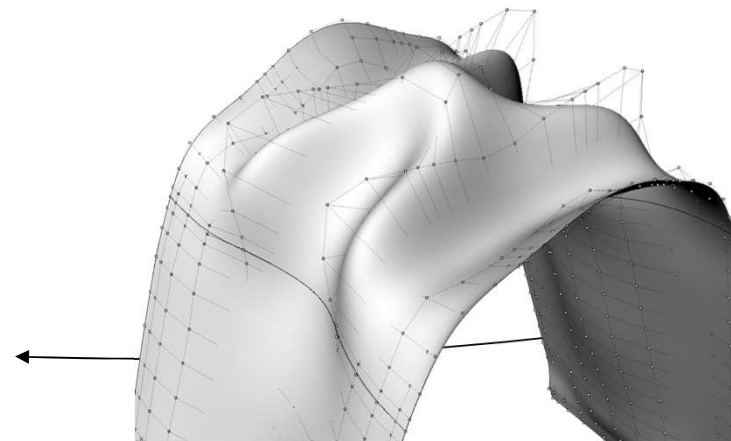
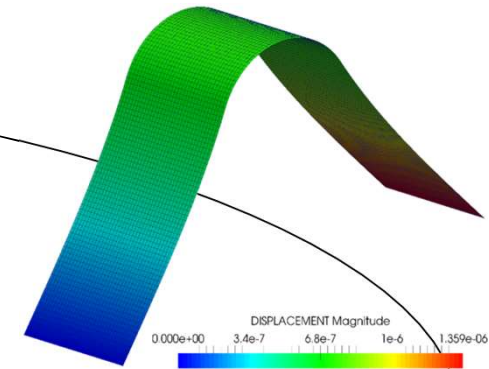
Optimized design



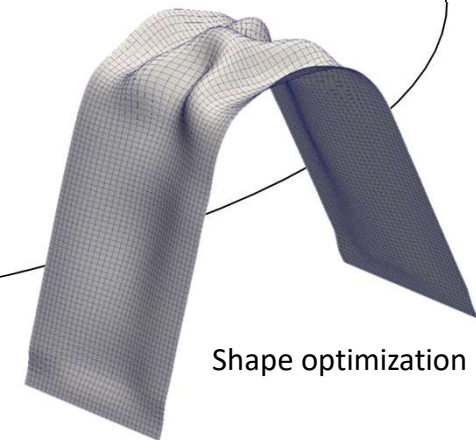
Initial CAD model



FE model & simulation

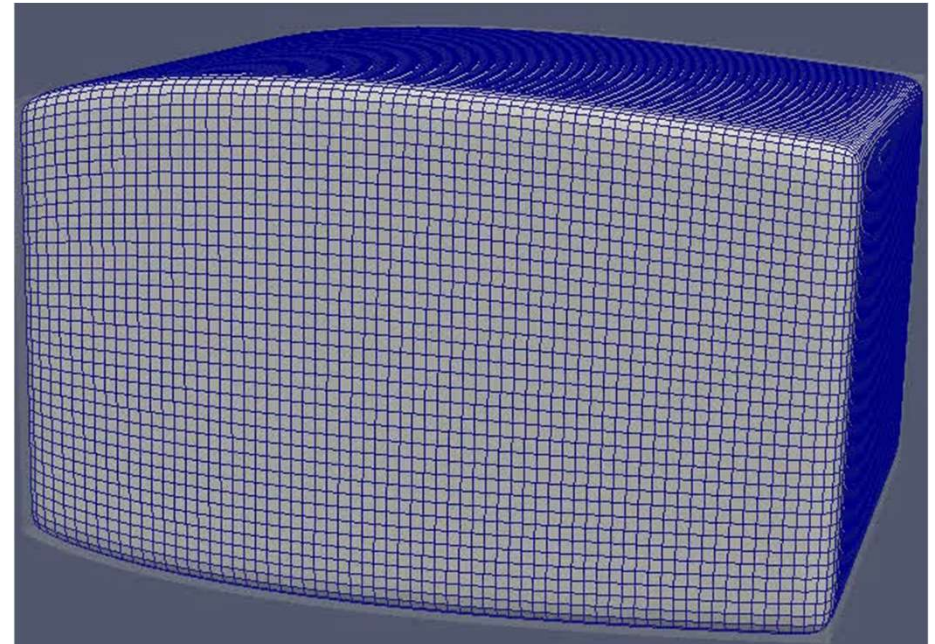
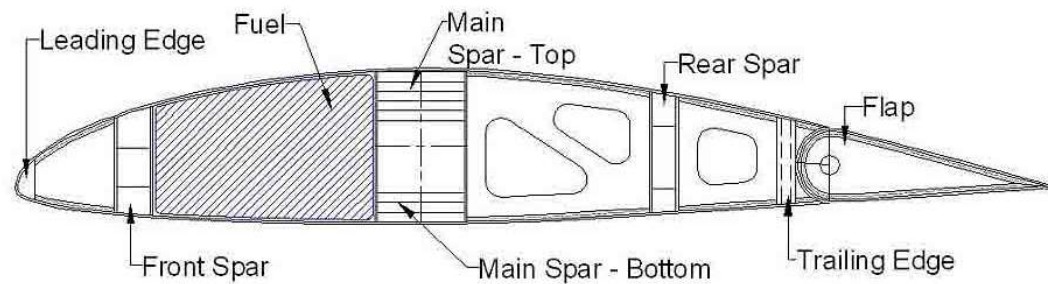


Reconstructed CAD (high quality prototyping input)



Shape optimization

Topology optimization



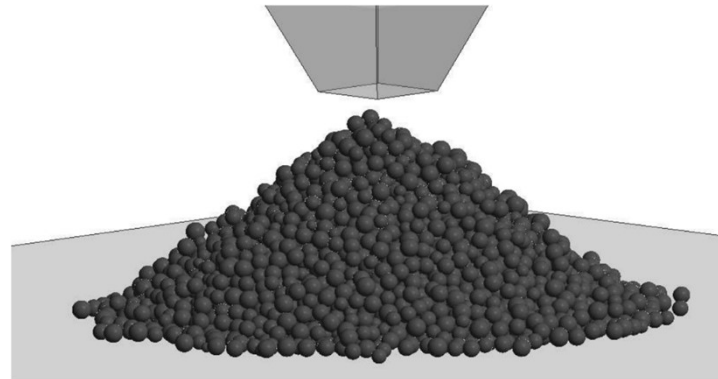
Wing box topology optimization

by Daniel Baumgärtner @ 

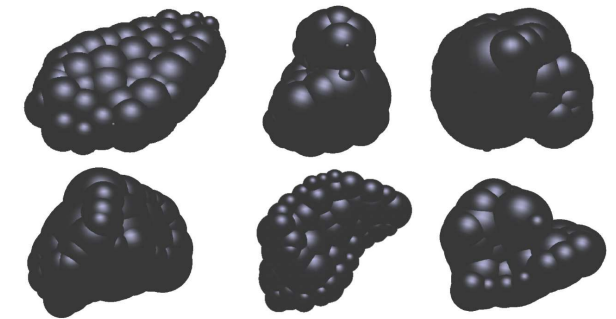
High speed train – ballast



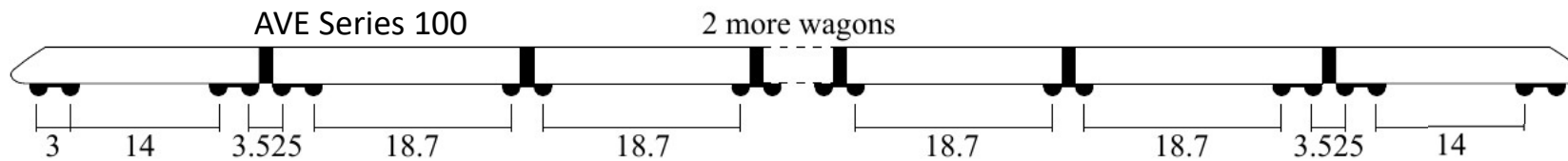
Spherical Particles with Rolling Friction



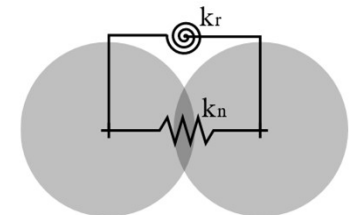
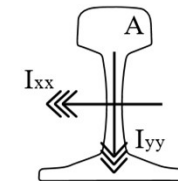
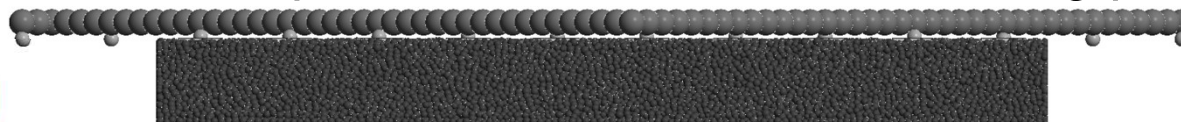
Clusters of spheres



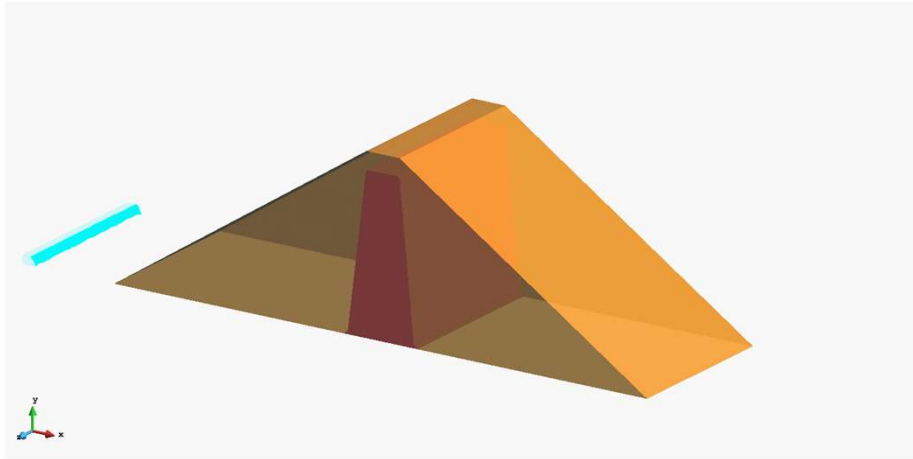
Detailed distribution of contacts



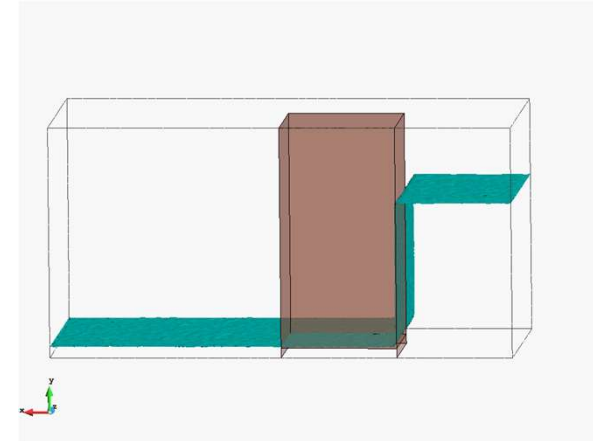
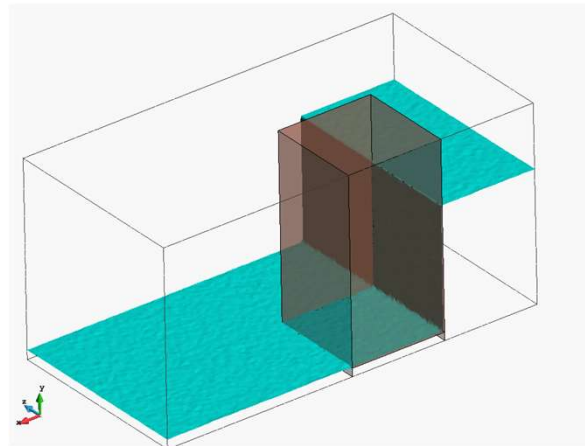
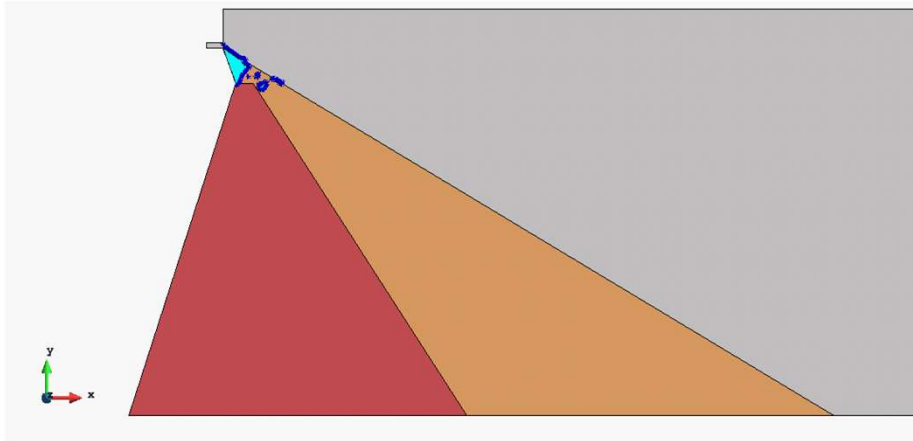
Numerical representation of rails and bearing plates



Rockfill dams or harbour dikes

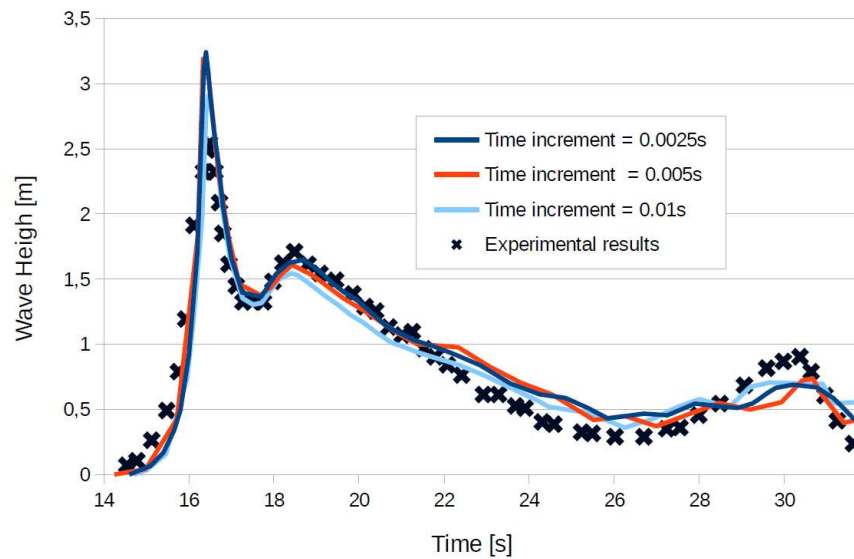


- Navier Stokes Eulerian solver
- Nonlinear resistance law
- Level set technique to track the evolution of the free surface and seepage line



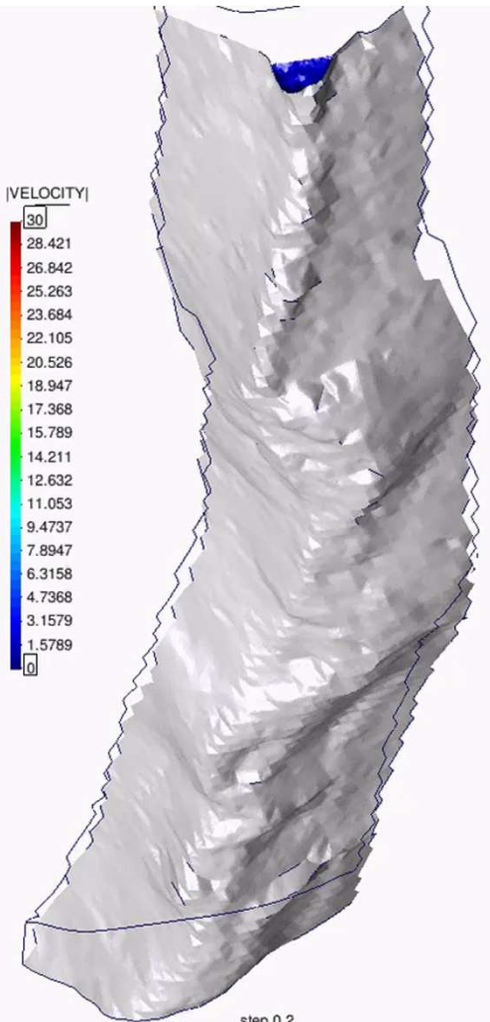
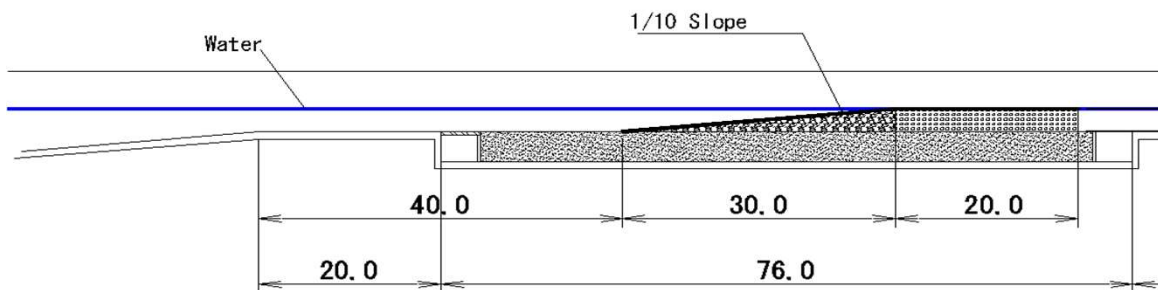
By: Antonia Larese

Surface flow and Waves

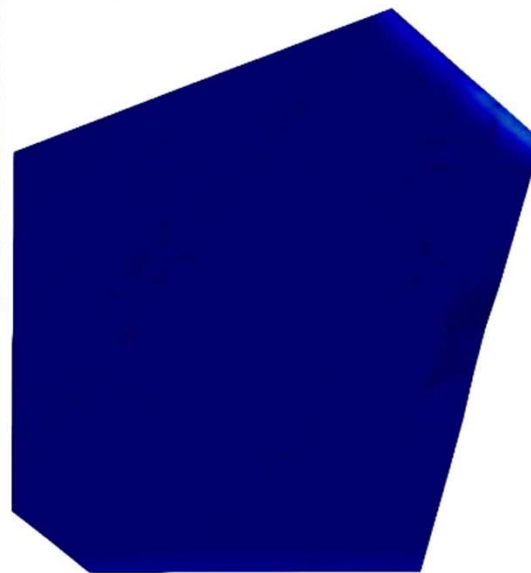
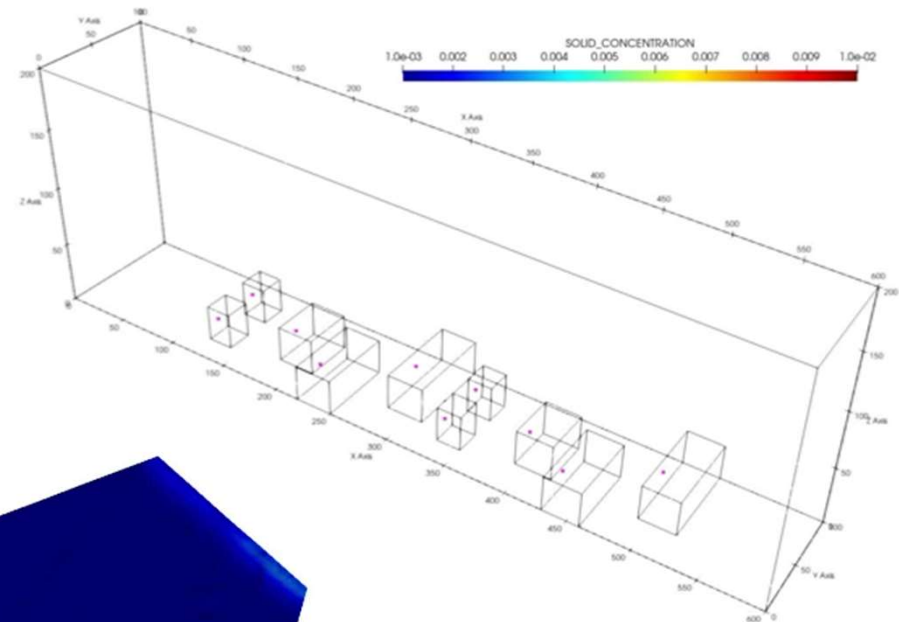


Computation data

- 84 processors
- computation time of 2 hours
- 700 000 elements
- 900 time steps



Simulation of pyroclastic flows

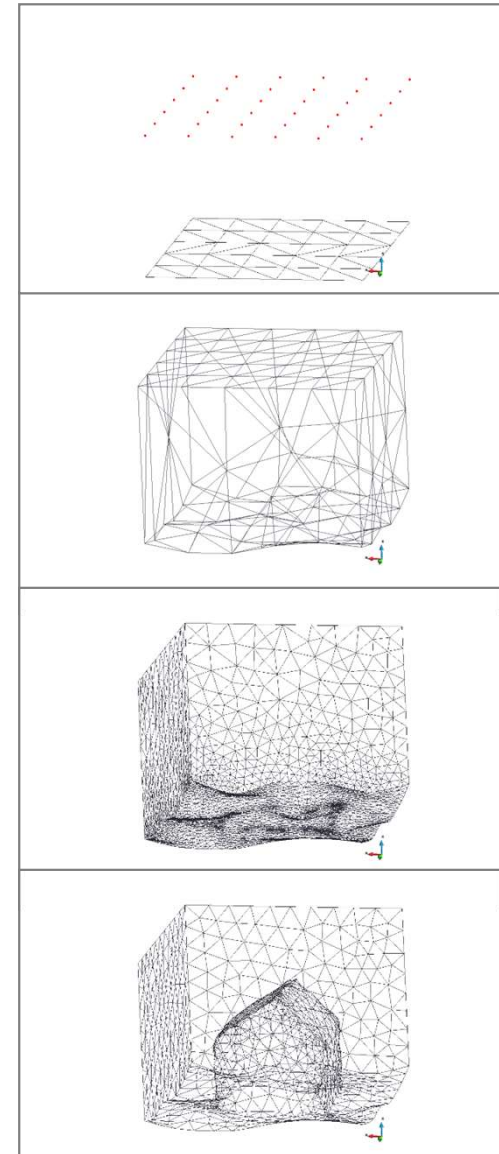
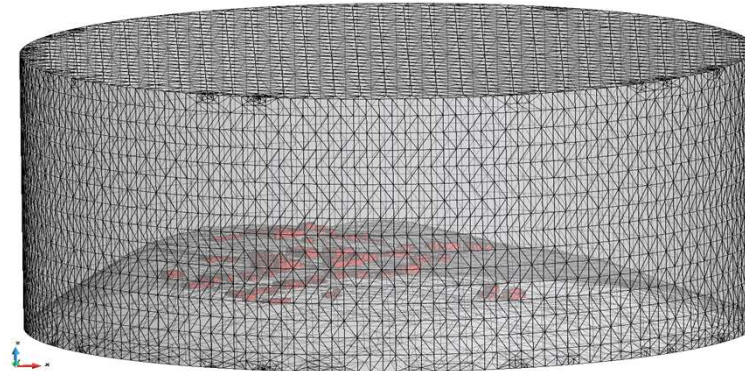
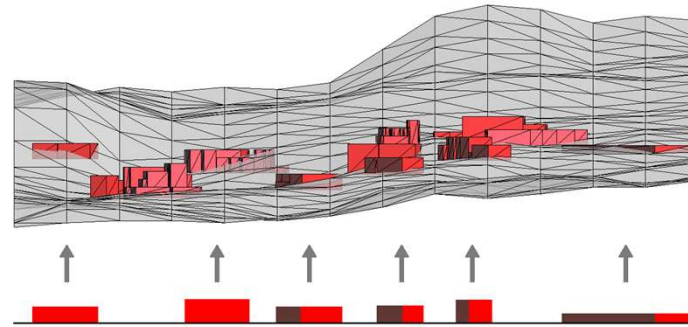
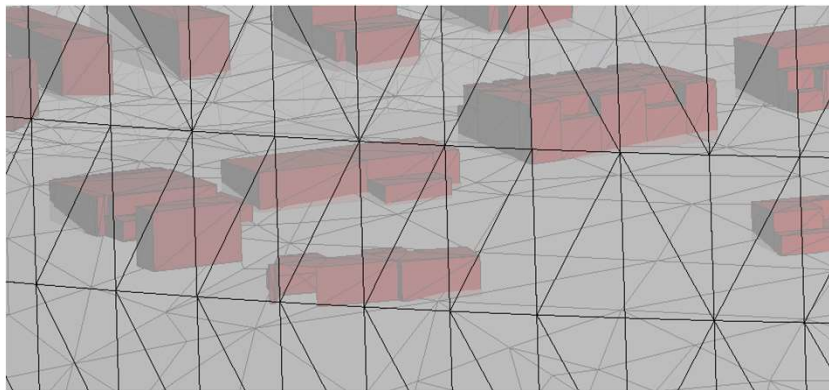


Dr. Andrea Montanino

Simulation of pyroclastic flows

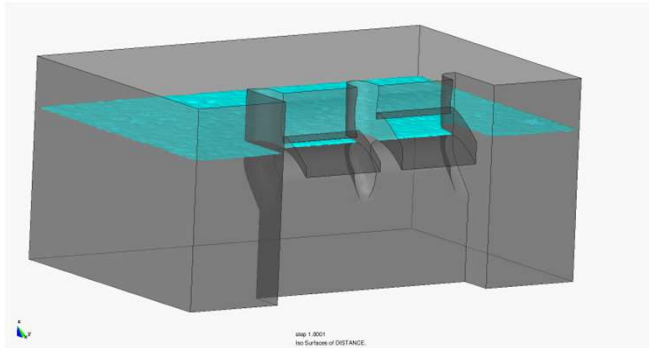
Dr. Andrea Montanino

Geodata Processing for Large-Scale Fluid Simulations

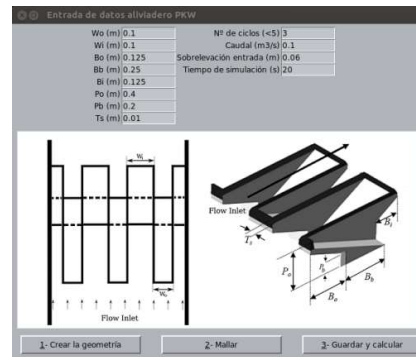


DAM ENGINEERING

GATED SPILLWAYS



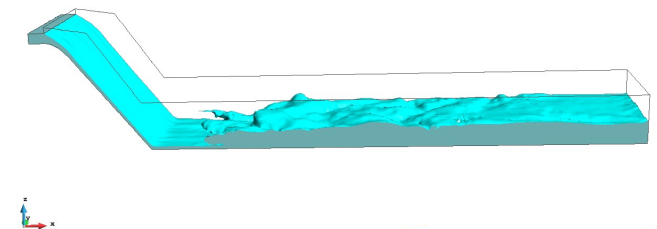
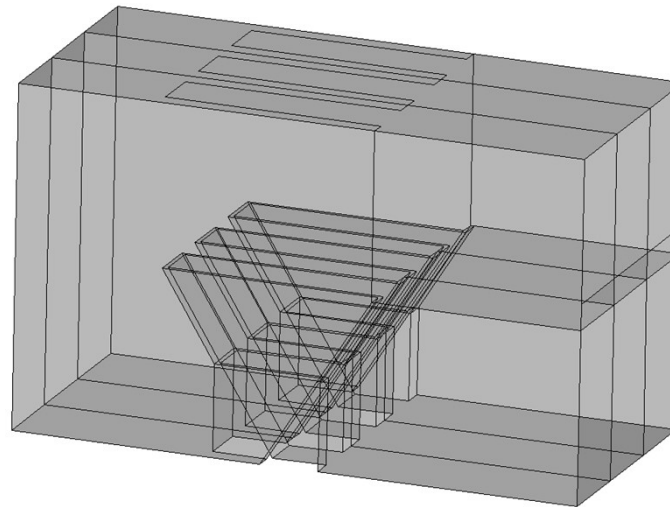
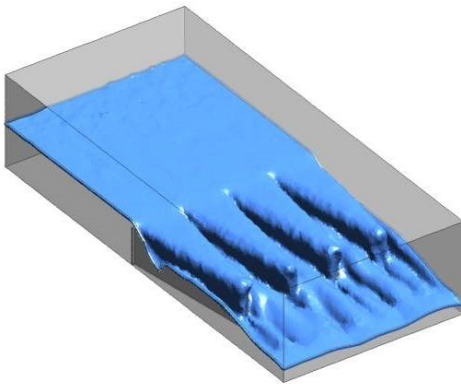
WIZARD



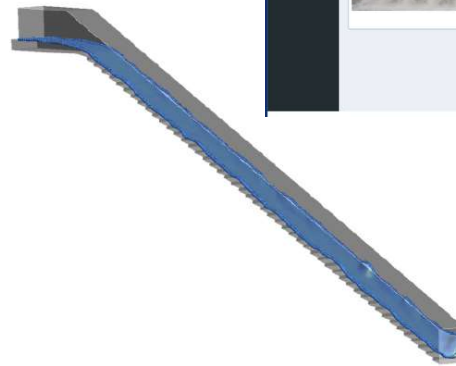
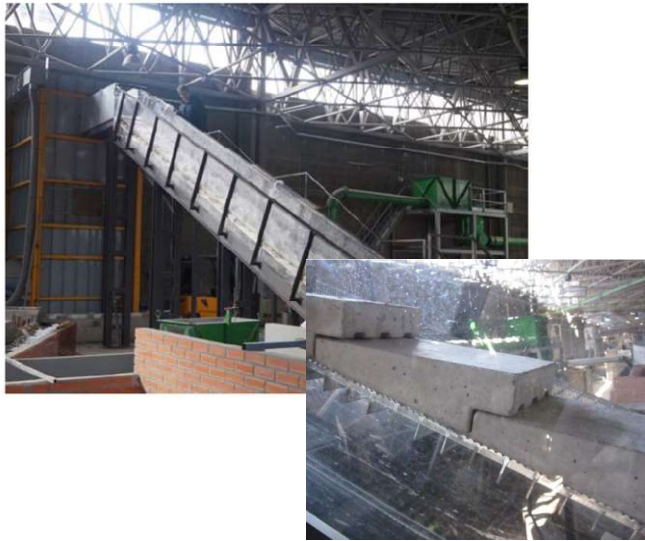
Stilling basin. Hydraulic jump



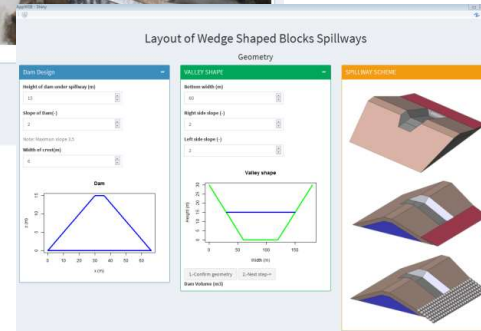
LABYRINTH AND PIANO-KEY SPILLWAYS



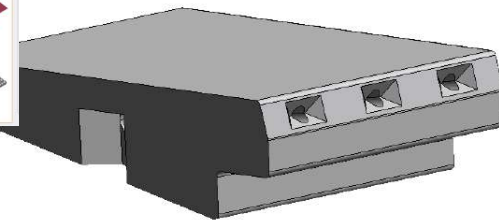
DAM ENGINEERING



1. Hydraulic performance and design

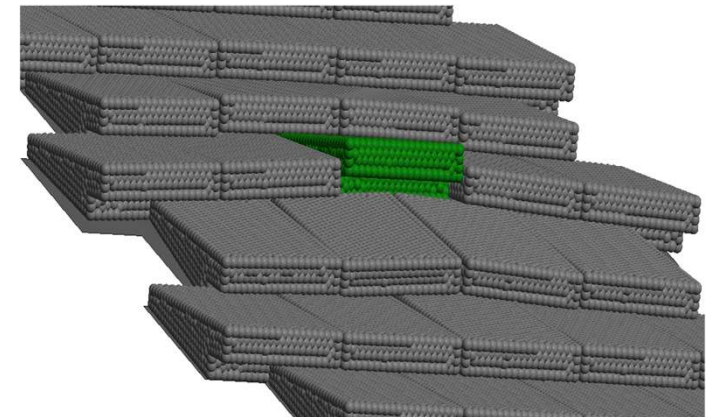
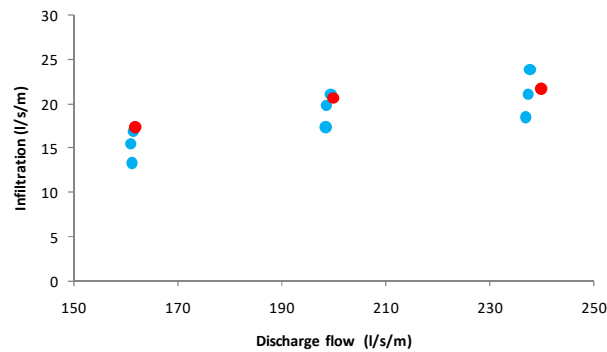


2. WIZARD FOR WSBs DESIGN



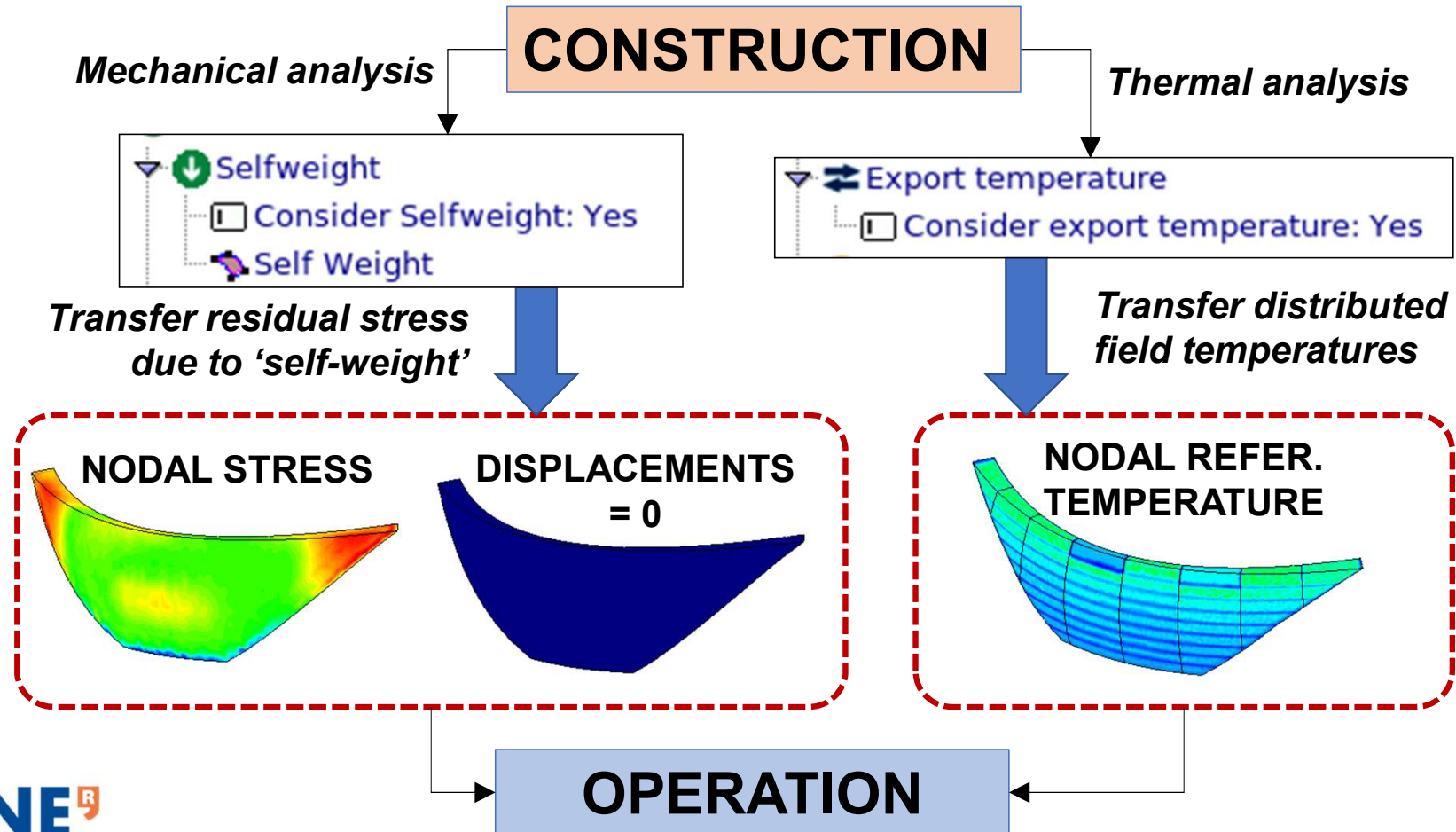
3. Security analysis to anthropic risk

Experimental-numerical tests



DAM ENGINEERING

Thermo-mechanical analysis of concrete dam



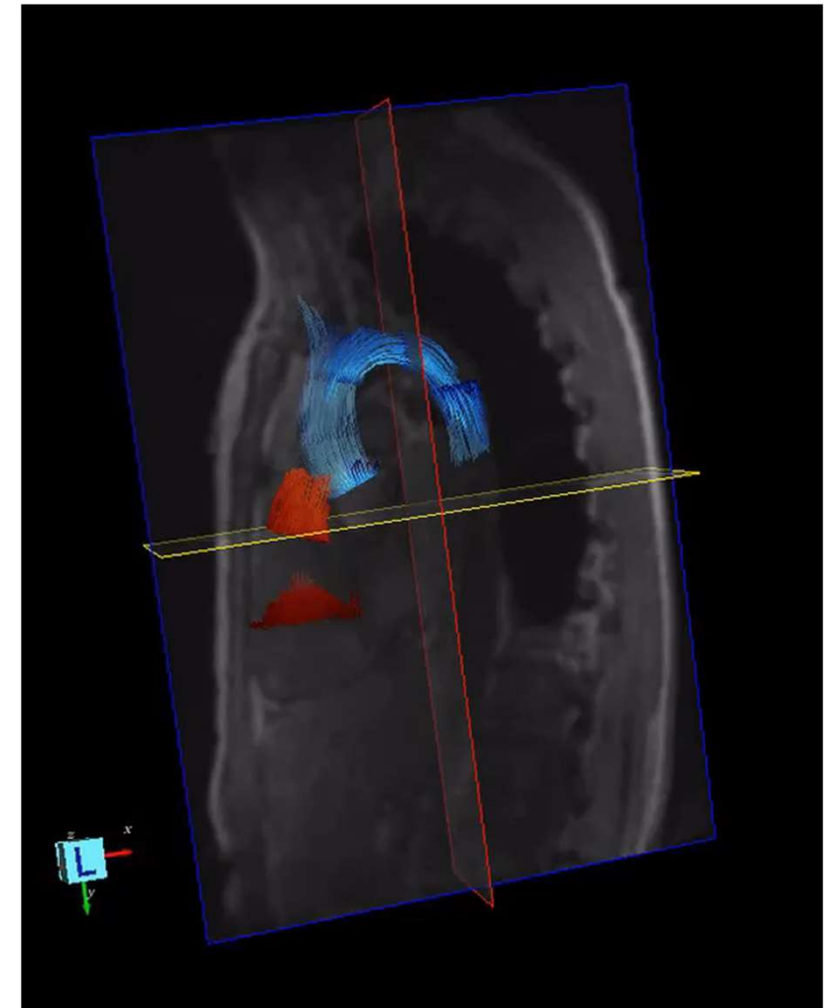
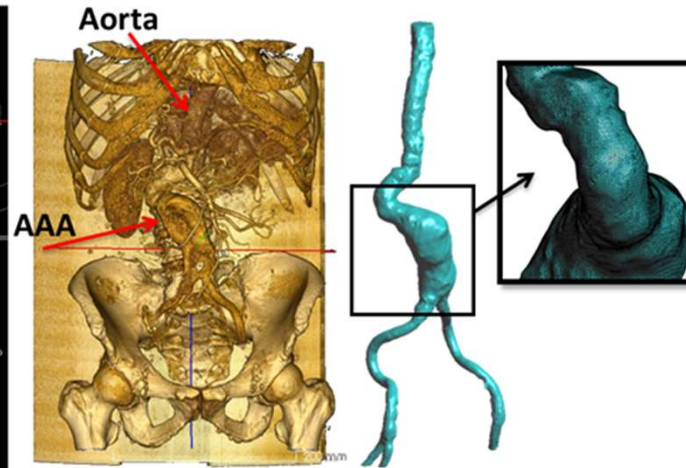
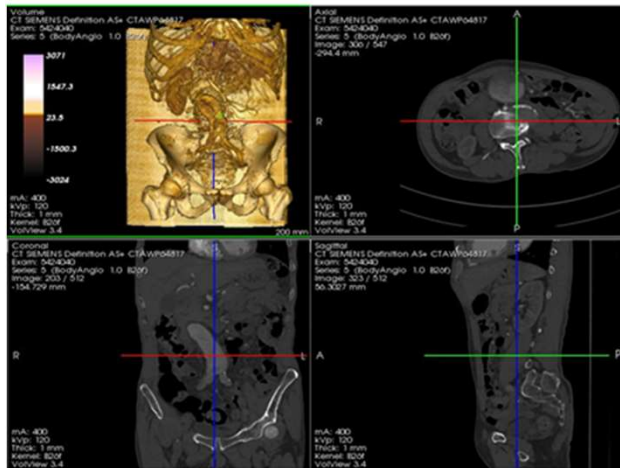
Bio Medical

COMPUTATIONA TOOL of the cardiovascular system

Image
Processing

Numerical
Model

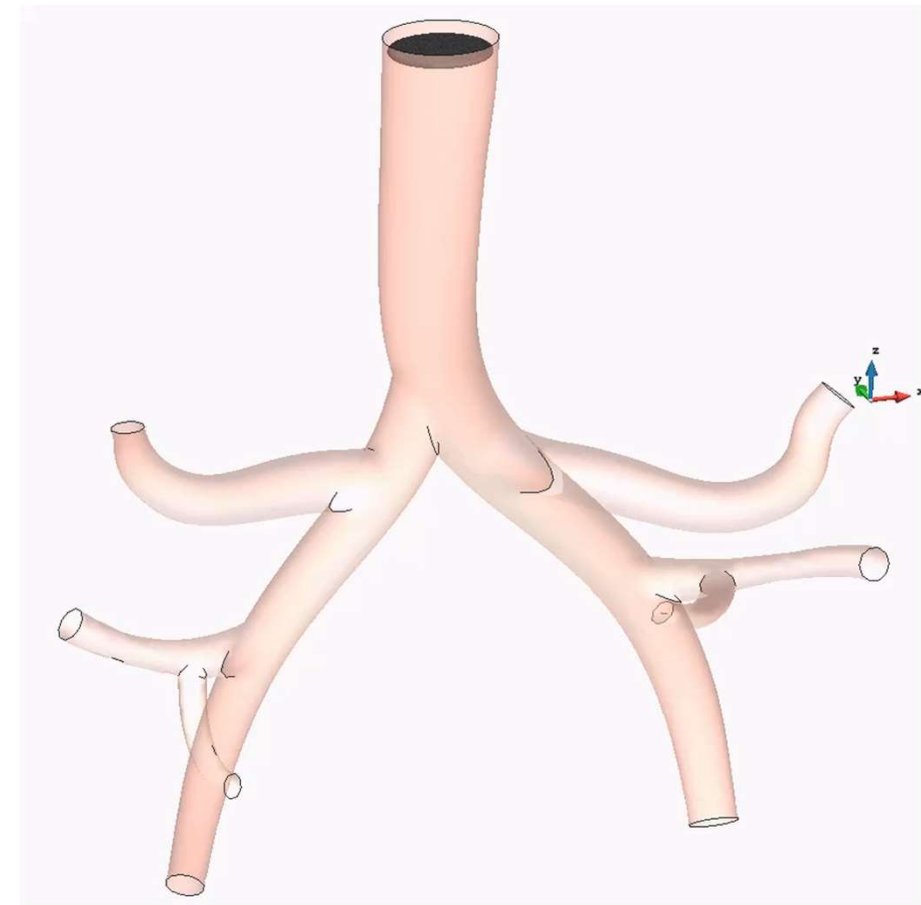
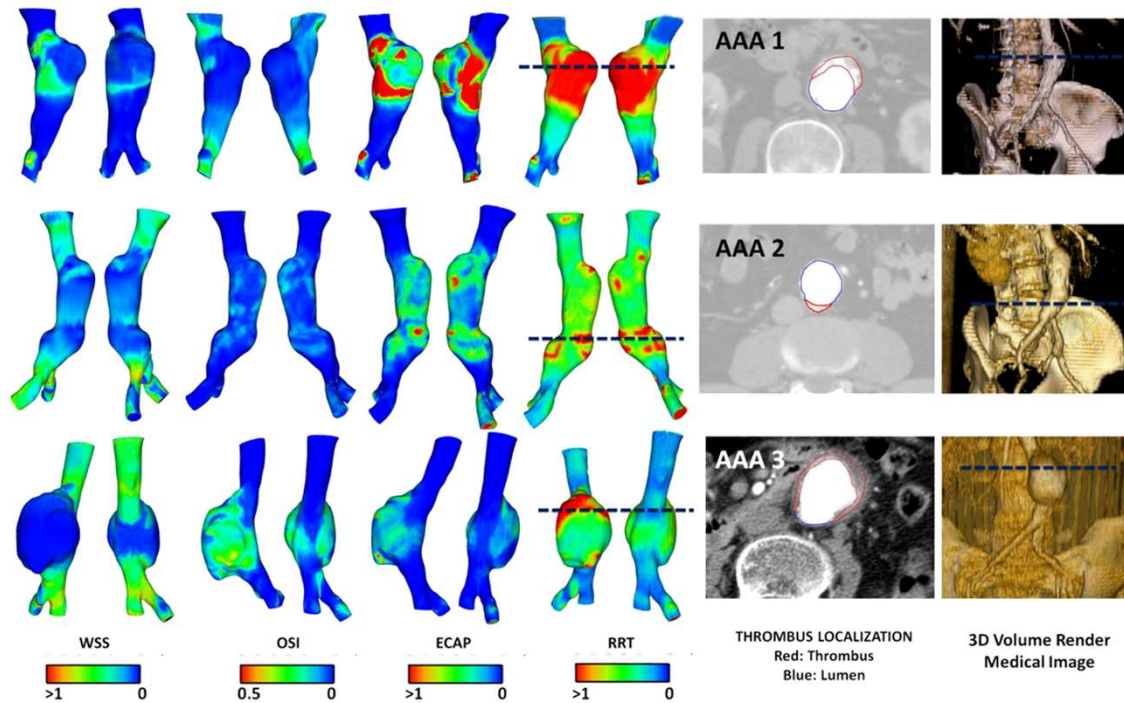
Numerical
Validation



By: Eduardo Soudah

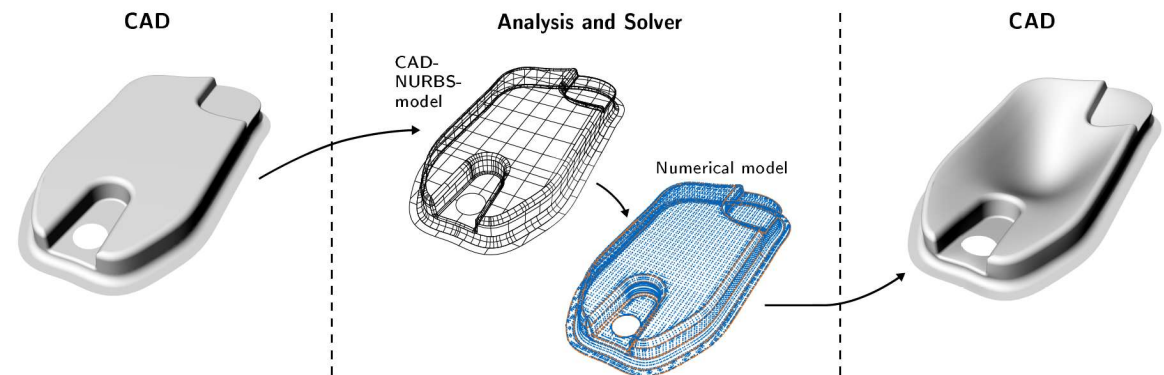
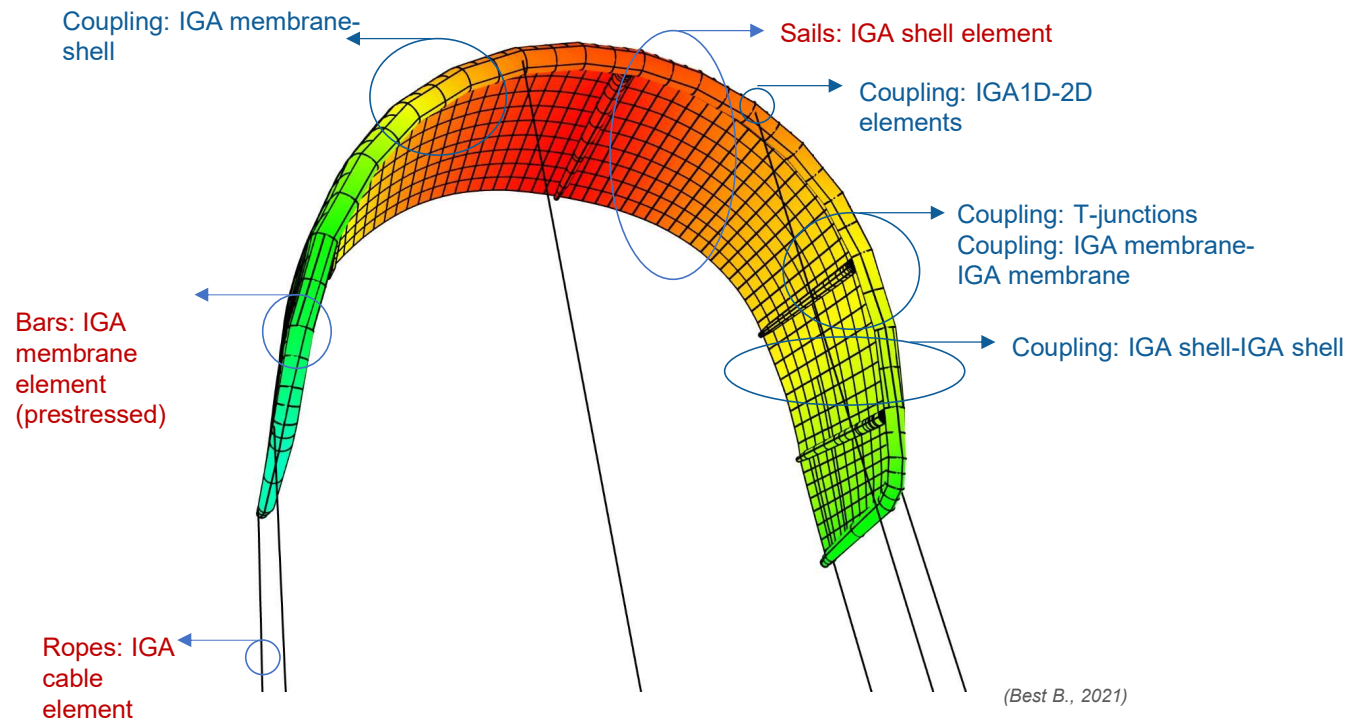
Bio Medical

Abdominal Aneurysm

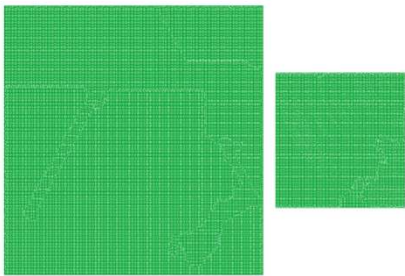


IGA-IBRA

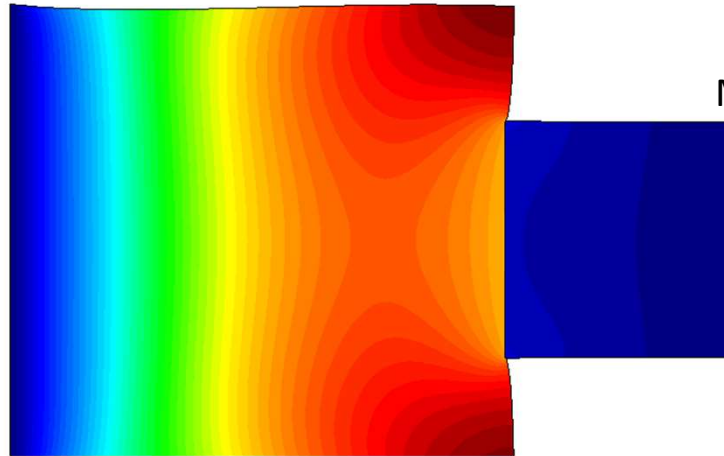
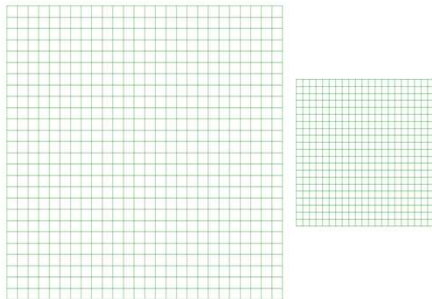
- Trimmed IGA capabilities
- Shell and Beams
- Weak constraints



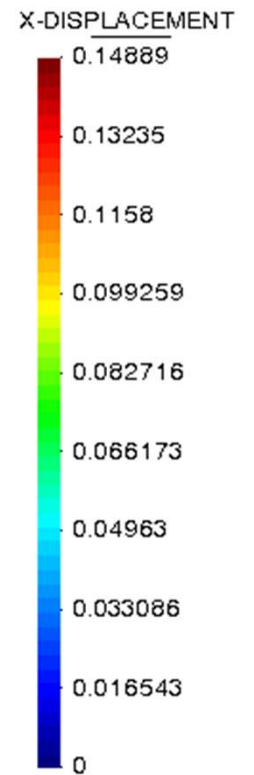
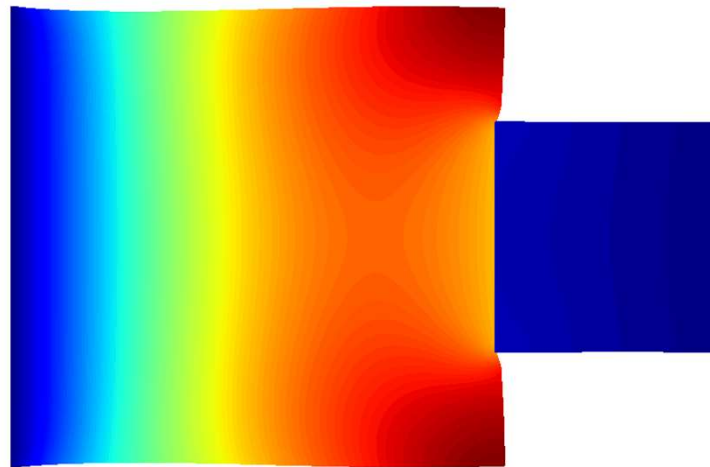
Contact in FEM



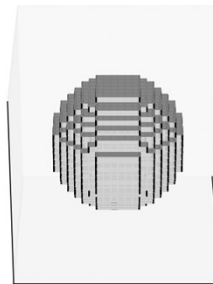
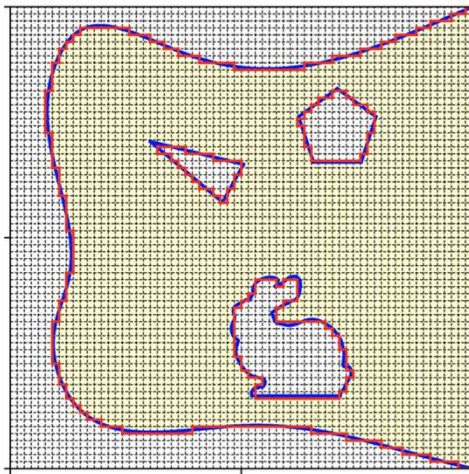
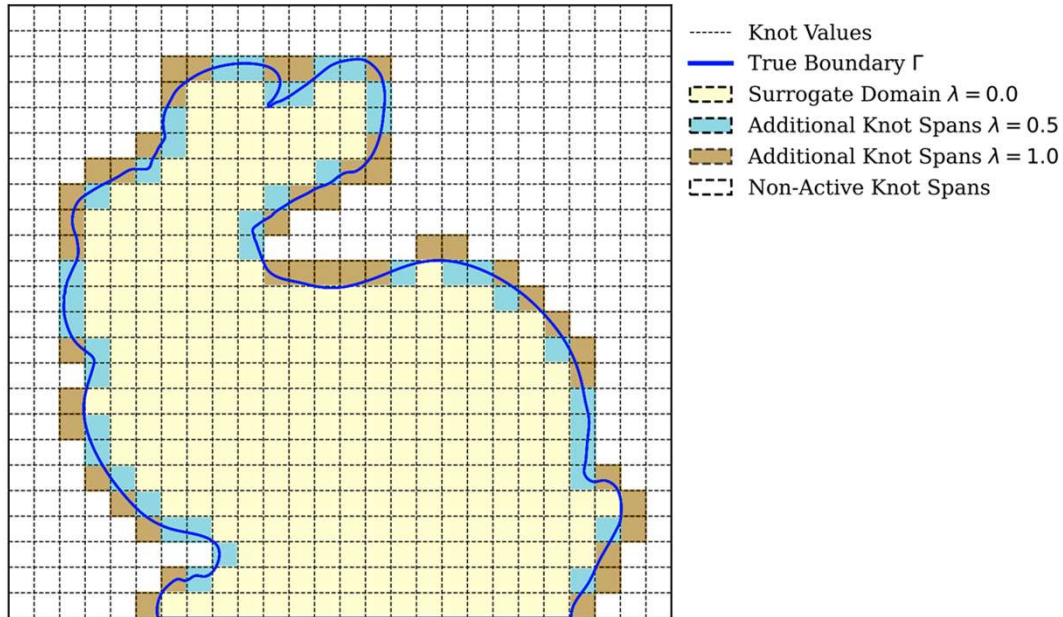
Contact in IGA



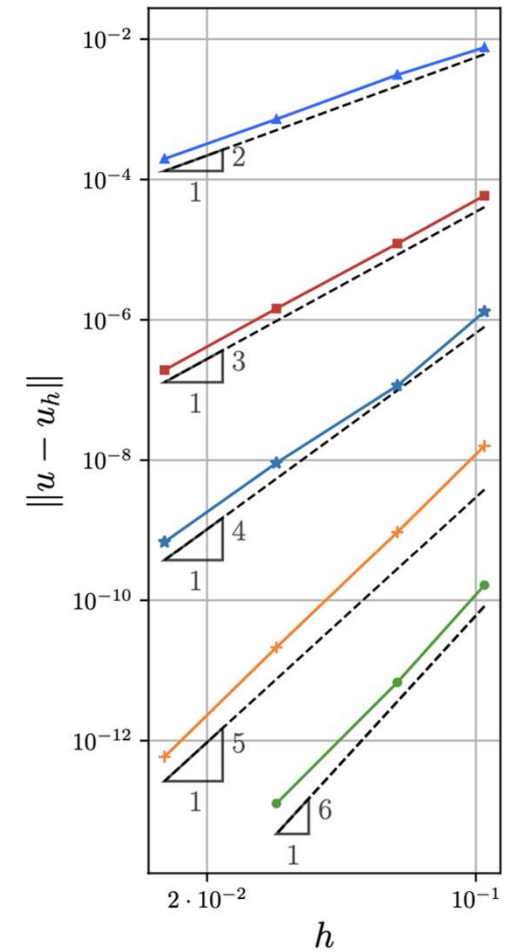
Mr. Andrea Gorgi



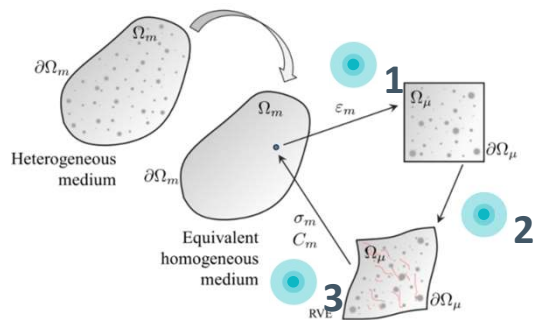
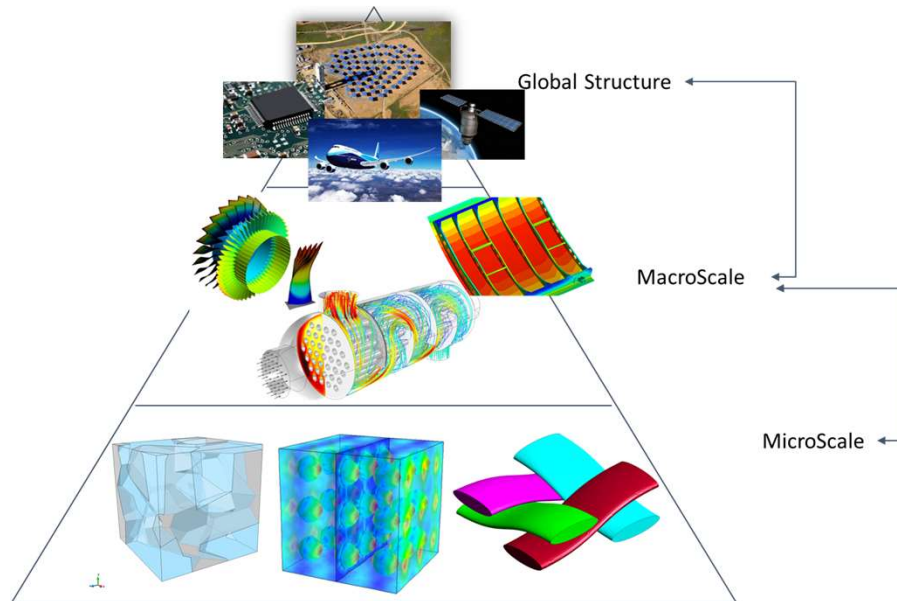
IGA+SBM in CFD



Mr. Nicolás Antonelli



MULTI SCALE APPROACHES

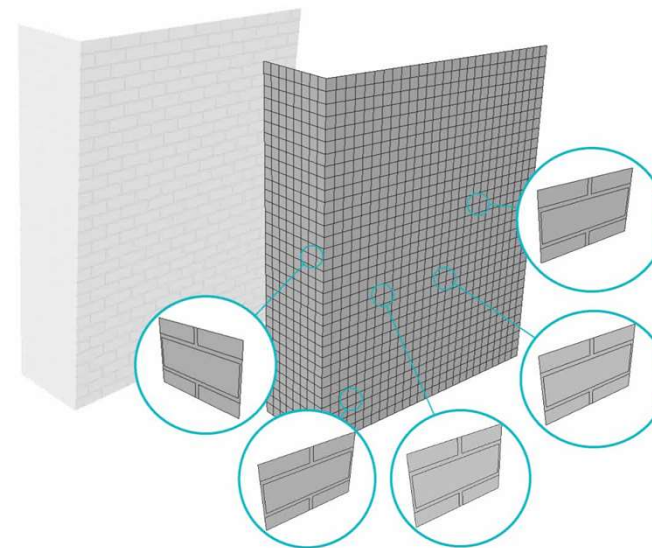


Based on computational homogenization methods for:

- 2D/3D 1° order continua
- Shell-like continua

Homogenization of:

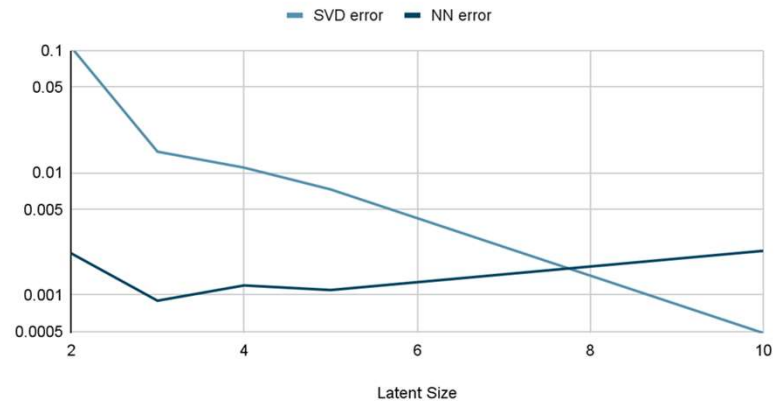
- Mechanical and ...
- Thermo-Mechanical problems



Out-of-plane failure of a brick masonry wall

Machine Learning

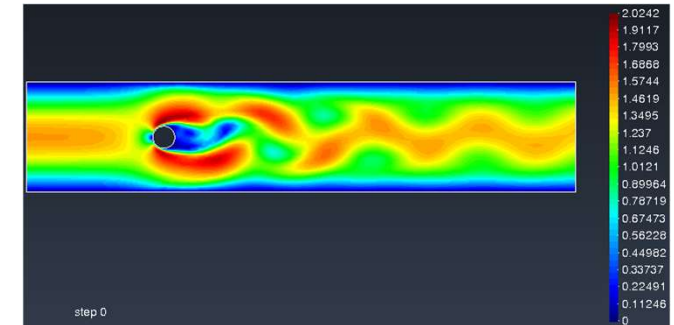
SVD error and NN error



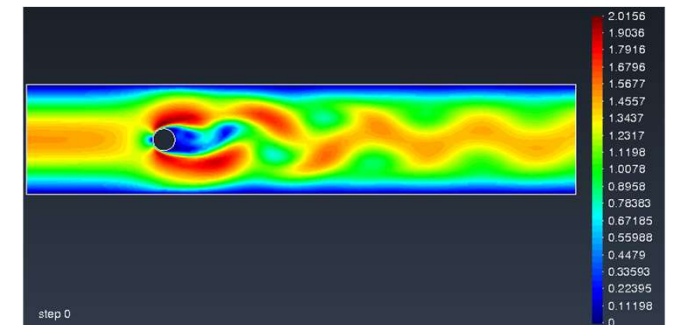
Latent Size	Training Time (s)	SVD error	NN error
2*	481	0.10672	0.0022
3*	420	0.01486	0.0009
4	475	0.01101	0.0012
5	893	0.00731	0.0011
10	878	0.00049	0.0023

* Reduced number of ReLU layers in the encoder

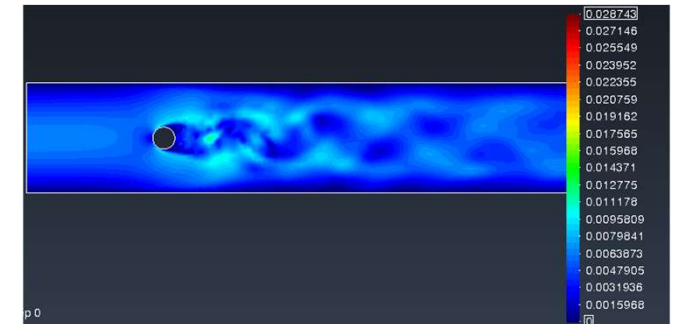
Original



Projected



Error

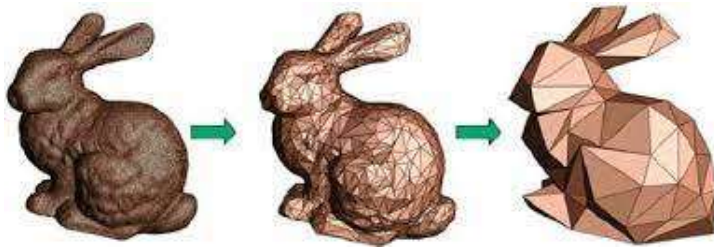


15:40 - 16:00

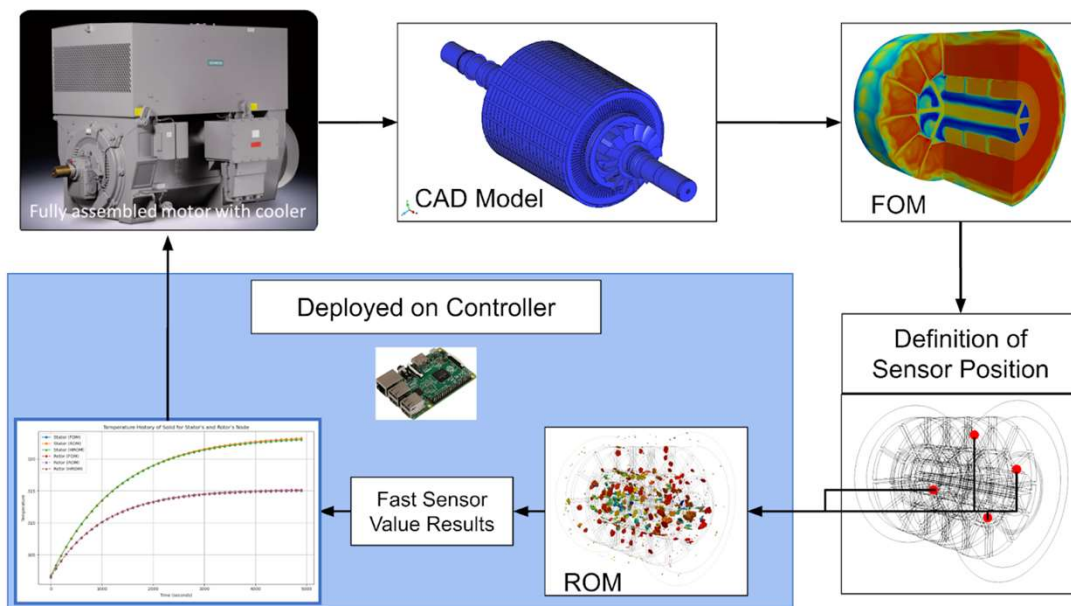
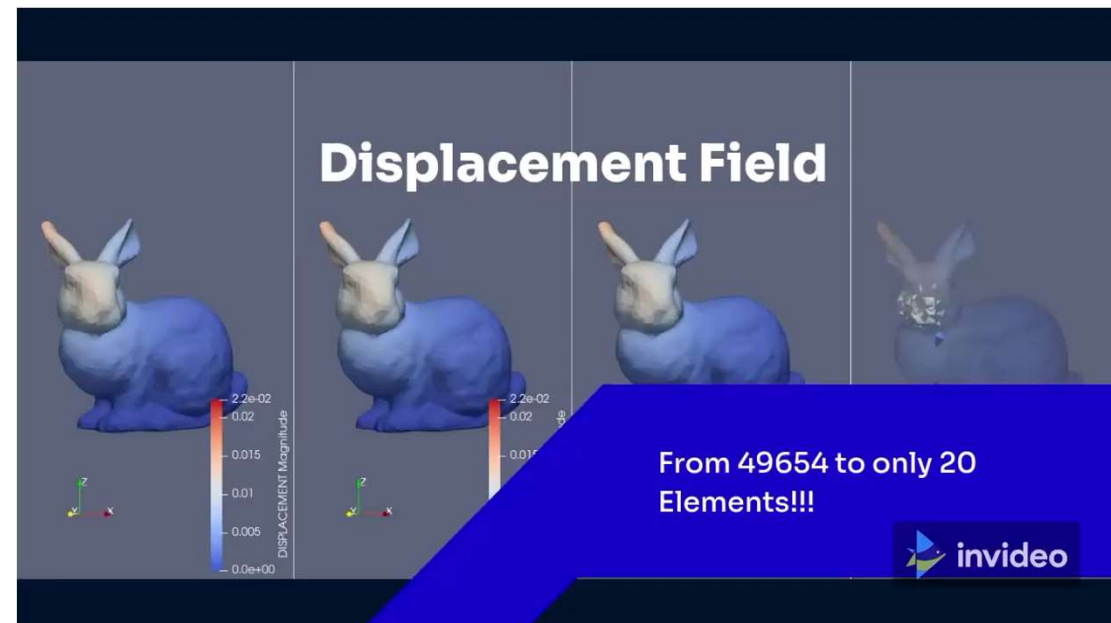
ROM Capabilities and Neural Networks in Kratos

Carlos Roig (CIMNE)

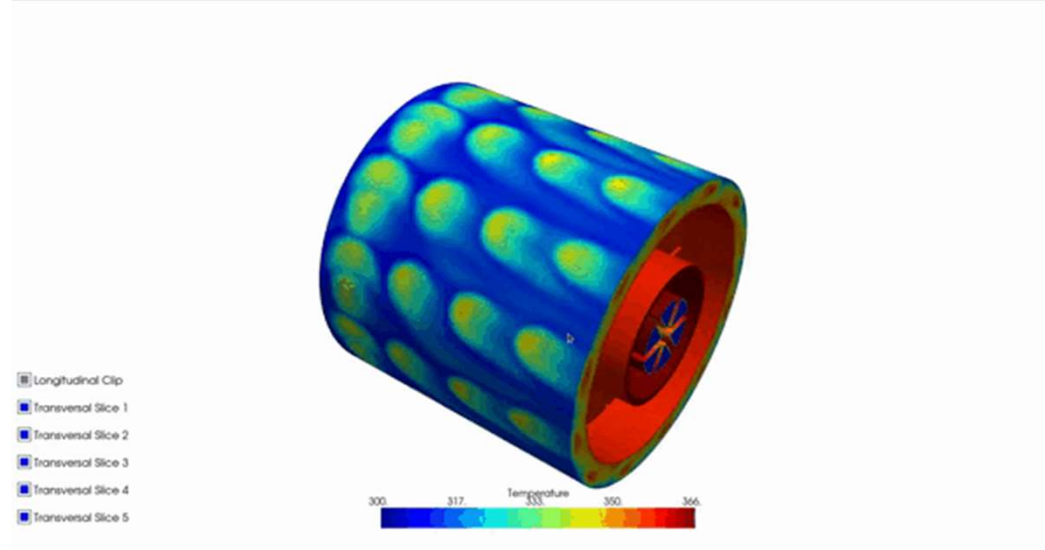
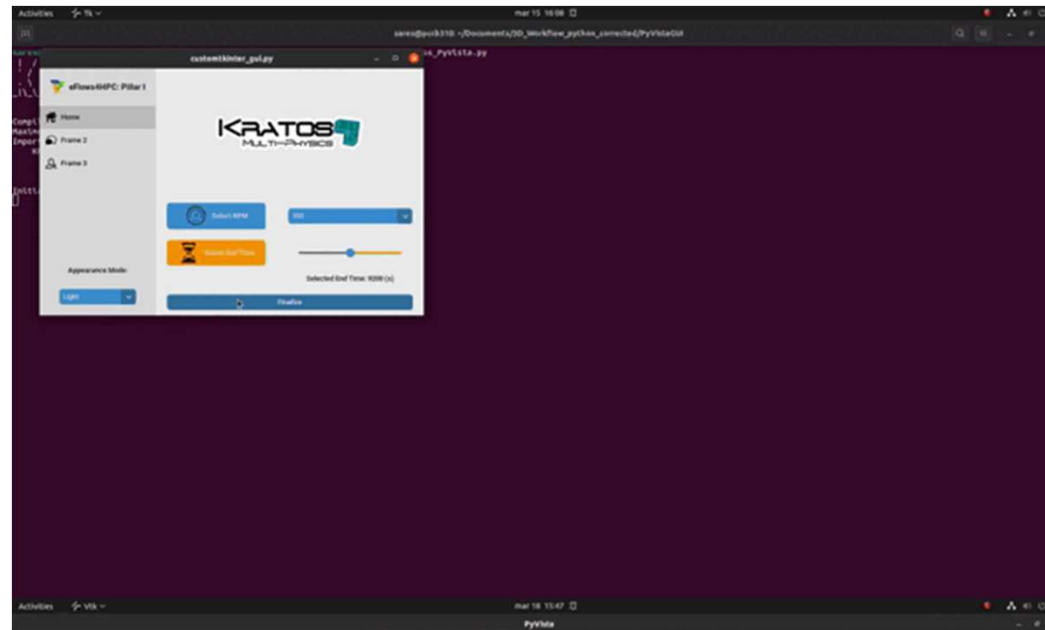
ROM MODELS



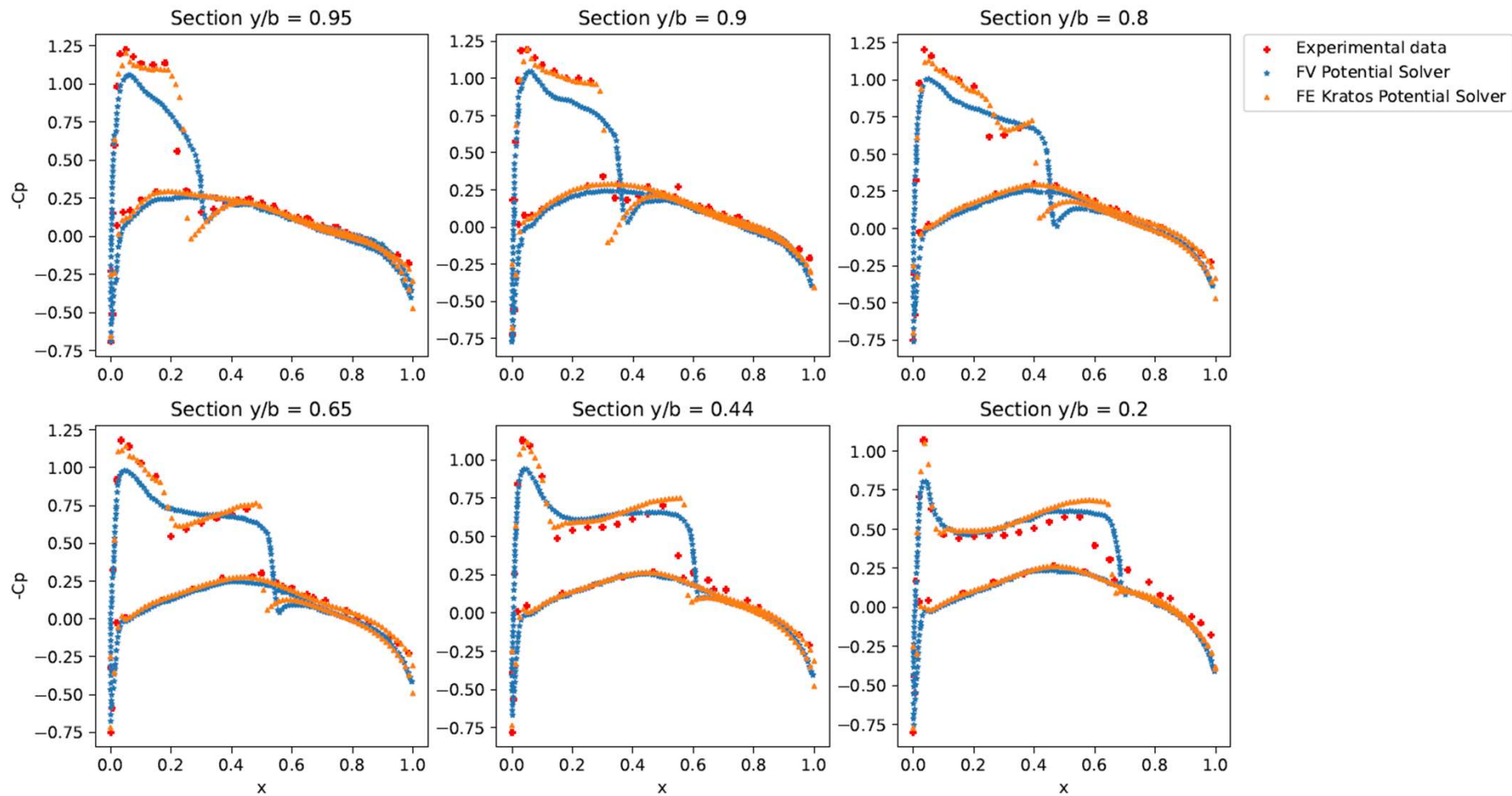
Raul Bravo
Sebastian Ares
Nicolas Sibuet
Marco Zuñiga
Carlos Roig



ROM

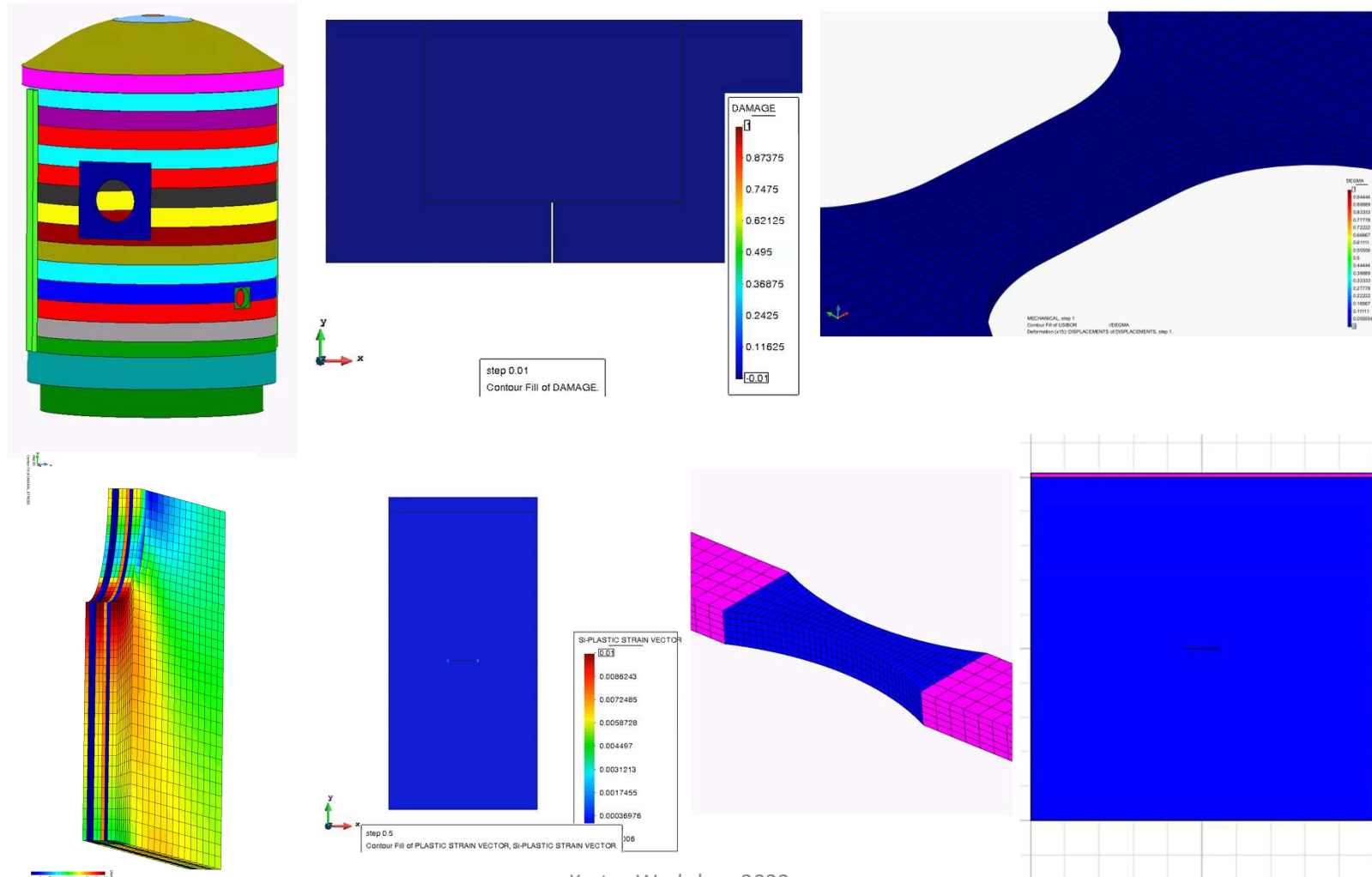


Compressible CFD



Applications in nonlinear solid mechanics

Prof. Alessandro Cornejo
& Dr. Vicente Mataix



9/11/2022

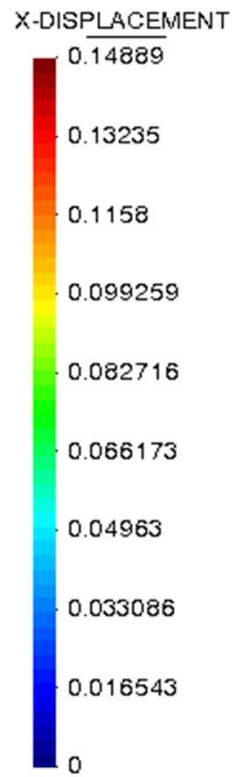
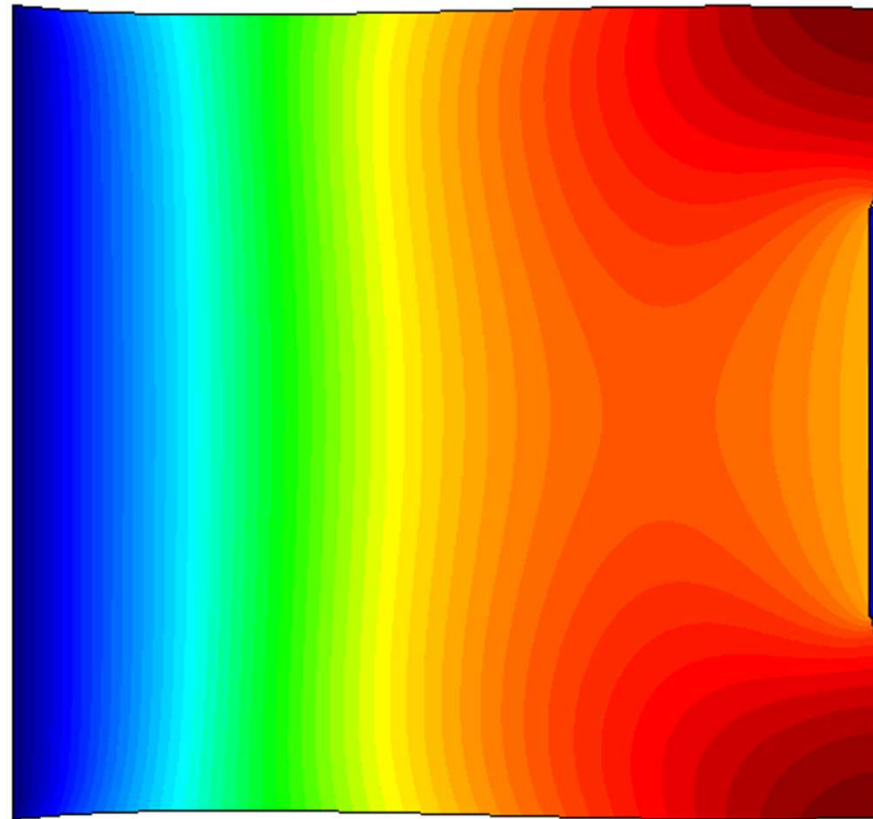
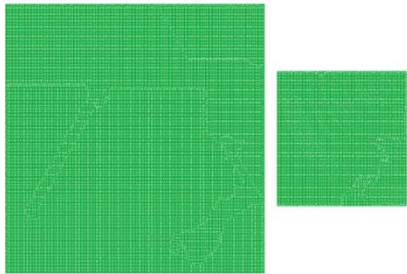
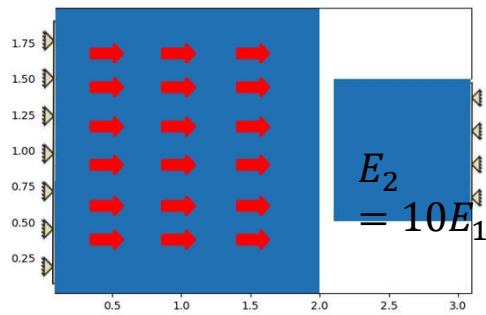
Kratos Workshop 2022

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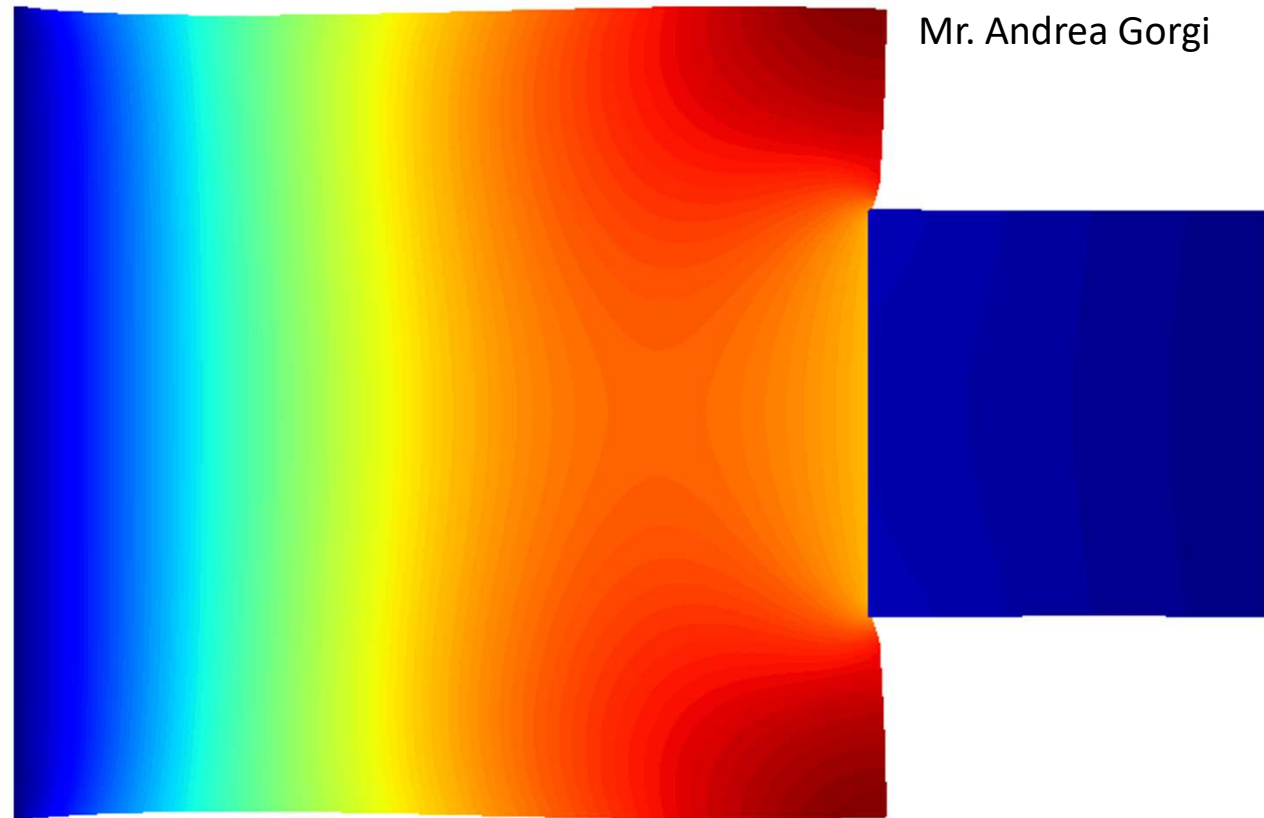
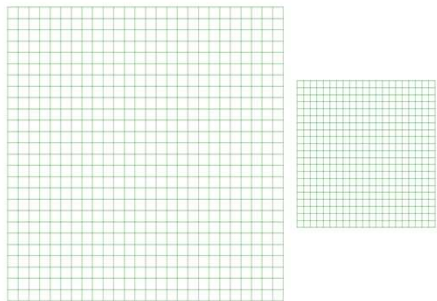
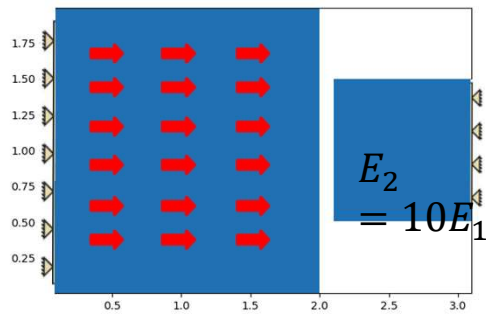


Contact in FEM

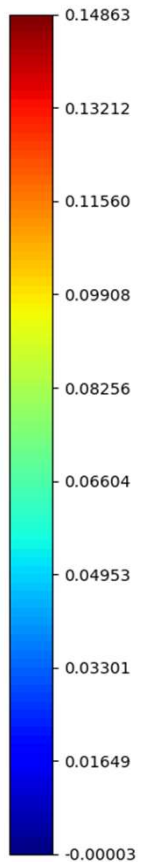
Mr. Andrea Gorgi



Contact in IGA



Mr. Andrea Gorgi



Roadmap of future changes

We should be:

- Faster
- Use less memory
- Easier to use
- Safer (make more difficult to make errors)
- Less “Bureocratic” in the revision process
- Have a good documentation
- ...

While

- Preserving (as much as possible) backward compatibility
- Allow using “old” systems (still in production in companies)
- ...



A tradoff is needed

Large changes in the pipeline

- Strategies V2 (linked to Core.Legacy and Core.Future)
- “PointerVectorSet” cleanup
- pointer constness
- dense linear algebra performance issues (getting rid of ublas). Should we wait for C++’s P1385 or shall we go for "A-matrix" kind of approach?
- python packages structure. Reformulating how wheels are packaged to prevent conflicts on installation/deinstallation
- default to vtu instead of vtk (deprecate vtk). check how gauss point values are written. Check for zip compression
- Kratos::StdVector (no_init by default, allocator ...) + mdspan + other goodies
- Vectorized Database Access
- Flowgraph

Large changes in the pipeline

Strategies V2 (linked to Core.Legacy and Core.Future)

LEAD: Ruben Zorrilla & Riccardo Rossi

The current version of “**SolvingStrategies**” shows its age:

- The SparseSpace layer does not really allow to abstract from the implementation
- Impossible to do FEM assembly without BuilderAndSolve
 - In particular: does not allow functional-type programming
- Thread Local Storage problematic in scheme
- MPCs ***not correctly implemented***
- ***VERY RIGID structure (essentially you need to construct elements/conditions to assemble something).***
- Depending on Trilinos for MPI

Large changes in the pipeline

Strategies V2 → HOW?

Introduce two namespaces (tentative names):

- **Legacy** → to contain stuff still needed but which we want to phase out
- **Experimental** → experimental changes, subjected to API breaking changes

Such namespaces **will map to submodules in Python.**

Large changes in the pipeline

“PointerVectorSet” cleanup

LEAD: Suneth Warnakulasuriya & Mate Kelemen

Currently Pointer Vector set is not (really) guaranteed to be Sorted.

- GOAL: guarantee the property without incurring in large performance penalties
- May introduce changes in modelpart & the use of concept of “authority” as well as the introduction of manipulation helpers

Large changes in the pipeline

“const Pointer” ...not really const

LEAD: ?

The interface of “Pointers” and the “const iterators” of Kratos do not really ensure the constantness of the content.

Large changes in the pipeline

python packages structure.

LEAD: Carlos Roig

Reformulating how wheels are packaged to prevent conflicts on installation/deinstallation

Large changes in the pipeline

default to vtu instead of vtk (deprecate vtk). check how gauss point values are written. Check for zip compression

LEAD: Suneth? Mate?

Large changes in the pipeline

dense linear algebra (big) performance issues (aka ... getting rid of ublas).

LEAD: ?

- Should we wait for C++'s P1385 (possible candidate for c++26)
- shall we go for "in-house" approach?
- Shall we make use of eigen or similar? (getting old, possible problems with license)

Large changes in the pipeline

More careful use of **“std” data structures**

- Kratos::StdVector (no_init by default, optimized allocators)
- Start using span & mdspan (backport)
- Use mdarray (backports from c++26) once they become available
- Parallel unordered maps?

Large changes in the pipeline

More careful use of “std” data structures

- Kratos::StdVector (no_init by default, optimized allocators)
- Start using span & mdspan (backport)
- Use mdarray (backports from c++26) once they become available
- Parallel unordered maps?

Large changes in the pipeline

Vectorized database Access.

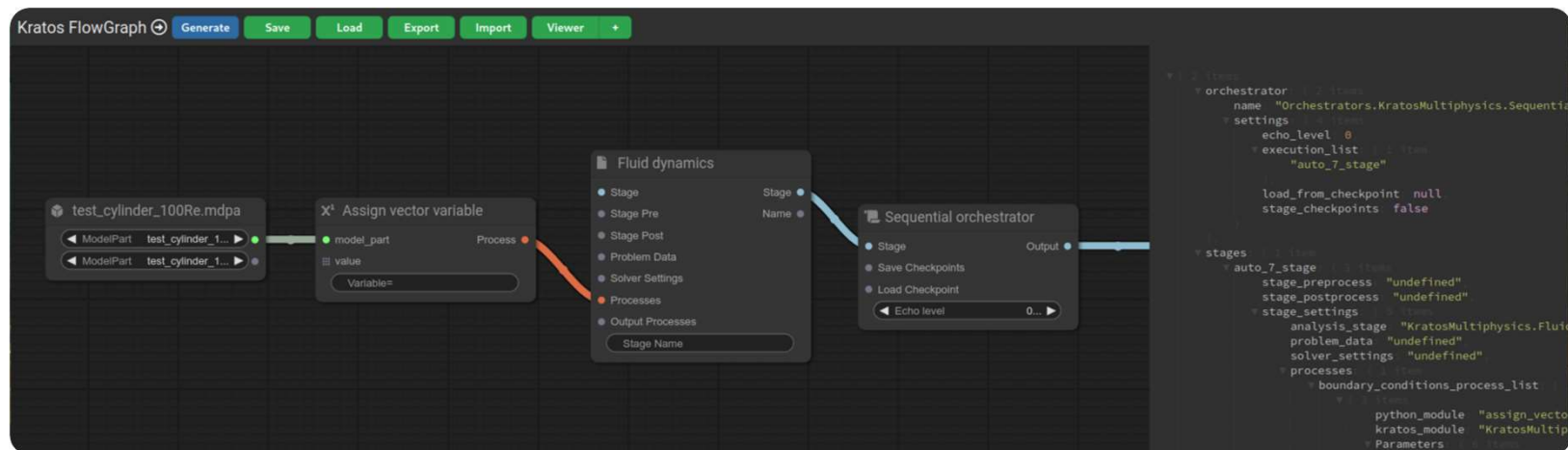
Essentially the capability of exchanging data from arrays (like numpy to understand) and the Kratos database

- Approach a la “variable_utils” (obtain a copy and then a view of it)
- Approach a la “Expressions” (more sophisticated. Deferred evaluation)

Prototype exist for both. Still some refinement in the works

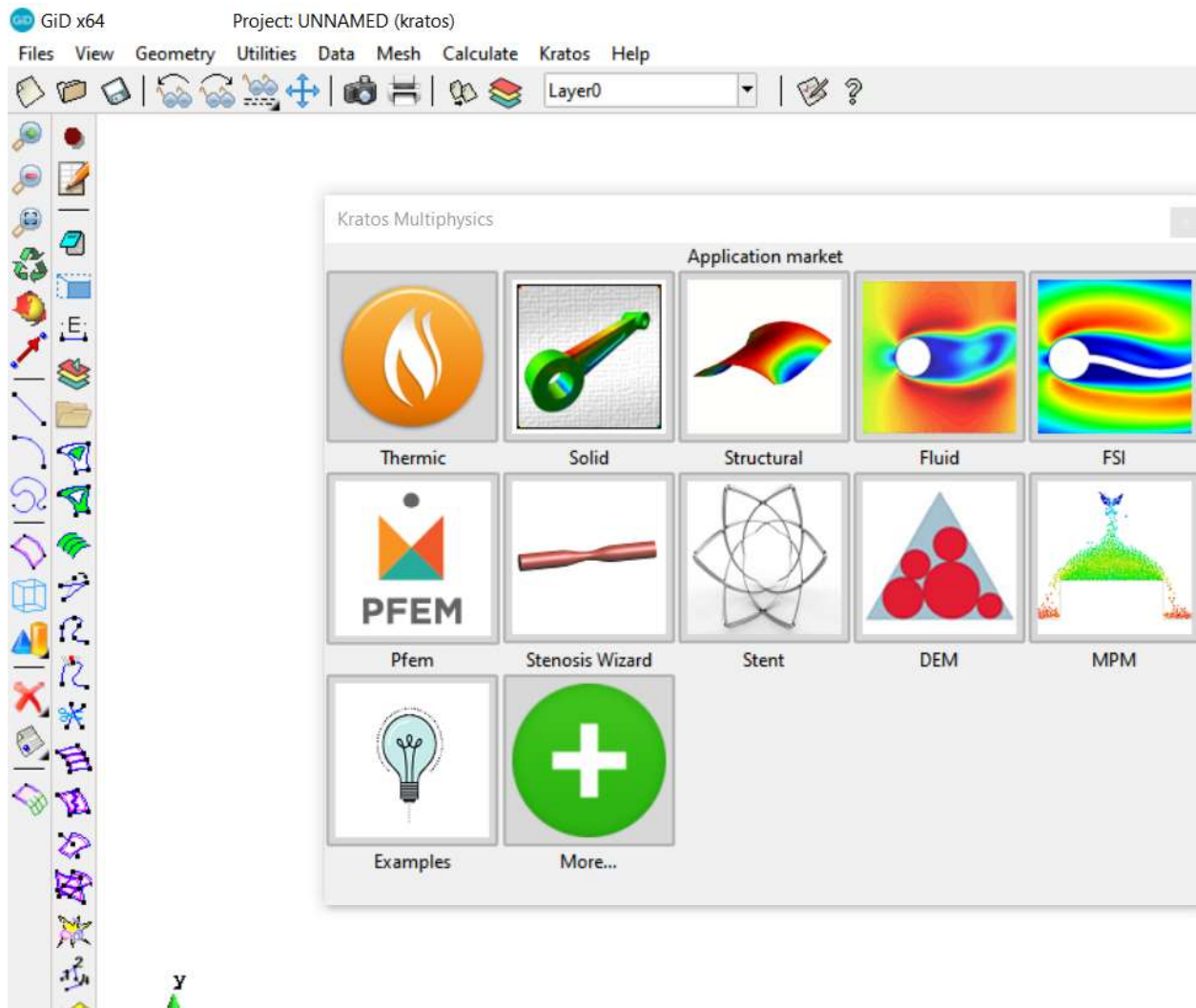
Large changes in the pipeline

“**Flowgraph**” ... a new way of constructing Project Parameters

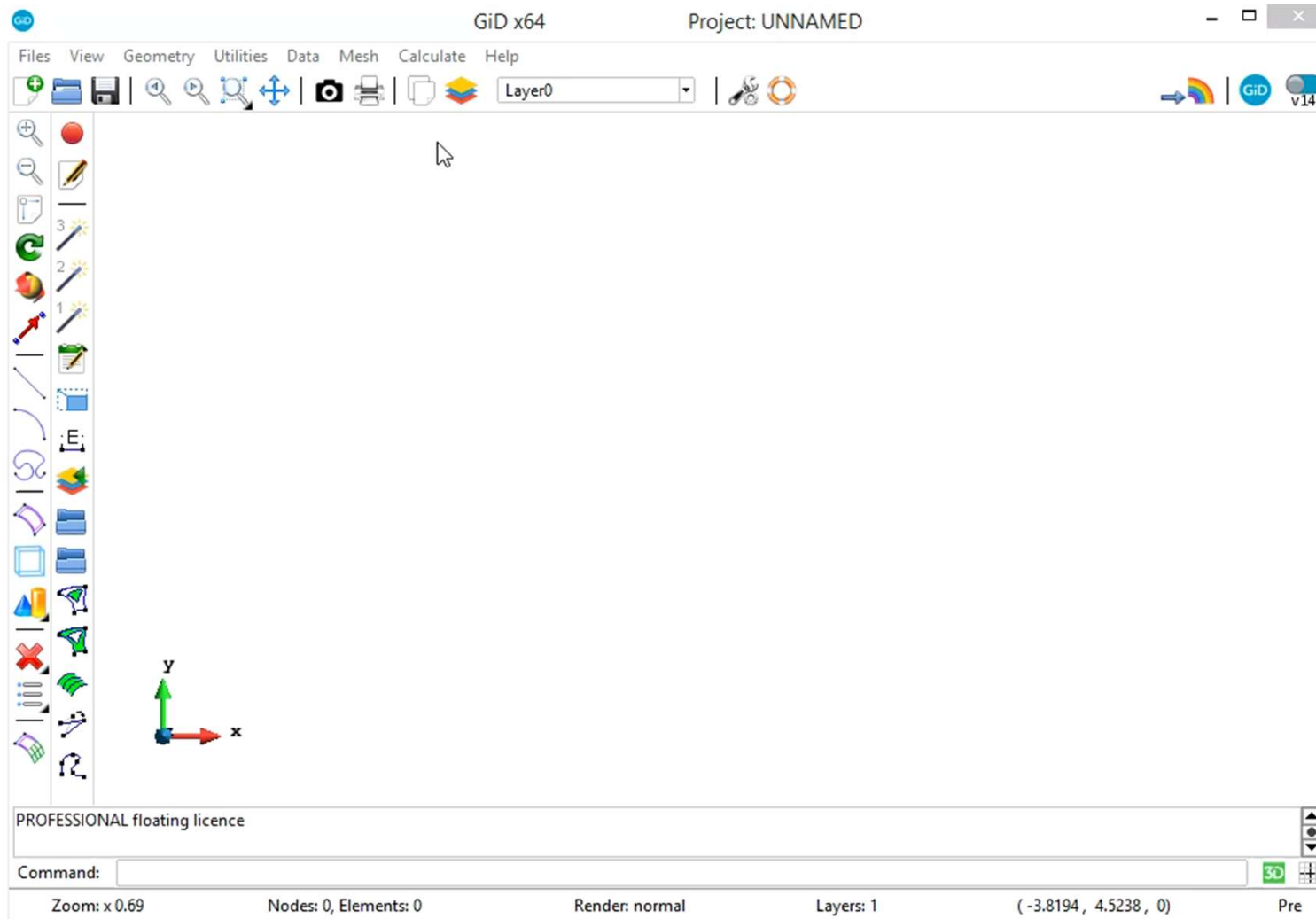


Kratos In Products

Applications: Kratos Multiphysics

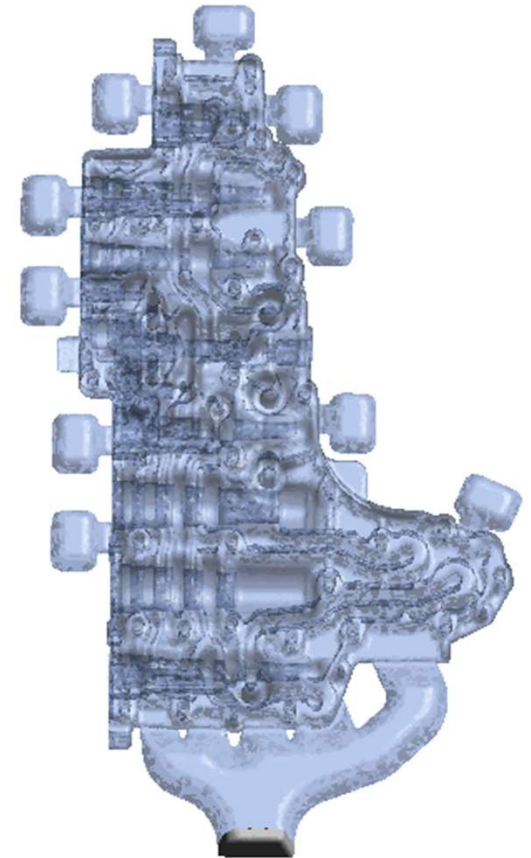
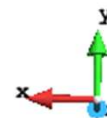
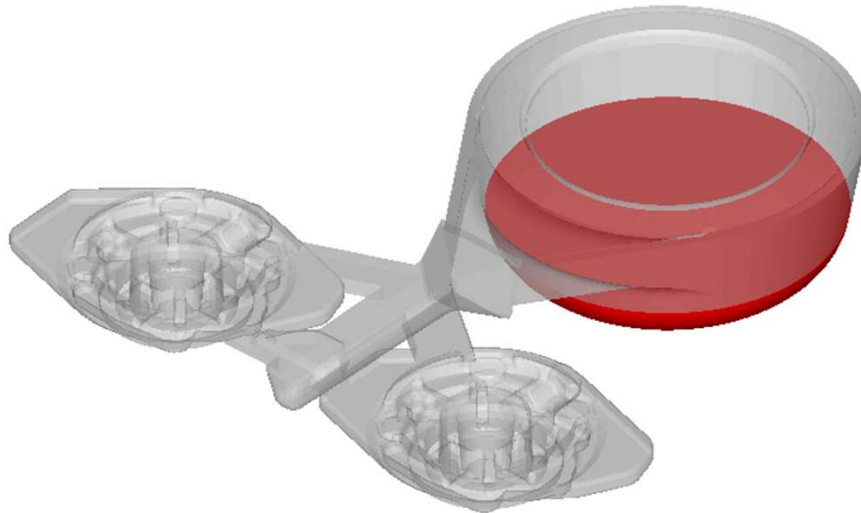


Applications: Kratos Multiphysics



Altair Inspire Cast

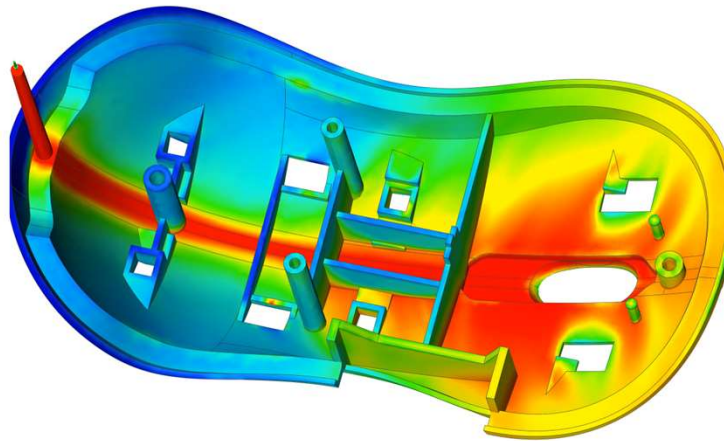
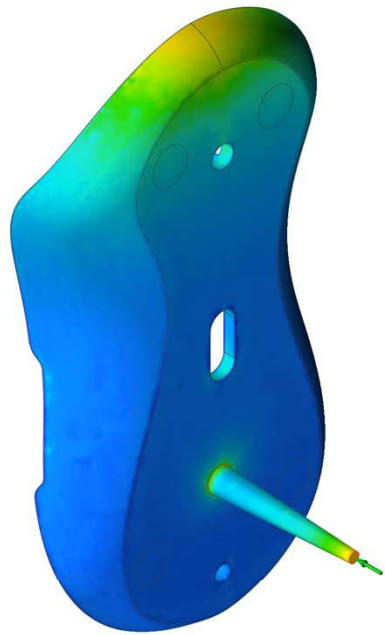
- Casting
- Solidification
- Cooling
- Cycling



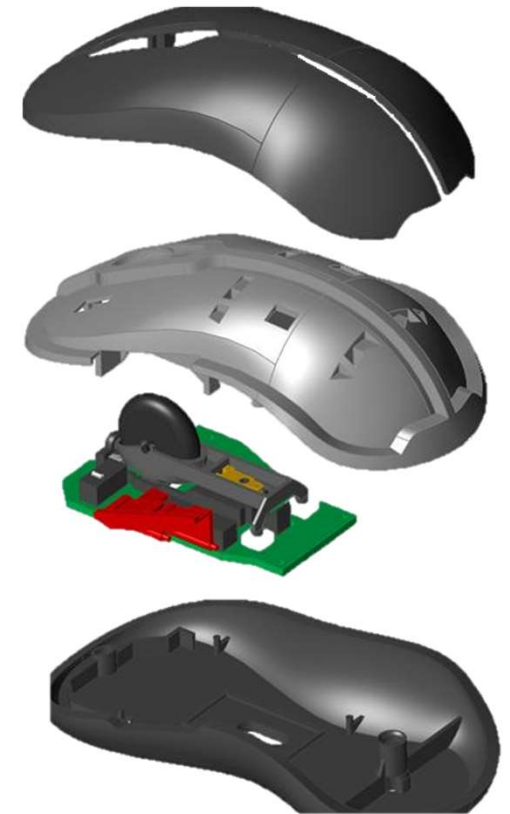
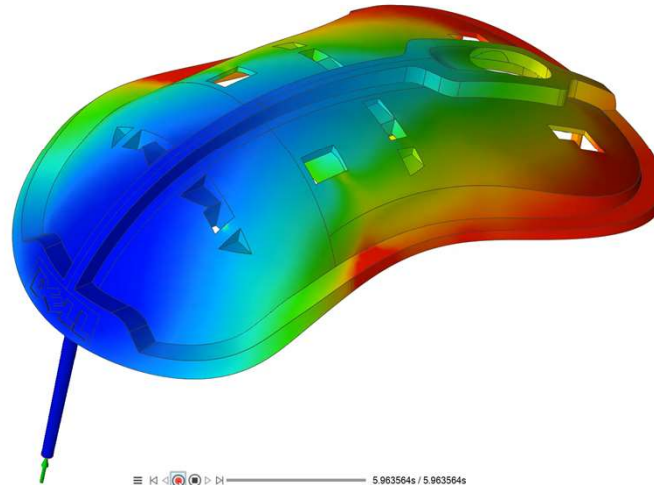
step 0.00060412
Contour Fill of MATERIAL.

Altair Inspire Mold

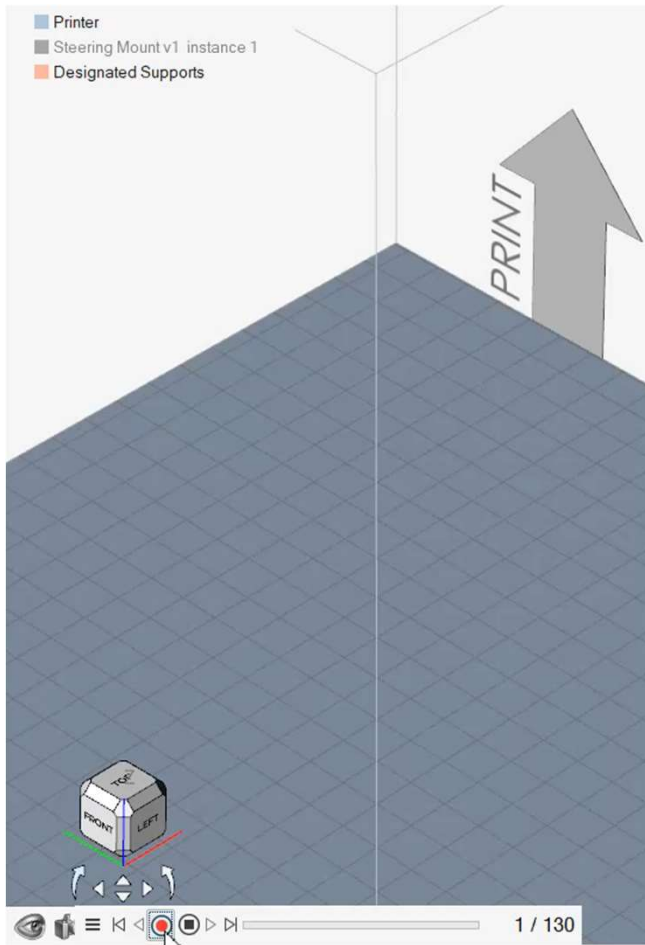
■ Part Cavity
■ Sprue
■ Undefined



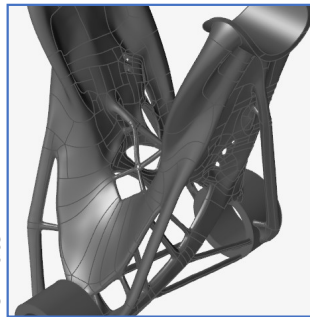
■ Part Cavity
■ Sprue
■ Undefined



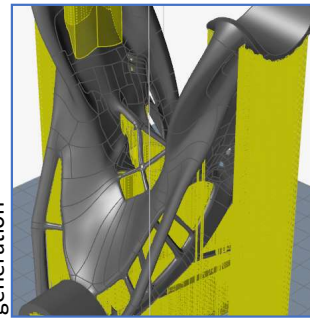
Inspire Print3D



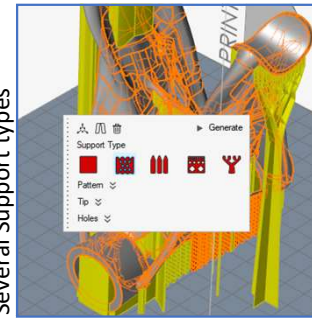
Support for many CAD formats



Automatic Support generation



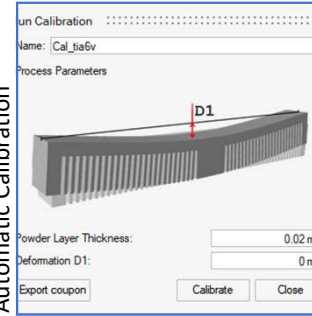
Several Support types



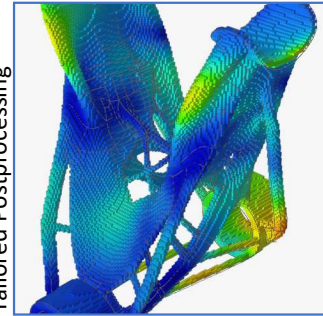
Fast and robust voxel mesh generation



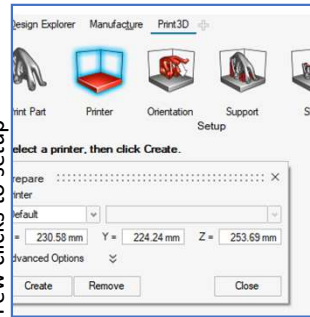
Automatic Calibration



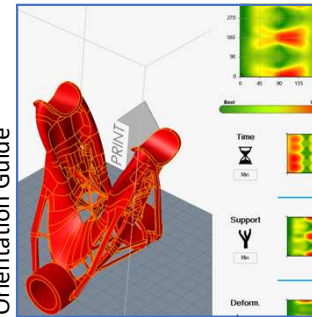
Tailored Postprocessing



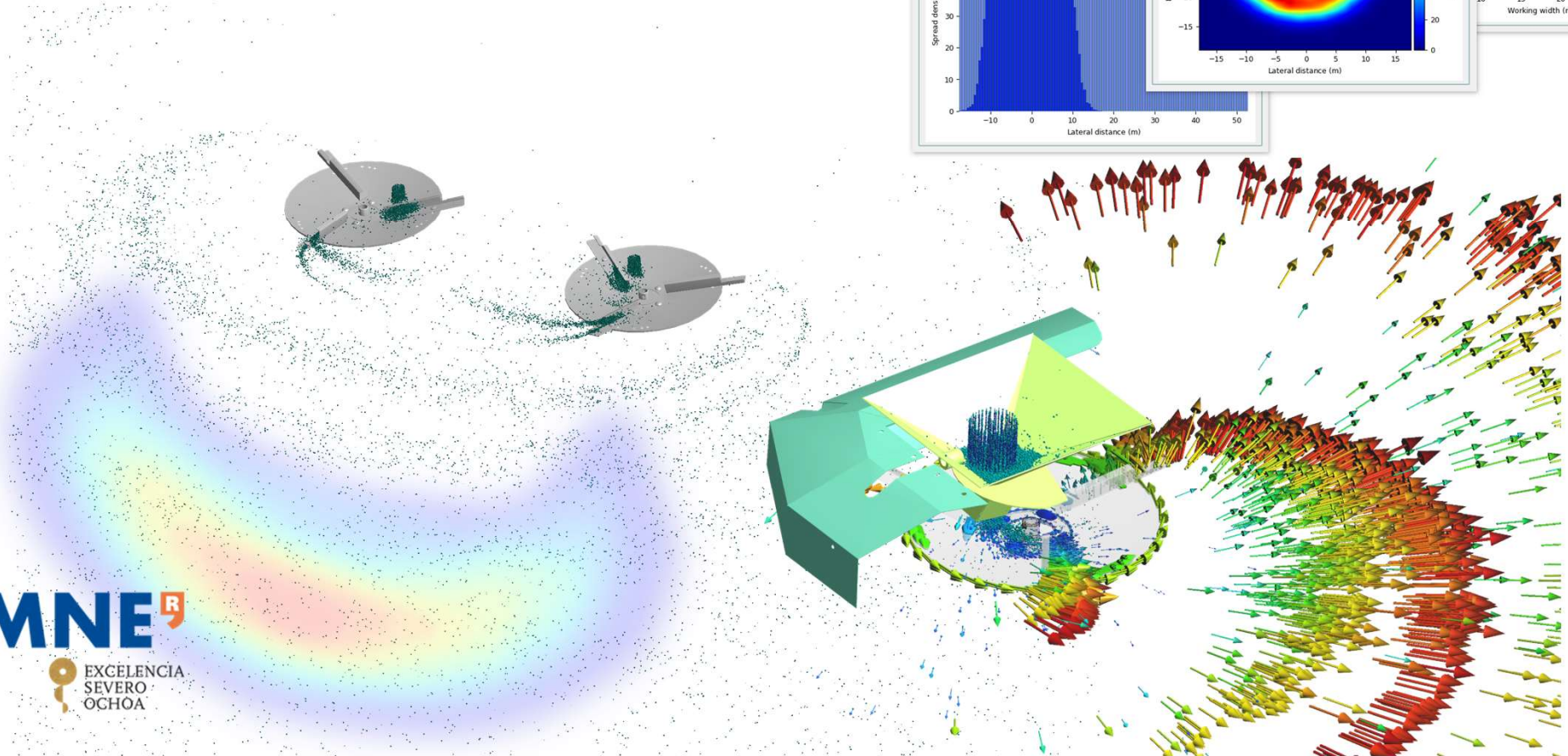
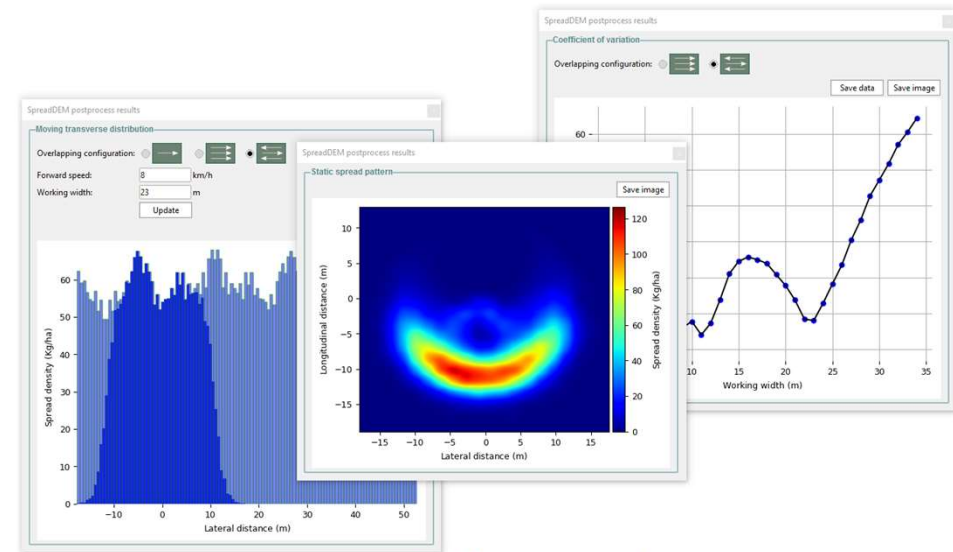
Few clicks to setup



Orientation Guide

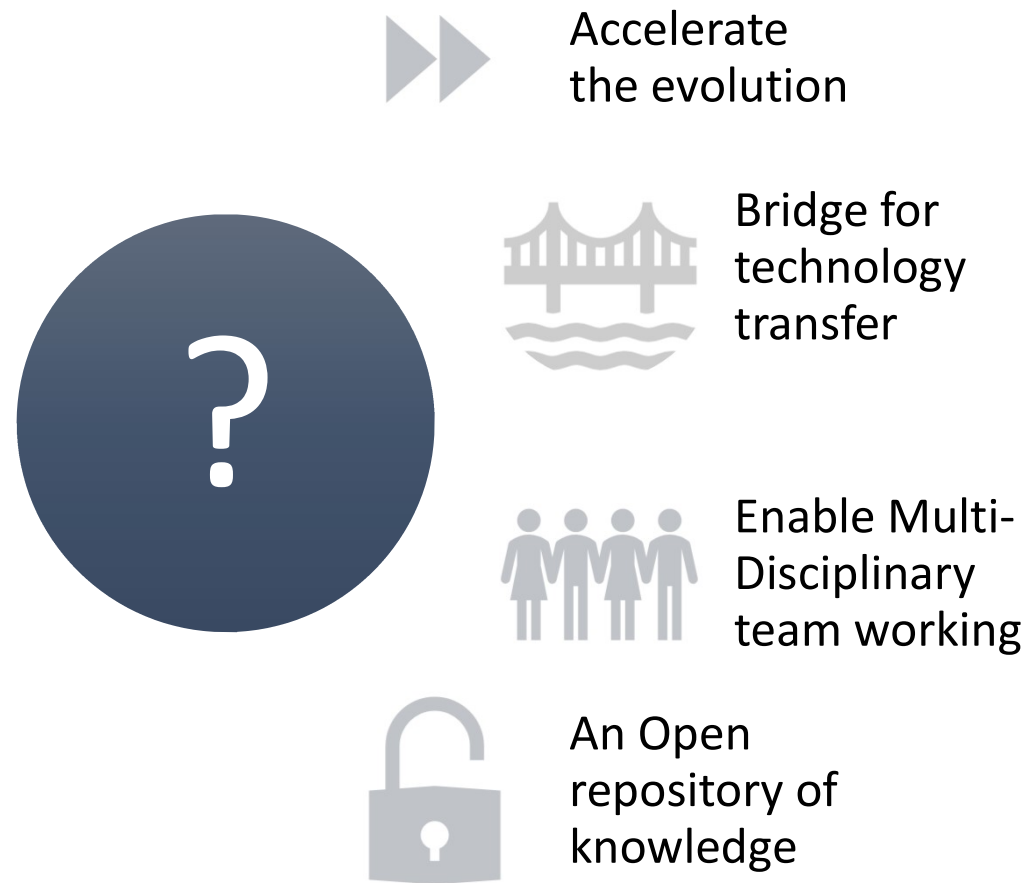


SpreadDEM

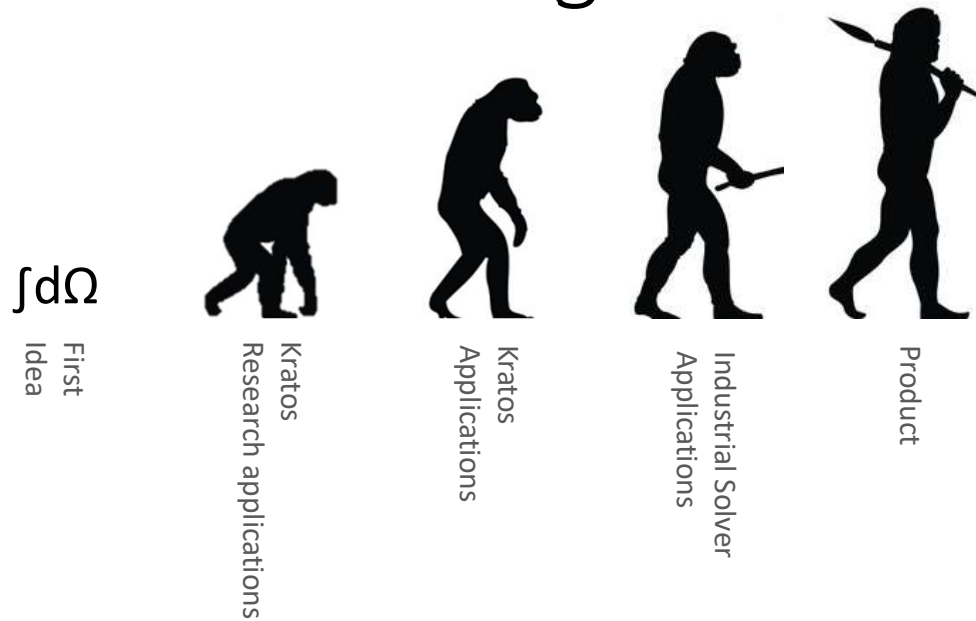


Final Words

Do we achieved our objectives?



Accelerating the Evolution



Kratos Research Applications

- Reusing already tested building blocks
- experimental development in Kratos



Kratos Applications

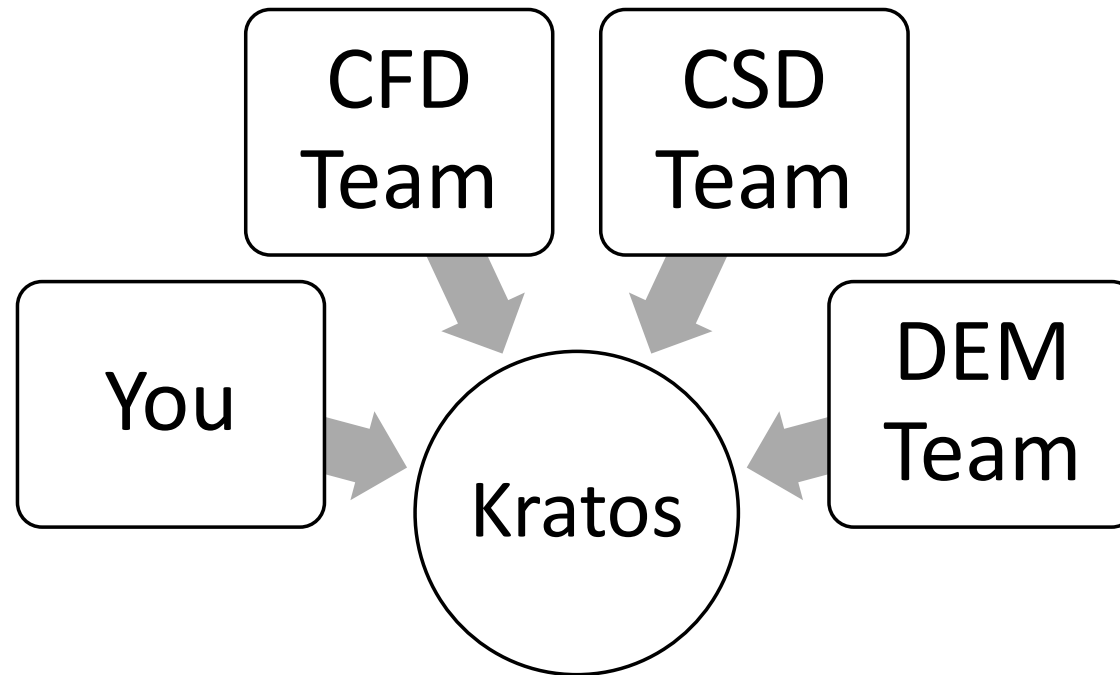
- Well maintained with community behind
- Assuring the behavior with QA



Variety of tools

- Many technologies are ready to use
- Several pre and post processors are adapted for Kratos

Team Working



In Kratos each developer focusses on his/her application and usually “talks” with Kratos or via Kratos to others

Bridge for Technology Transfer

Academy and Companies
can use it effectively to
interchange code,
knowledge and experts

No need to
have NDA
for code
sharing

Open Source

direct use
of the code
into
commercial
software

Permissive License

Already
trained
developers
to work in
industry

Community

Open Repository of Knowledge



Know How

- Kratos Community includes many experts in different areas of numerical methods:
 - In several physics like structural, cfd, particles, mpm, etc.
 - In many types of couplings like FSI, Thermo-mechanical, particles with FEM, etc.



Reusable Code

- Several solvers and Coupling tools
- Modular and generic
- Open and free



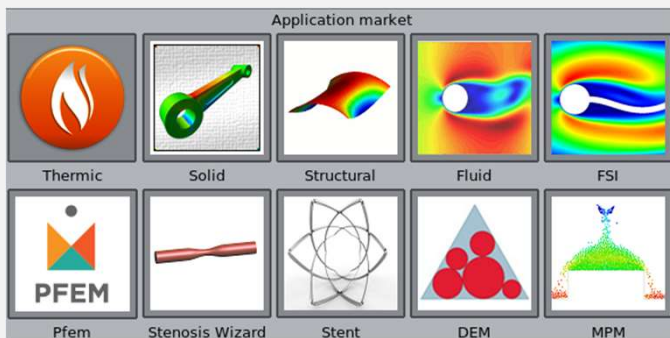
Solved Problems

- Large amount of working examples
- Several benchmarks
- Many examples from the literature

Where do we stand?

A consolidated and very active community
Collaborating from around the world

A flexible and modular framework with
extension to many different solutions

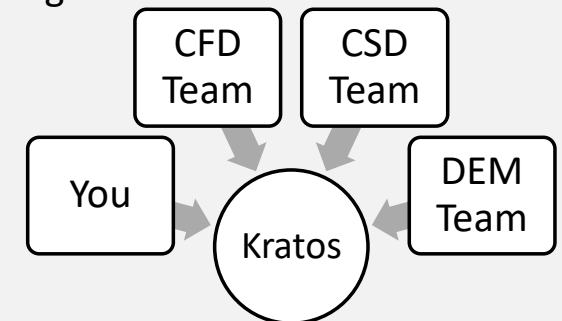


Modern development practices to
improve the quality of the code

- GitHub
- Code review
- Testing
- Hierarchy of committees ensuring the effectiveness of decisions

Prepared for multi-disciplinary team working

- The modularity avoids unnecessary dependency which results in less conflicts
- Kratos defines the standards and protocols of communication not persons in charge of coupling

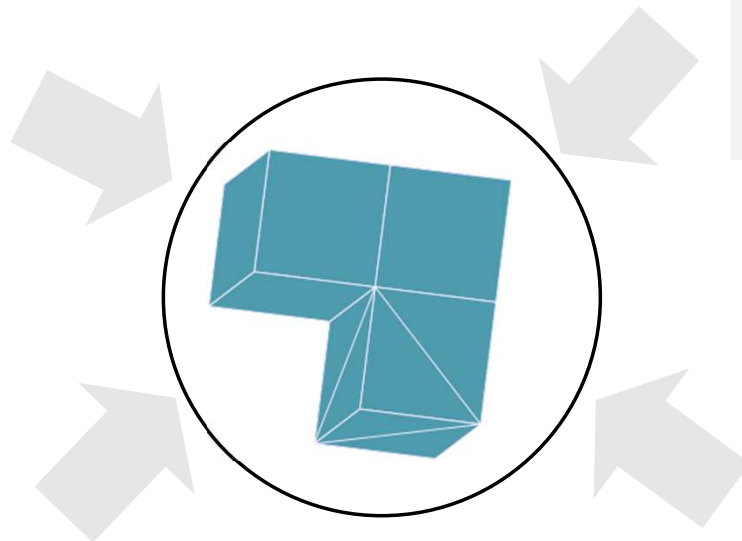


Why we are open to collaborate?

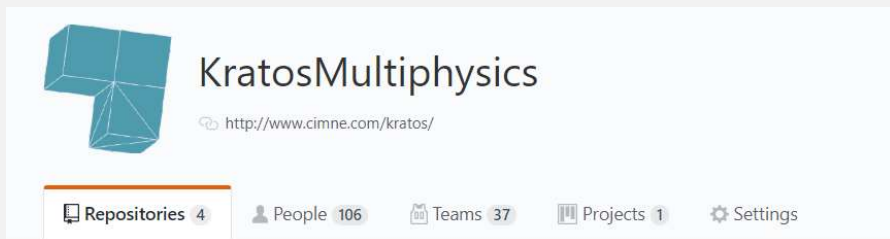
Different expertise



Innovative ideas



More contributors



More fun!



Hoping to have you on board!

<https://github.com/KratosMultiphysics/Kratos>

